

Preferences & Utility

2100010850-姜峻赫

2022.9.17

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1 Analyzing Economic Problems

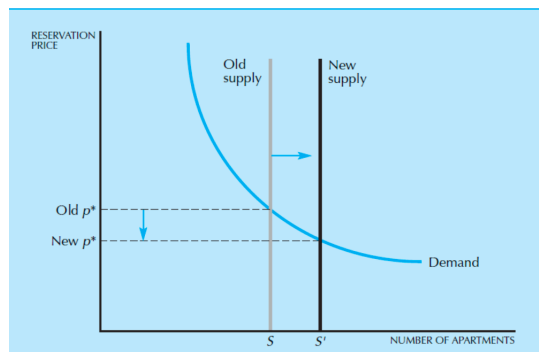
1.1 Questions micro seeks to answer

- What goods & services will be produced and how much?
- Who will produce them and how?
- Who will get them?

1.2 Three Analytical Tools

- Constrained Optimization
- Equilibrium Analysis
- Comparative Statics

An Example: Supply-Demand in Apartment Market



If the government collects taxes, in the short term, the supply-demand curve will not change from the Equilibrium. but in the long term, the curve will shift to the left (the total price contains the taxes)

2 Preferences

2.1 Concepts

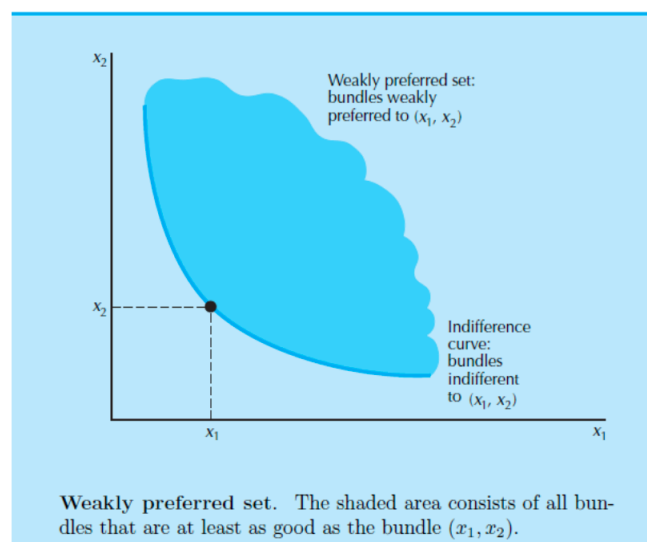
- Suppose there are two goods.
- Let $\mathbf{X} = \mathbb{R}_x^2$ be the consumption space
 - * each point in $\mathbf{X}$ is a pair (x_1, x_2)
 - * x_1 is the amount of good 1 and x_2 is the amount of good 2
- (x_1, x_2) is called a consumption bundle
- so $\mathbf{X}$ contains all possible consumption bundles
- Ultimately, we want to understand how consumers choose their best consumption bundles from those that are affordable
 - we must understand how consumers rank consumption bundles as to their desirability
 - such a ranking is called a consumer's Preference bundles in $\mathbf{X}$
- A preference, denoted by $\succsim$, is an ordering of consumption
- weak Preference

- $(x_1, x_2) \succeq (y_1, y_2)$
- the consumer finds (x_1, x_2) is at least as good as (y_1, y_2)
- indifference
 - $(x_1, x_2) \sim (y_1, y_2)$
 - both $(x_1, x_2) \succeq (y_1, y_2)$ and $(y_1, y_2) \succeq (x_1, x_2)$ hold
- strict Preference
 - $(x_1, x_2) \succ (y_1, y_2)$
 - $(x_1, x_2) \succeq (y_1, y_2)$ but not $(y_1, y_2) \succeq (x_1, x_2)$
- These are ordinal relations, only rank alternative bundles by order but don't specify magnitude of Preference differences

2.2 Basic Properties

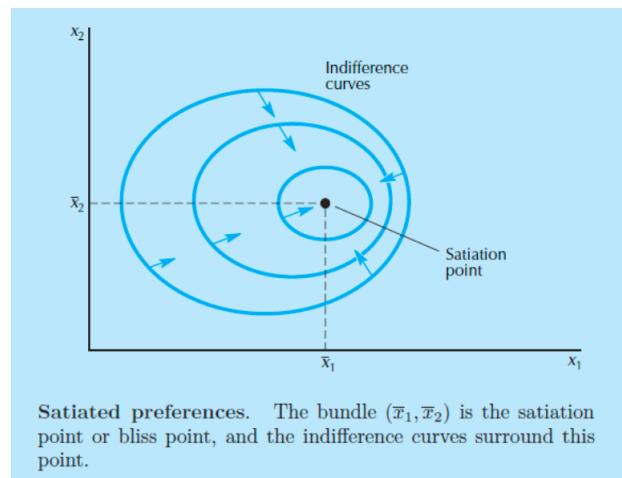
- *Complete*: for any pair of consumption bundles, (x_1, x_2) and (y_1, y_2) , either $(x_1, x_2) \succeq (y_1, y_2)$, or $(y_1, y_2) \succeq (x_1, x_2)$, or both hold.
- *Reflexive*: any bundle is at least as good as itself: $(x_1, x_2) \succeq (x_1, x_2)$
- *Transitive*: if $(x_1, x_2) \succeq (y_1, y_2)$ and $(y_1, y_2) \succeq (z_1, z_2)$, then $(x_1, x_2) \succeq (z_1, z_2)$, consumers' preferences are logically consistent. (otherwise he cannot decide how to choose the favorite goods)

2.3 indifference curve



- the consumer is indifferent between all bundles on that curve.
- Without additional properties on preferences, the indifference curves may behave badly.

An example of badly behaved indifference curves



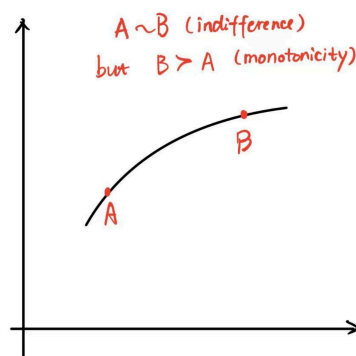
2.4 Additional Properties of Preferences

2.4.1 Monotonicity

- for all bundle (y_1, y_2) such that $y_1 \geq x_1$ and $y_2 \geq x_2$ with at least one inequality being strict, we have

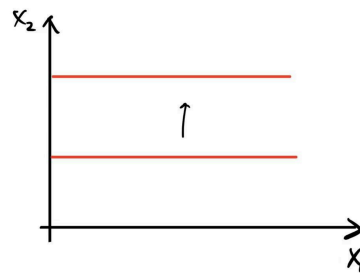
$$(y_1, y_2) \succ (x_1, x_2), \text{ more is better}$$

- is also called the assumption of non-satiation (非饱和性假设)
- implies that
 - negative slope for each indifference curve



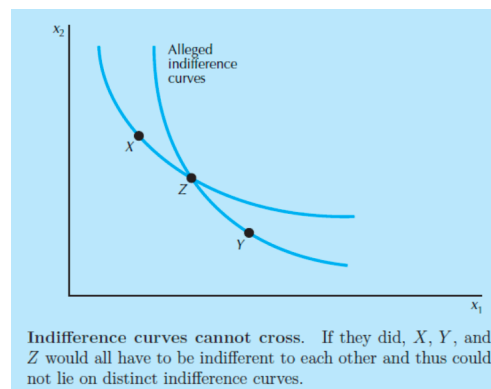
(可以导出矛盾! 所以不能向上倾斜!)

特殊情况: 中性商品 (可有可无)



这个图就可以表明 x_1 对我们是可有可无的

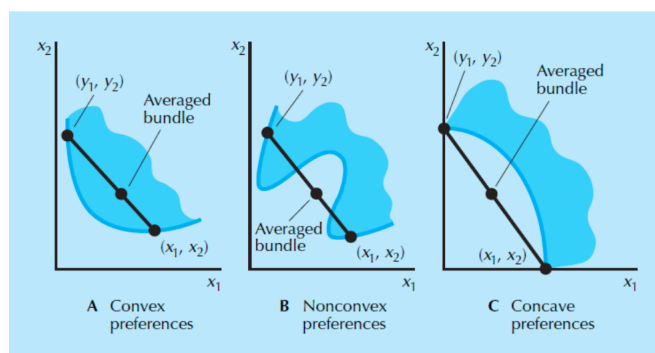
2. indifference curves are not “thick”(只可能是一条线, 而非一个区域, 区域中必然会违背越多越好的原则)
3. difference curves cannot cross (with transitivity)



违背了传递性和单调性!

2.4.2 Convexity (凸性)

- if $(y_1, y_2) \succeq (x_1, x_2)$, and $(z_1, z_2) \succeq (x_1, x_2)$, then $(ty_1 + (1-t)z_1, ty_2 + (1-t)z_2) \succeq (x_1, x_2)$ for any $t \in [0, 1]$
- 几个例子



- consumers want to diversify their consumption(人们往往喜欢平均而非极值)

3 Utility Functions

Utility function is a function that measures the level of satisfaction a consumer receives from any consumption bundle

3.1 Basic Properties

- basic idea: assign a real number to each consumption bundle

$$(x_1, x_2) \succeq (y_1, y_2) \Leftrightarrow u(x_1, x_2) \geq u(y_1, y_2)$$

$$(x_1, x_2) \sim (y_1, y_2) \Leftrightarrow u(x_1, x_2) = u(y_1, y_2)$$

$$(x_1, x_2) \succ (y_1, y_2) \Leftrightarrow u(x_1, x_2) > u(y_1, y_2)$$

- Note that if $\succeq$ can be represented by u , it satisfies completeness, reflexivity and transitivity
- 由于 utility 是序数关系，从而可以有不止一个效用函数来描述这种偏好。

- We can express the same preference relations by the different utility functions

Different ways to assign utilities.

Bundle	U_1	U_2	U_3
A	3	17	-1
B	2	10	-2
C	1	.002	-3

- Each of the three utility functions ranks the bundles in same way
- 因而通过单调变换得到的效用函数之间的效果是等价的，表示的是同一种效用偏好关系

if u represents $\succsim$, then the composite function $v : \mathbf{X} \rightarrow \mathbb{R}$ with

$$v(x_1, x_2) = f(u(x_1, x_2))$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function, is a utility function that represents the same preference.

f is usually referred to as a (positive) monotonic transformation

Example:

- ▶ $u(x_1, x_2) = x_1 x_2$
- ▶ $v_1(x_1, x_2) = x_1^2 x_2^2$
- ▶ $v_2(x_1, x_2) = 2x_1 x_2 + 10$

- Utility is an ordinal (ordering) concept, the only property of a utility assignment that is important is how it orders the bundles of goods

- If $u(x_1, x_2) = 6$ and $u(y_1, y_2) = 2$, then $(x_1, x_2) \succ (y_1, y_2)$, but (x_1, x_2) is not preferred three times as much as (y_1, y_2)
- The indifference curve is defined by the implicit function

$$u(x_1, x_2) = k$$

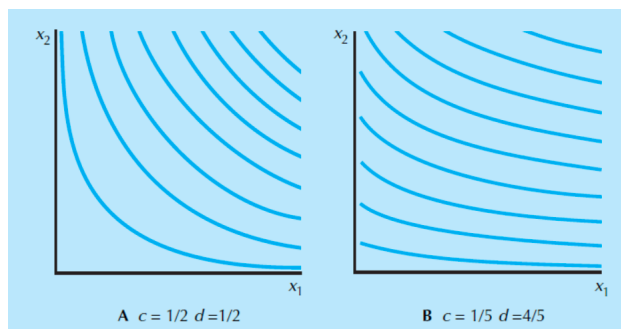
3.2 Some Examples

3.2.1 Cobb-Douglas preferences (科布道格拉斯效用函数)

The utility function is:

$$u(x_1, x_2) = x_1^c x_2^d, \quad c, d > 0$$

This is the most widely used utility form in economics



由于做单调性变换效用函数的效果不变，从而 CD 函数经常做单调变换来帮助运算：比如取对数（变成线性函数），或者进行某个幂次运算（把某个指数变成 1）。

3.2.2 Perfect substitutes (完全替代品)

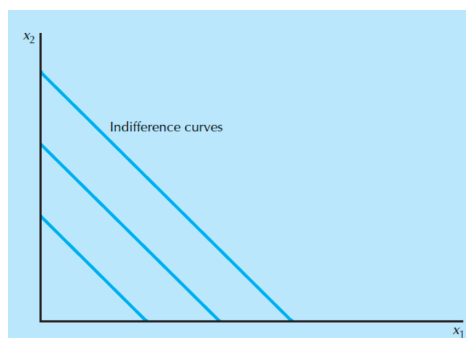
Two goods are perfect substitutes if the consumer is willing to substitute one good for the other at a constant rate.

往往是相同类的物品，两者之间基本完全一致，比如不同品牌的矿泉水，百事可乐和可口可乐。

- The utility function is

$$u(x_1, x_2) = ax_1 + bx_2$$

- $a > 0$ and $b > 0$ measure the “value” of goods 1 and 2 to the consumer
- the consumer is willing to substitute $\frac{a}{b}$ units of good 2 for one unit of good 1
- or the consumer is willing to substitute $\frac{b}{a}$ units of good 1 for one unit of good 2
- 也就是为了保持 u 不变，good 1 变动一个单位的时候，good 2 要变动多少个单位保持不变呢？

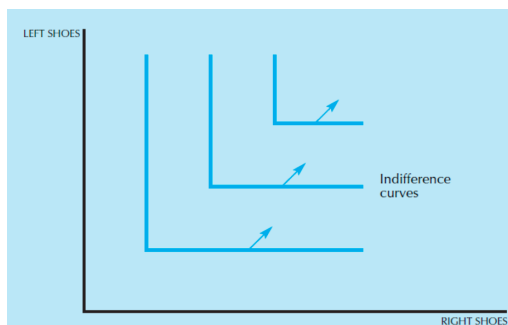


3.2.3 Perfect complements (完全互补品)

- Two goods are perfect complements if the consumer always wants to consume the goods in fixed proportions to each other
- 比如自行车的前轮和后轮，左脚鞋子和右脚鞋子
- The utility function is

$$u(x_1, x_2) = \min\{ax_1, bx_2\}$$

- $a > 0$ and $b > 0$ indicate the proportions in which the goods are consumed. The consumer always consumes goods 1 and 2 in ratio $\frac{b}{a}$
- 你给我两只右脚鞋和一只左脚鞋，跟一双鞋的效用是一样的（有点像木桶效应）



4 Marginal Utility

4.1 Concepts

- Marginal utility is the rate at which total utility changes as the level of consumption rises
- When the utility function $u(x_1, x_2)$ is differentiable, the marginal utility of good 1 is defined as the partial derivative of u with respect to x_1 :

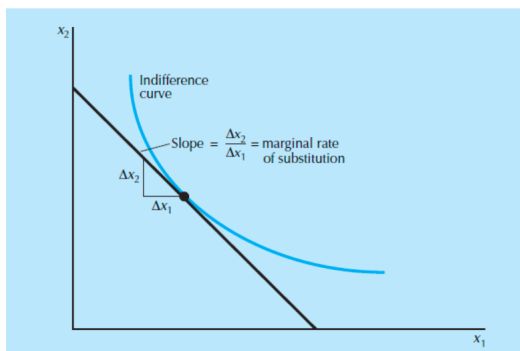
$$MU_i(x_1, x_2) = \frac{\partial u(x_1, x_2)}{\partial x_i}$$

4.2 Marginal Rate of Substitution(MRS)

- Marginal rate of substitution (MRS, 边际替代率) is the rate at which the consumer will give up one good to get more of another, holding the level of utility constant
- The $MRS_{12}(x_1, x_2)$ is the number of x_2 the consumer is willing to give up in order to obtain one more unit of x_1 at (x_1, x_2)
- Mathematically, it is the “slope” of the consumer’s indifference curve at (x_1, x_2)
- 直觉上来说就是：为何是边际替代函数之比？牺牲多少的 MU_2 才能弥补 MU_1 的损失呢？
- prove it: Taking total derivative of the function of the indifference curve that passes through (x_1, x_2) (the total utility doesn’t change)

$$MU_1 dx_1 + MU_2 dx_2 = 0 \Rightarrow \frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2}$$

- We use the absolute value of the negative slope as the MRS₁₂
- 如果无差异曲线下凸（通常情况,convex）则由于下凸，MRS₁₂ 会越来越少，也就是通常所说的边际效用递减？（物以稀为贵，水便宜钻石贵）



- e.g. Consider the Cobb-Douglas utility function $u(x_1, x_2) = x_1 x_2 \Rightarrow MU_i \equiv \frac{\partial u(x_1, x_2)}{\partial x_i} \Rightarrow$
 $MRS_{12}(x_1, x_2) = \frac{MU_1}{MU_2} = \frac{x_2}{x_1}$
- 检查上式的方法： x_1 换 x_2 ，边际效用递减， x_1 增加则 MRS_{12} 绝对值下降（只是检查，因为实际上 x_2 也在变）

Budget Constraint & Consumer Choice

2100010850-姜峻赫

2022.9.24

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1 Budget Constraint

Economists assume that when a person is faced with a choice among several possible options, he or she will choose the one that yields the highest utility (utility maximization)

- economists assume that people know their own minds and make choices consistent with their preferences
- people are constrained in their choices by the size of their incomes and other resources

1.1 Budget Constraint

- Suppose the price of good 1 is $p_1 > 0$ and the price of good 2 is $p_2 > 0$. Suppose the total amount of money that the consumer can spend on goods 1 and 2 is m (called the consumer's income)
- If you use the credit card to buy things, actually you are buying things with the income you get in the future.

1.2 Budget set

- if the consumer can afford the bundle $(x_1, x_2) \in R_+$, we have

$$p_1x_1 + p_2x_2 \leq m$$

- is called the consumer's Budget constraint
- The Budget set is called the set of all bundles that are affordable to the consumer
- It can be expressed as :

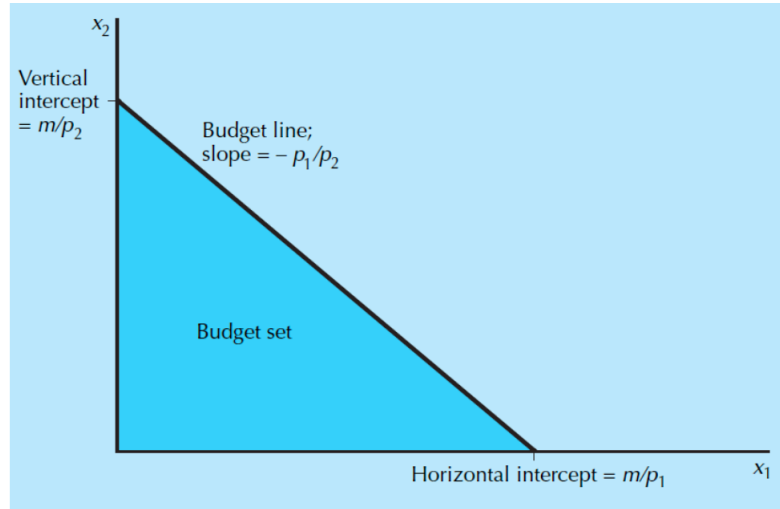
$$\{(x_1, x_2) | p_1x_1 + p_2x_2 \leq m\}$$

- The Budget line is the set of all bundles that just exhaust all the consumer's money

- it can be expressed as

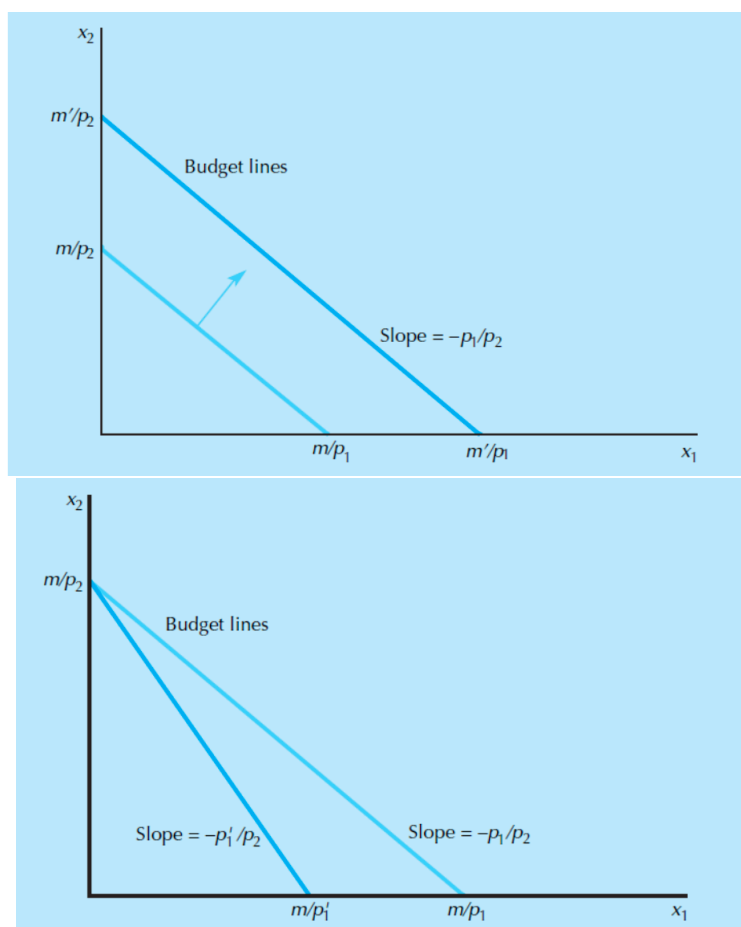
$$\{(x_1, x_2) | p_1 x_1 + p_2 x_2 = m\}$$

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1$$



- The budget line intersects the x_1 - axis (or x_2 - axis) at $\frac{m}{p_1}$, the amount of good 1 that the consumer can afford if he spends all his money on good 1
- The slope of the budget line infers the price ratio between goods 1 and 2
- An increase in m will increase the intercepts and not affect the slope of the line
- parallel shift outward of the budget line (budget set expands)
- 比如爸妈心情好突然给你多打钱了
- An increase in p_1 will not change the vertical intercept, but p_1/p_2 becomes larger
- The budget line is steeper (电影涨价了, 那我们要用更多的奶茶去换电影, 从斜率上来看 p_1 变大, 要变陡; 从截距上来看, income 不变, 但是能买的电影少了, 所以截距点要向着原点平移)

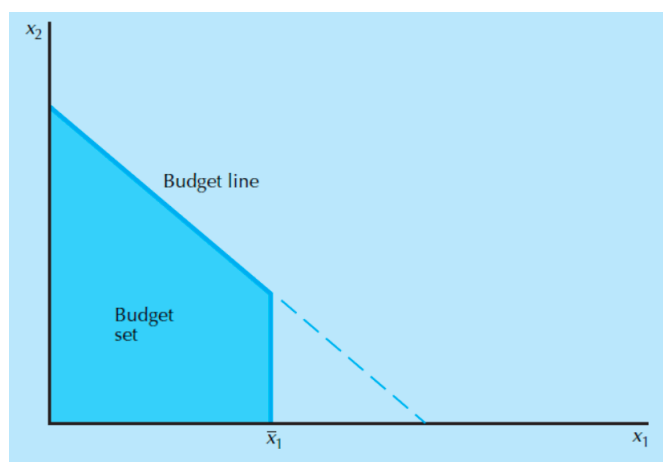
- 如果两个商品同时上涨同样的幅度 m ，则斜率不变，这跟我的 income 缩小 $1/m$ 是同等的情况 (同一条线)(呜呜，穷死了)
- examples



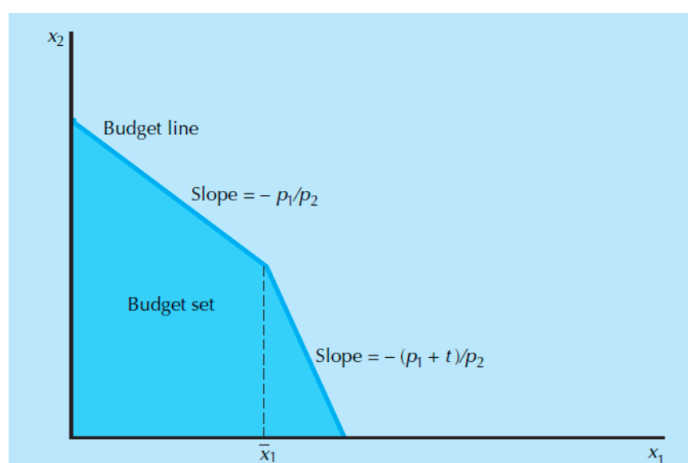
- 有时候我们会把一个商品的价格单位化为 1，称作标准计价物 (x_1 价格变成 p_1/p_2 , m 变成 m/p_2)，这是因为我们关注的是两者的相对关系，现实生活中钱就充当了计价物，只不过我们不在乎计价物自身的价值，而是关注别的东西和它的比率是多少。
- 如果我们的 income 增加，同时一个物品的价格提高了，那么预算约束线就会先向外平移，再进行倾斜。(比如我们充会员花了 income 但是我们后续的商品价格变低了，就是如上所述)

2 Rationing

- Suppose good 1 were rationed so that no more than $\bar{x}_1$ could be consumed by a given consumer



- Suppose a consumer had to pay a tax t on all consumption in excess of $\bar{x}_1$

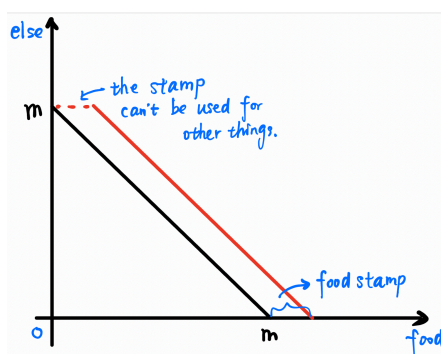


- 搭售策略也是特殊的预算限制线。

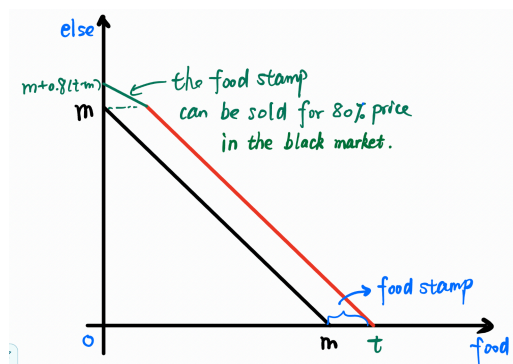
2.1 Special situation

2.1.1 Food Stamps(定向消费券)

popular income support program, coupons given to poor, can be legally exchanged only for food, so it will look like:

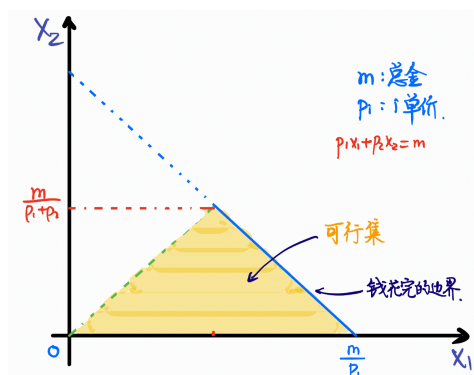


But if you can trade it in the black market with a 80% price, then it will look like:



2.1.2 Bundling(捆绑销售)

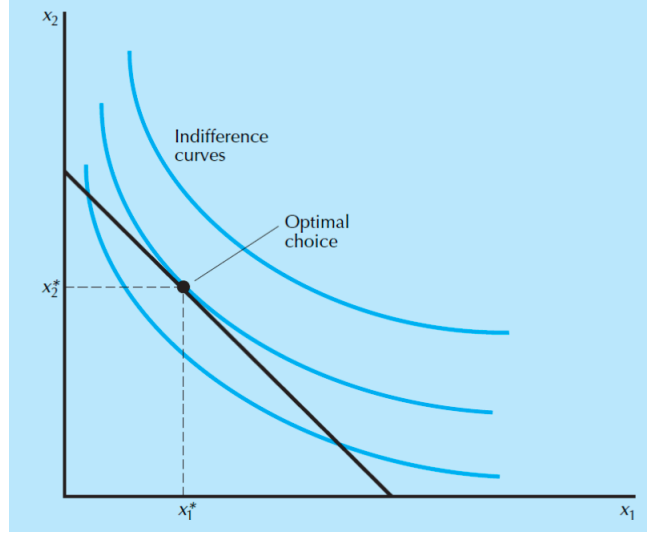
x_2 could and only could be sold to a costumer when the costumer buy x_1 , then we cannot buy more x_2 than x_1 , so we can only buy things in the yellow area:



3 Utility Maximization

- We now put together the budget set and the theory of preferences to examine the optimal choice of consumers
- consumers choose the most preferred bundle from their budget sets
- Suppose the utility function of a consumer is $u : \mathbb{R}^2_+ \rightarrow \mathbb{R}$. Let p_1 and p_2 be the prices and m be his wealth. Then
- If the (x_1^*, x_2^*) , solves the above utility maximization problem, we call (x_1^*, x_2^*) an optimal choice
- interior optimum (内部最优): $x_1^* > 0, x_2^* > 0$
- boundary optimum (边界最优, 边角解) (otherwise)

3.1 Graphic Demonstration



- At point (x_1^*, x_2^*) indifference curve is tangent to the budget line so that

$$\frac{p_1}{p_2} = MRS_{12}(x_1^*, x_2^*)$$

- $MRS_{12}(x_1^*, x_2^*)$ measures the amount of good 2 that the consumer is willing to exchange for one unit of good 1 without changing his utility level
- $MRS_{12}(x_1^*, x_2^*)$ measures the amount of good 2 that the consumer can exchange in the market for one unit of good 1
- if $MRS_{12}(x_1^*, x_2^*) < \frac{p_1}{p_2}$, then the consumer would like to consume more of good 2
- if $MRS_{12}(x_1^*, x_2^*) > \frac{p_1}{p_2}$, then the consumer would like to consume more of good 1

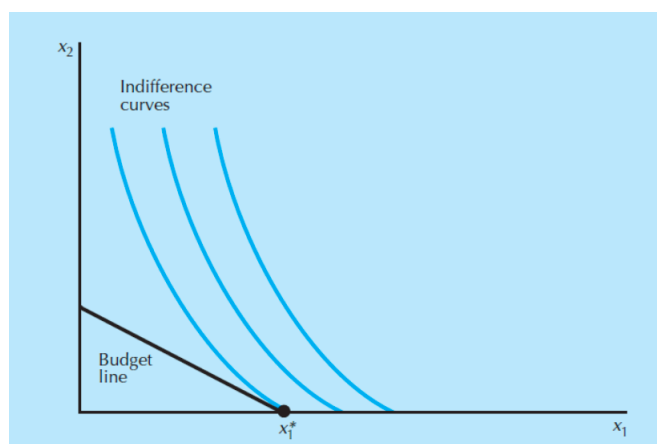
3.1.1 Another way

•

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$$

- The marginal utility of the last penny spent on each good is the same(花在两件商品的每块钱带来的效用是相等的)
- consumer is indifferent about how to spend the extra one unit of money
- if $\frac{MU_1}{p_1} > \frac{MU_2}{p_2}$, the consumer would like to save one unit of money on good 2 and spend it on good 1
- if $\frac{MU_1}{p_1} < \frac{MU_2}{p_2}$, the consumer would like to save one unit of money on good 1 and spend it on good 2
- 这其实是一个很有意思的事情：两个从商店里面出去的两个“满意的”消费者，事实上每件商品的价值恰好等于了他心目中的价值，那么在商店之外这两个消费者之间是不会发生交换的。

3.1.2 Boundary optimum



- 这个人为了 x_1 可以付出很多很多 x_2 ，从而陡，从而最优点是在 x_1 的角点
- 遇到角点解就去分析两个角点

3.2 Method of Lagrange Multipliers

The consumer's optimization problem:

$$\max_{x_1, x_2 \geq 0} u(x_1, x_2), \quad \text{when } p_1 x_1 + p_2 x_2 \leq m$$

The Lagrange function (拉格朗日函数) is

$$\mathcal{L} = u(x_1, x_2) + \lambda(m - p_1x_1 - p_2x_2)$$

where λ is the Lagrange multiplier (拉格朗日乘数), 其实 λ 是影子价格, 衡量的是金钱的边际效用, 指的是预算总额变动的一单位 (约束放松一点点), 那么目标函数 L 最大值的改变量。(其实也是增加一单位 x_1 所得到的效用和在市场上消耗的金钱之比)

If (x_1^*, x_2^*) is the optimal choice, it must satisfy:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_1} &= \frac{\partial u(x_1^*, x_2^*)}{\partial x_1} - \lambda p_1 = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} &= \frac{\partial u(x_1^*, x_2^*)}{\partial x_2} - \lambda p_2 = 0 \\ \frac{\partial u(x_1^*, x_2^*)}{\partial x_1} &= \lambda p_1 \\ \frac{\partial u(x_1^*, x_2^*)}{\partial x_2} &= \lambda p_2 \\ \Rightarrow \frac{p_1}{p_2} &= MRS_{12}(x_1^*, x_2^*)\end{aligned}$$

One obvious advantage of the method of Lagrange multipliers is that it is general enough to deal with choices with more than two goods

Consider

$$\max_{x_i > 0} u(x_1, x_2, \dots, x_n), \quad \text{when} \quad \sum_{i=1}^n p_i x_i \leq m$$

If $(x_1^*, \dots, x_n^*)$ is the optimal choice, then it satisfies:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u(x_1^*, \dots, x_n^*)}{\partial x_i} - \lambda p_i = 0$$

3.2.1 Some General Properties

钱花完定理!

- Note the optimal choice appears exactly on the consumer's budget lines.
- at the optimal choice, the consumer's budget constraint is binding:
 $p_1 x_1^* + p_2 x_2^* = m$

- an implication of the monotonicity of preference
- 也就是一定把钱花完了
- To see why, let's suppose $p_1x_1^* + p_2x_2^* < m$ by contradiction
 - there must exist a small $\epsilon > 0$ such that

$$p_1(x_1^* + \epsilon) + p_2(x_2^* + \epsilon) < m$$

- $(x_1^* + \epsilon, x_2^* + \epsilon)$ is also affordable for small enough $\epsilon > 0$
- by monotonicity, $(x_1^* + \epsilon, x_2^* + \epsilon)$ is strictly preferred to (x_1^*, x_2^*) , contradicting the optimality of (x_1^*, x_2^*)

解唯一定理!

Another feature of the previous graph is that the optimal choice is unique

- not true in general. e.g.: $u(x_1, x_2) = 0$ for all (x_1, x_2) , 啥都不想买, 那还不如不买!
- is true if the preference is strictly convex (严格凸)
- To see why, let's suppose, by contradiction, both (x_1^*, x_2^*) and (y_1^*, y_2^*) are optimal and $(x_1^*, x_2^*) \neq (y_1^*, y_2^*)$.
- $(\lambda x_1^* + (1 - \lambda)y_1^*, \lambda x_2^* + (1 - \lambda)y_2^*)$ is affordable for any $\lambda \in (0, 1)$
- This is because that:

$$x_1^* + x_2^* \leq m, \quad y_1^* + y_2^* \leq m$$

$$\text{then } \lambda(x_1^* + x_2^*) + (1 - \lambda)(y_1^* + y_2^*) \leq \lambda m + (1 - \lambda)m = m$$

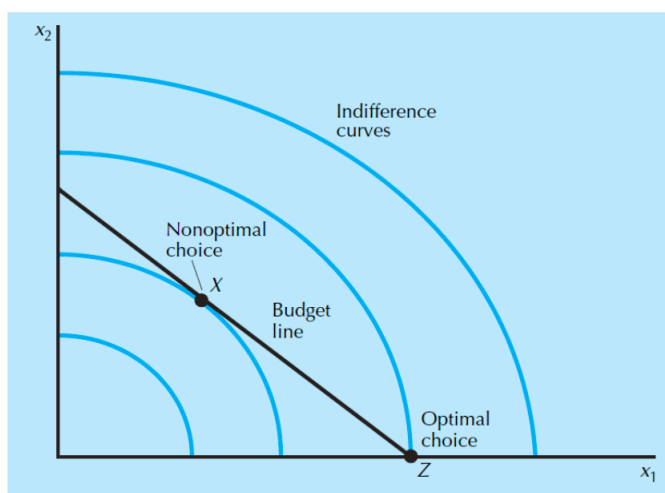
$$\text{that's } \lambda x_1^* + (1 - \lambda)y_1^* + \lambda x_2^* + (1 - \lambda)y_2^* \leq m$$

- strict convexity implies $(\lambda x_1^* + (1 - \lambda)y_1^*, \lambda x_2^* + (1 - \lambda)y_2^*) \succ (x_1^*, y_1^*)$, contradicting the optimality of (x_1^*, y_1^*) (这是由下凸函数定义要求的)

Attention

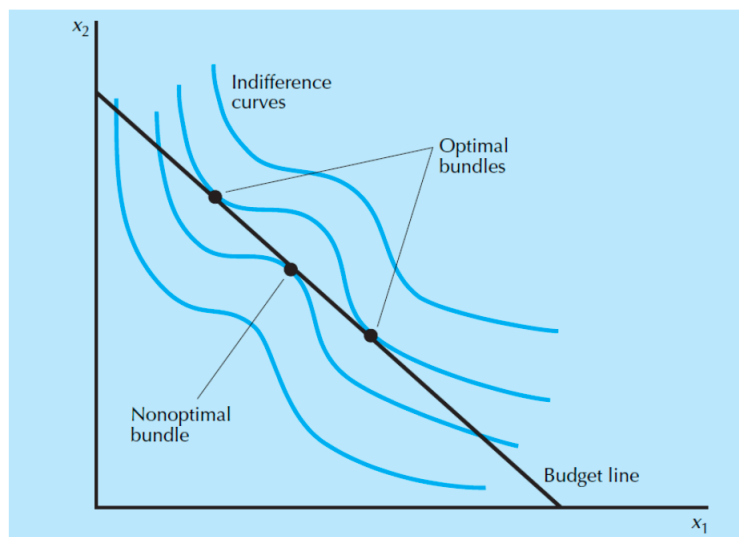
- Note that the first order condition is a necessary condition: a condition that any interior optimum must satisfy. (必要条件而不一定充分)
- Under our assumption that u is concave, it is also sufficient: any interior bundle that satisfies it must be optimal.
- For non-concave utility functions, we should use the first order condition with caution
- 反例

• Optimal choice with concave preferences



- the actual optimal choice is point Z
- 检验角点解要看是否符合经济学直观理解：这个曲线随着 x_1 增加而向下弯，这说明随着 x_1 增加，交易者愿意付出更多的 x_2 来交换 x_1 ，因而最大效用点应该就是 x_1 最多的时候

Non-convex preferences may lead to more than one tangency



- we have 2 optimal choice in this gram

4 Some Examples

4.1 Cobb-Douglas Utility

- Consider the Cobb-Douglas utility function

$$u(x_1, x_2) = c \log x_1 + d \log x_2$$

- The $MRS_{12}(x_1, x_2)$ is $\frac{cx_2}{dx_1}$
- If prices are p_1 and p_2 , then the first order condition is

$$\frac{cx_2^*}{dx_1^*} = \frac{p_1}{p_2}$$

- Because the budget constraint must be binding, we know

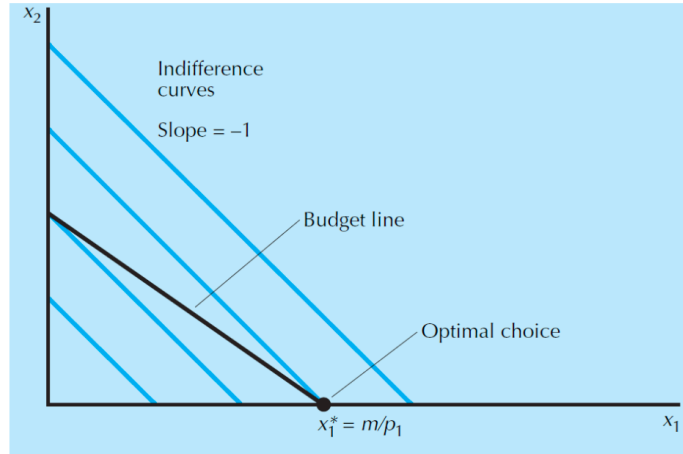
$$p_1 x_1 + p_2 x_2 = m$$

- we have

$$x_1^* = \frac{cm}{(c+d)p_1}, x_2^* = \frac{dm}{(c+d)p_2}$$

4.2 Perfect Substitutes

$$u(x_1, x_2) = x_1 + x_2$$



- More generally, if $u = ax_1 + bx_2$, we have

$$MU_1 = a, MU_2 = b$$

- if $\frac{a}{b} \geq \frac{p_1}{p_2}$ (也就是消费者心中 x_1 的价格要高于市场上 x_1 的价格), consumer will only consume x_1

$$x_1^* = \frac{m}{p_1}, \quad x_2^* = 0$$

- if $\frac{a}{b} \leq \frac{p_1}{p_2}$ (也就是消费者心中 x_2 的价格要高于市场上 x_2 的价格), consumer will only consume x_2

$$x_2^* = \frac{m}{p_2}, \quad x_1^* = 0$$

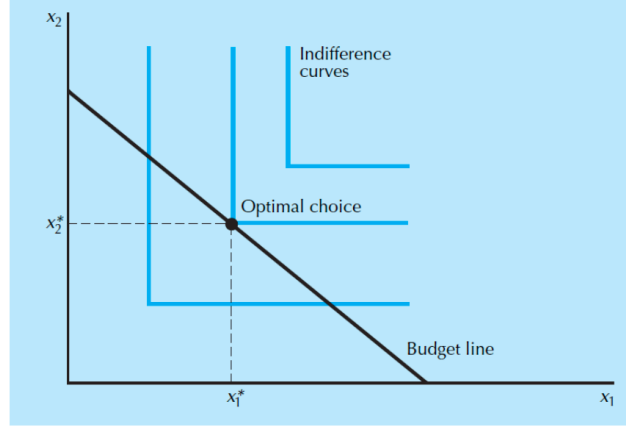
- if $\frac{a}{b} = \frac{p_1}{p_2}$ (也就是消费者心中此时二者按比例持有完全相同时), the consumer will be indifferent between any combination of the goods as long as he spends all his money

$$x_1^* = \begin{cases} \frac{m}{p_1} & \frac{a}{b} \geq \frac{p_1}{p_2} \\ \text{any number in } \left(0, \frac{m}{p_1}\right) & \frac{a}{b} = \frac{p_1}{p_2} \\ 0 & \frac{a}{b} \leq \frac{p_1}{p_2} \end{cases}$$

- 完全相互替代的产品对于价格变动非常敏感, 变动十分迅速

4.3 Perfect complements

$$u(x_1, x_2) = \min\{x_1, x_2\}$$



- More generally, if $u(x_1, x_2) = \min\{ax_1, bx_2\}$, tangency condition will not hold since indifference curves are not continuous (slope is either zero, infinity or not defined)
- Consumer will always be on the corner of the indifference curve consuming in fixed proportions

$$ax_1 = bx_2 \quad \& \quad p_1x_1 + p_2x_2 = m$$

$$\Rightarrow x_1^* = \frac{bm}{bp_1 + ap_2}, \quad x_2^* = \frac{am}{bp_1 + ap_2}$$

4.4 Method of Lagrange Multipliers

The consumer's optimization problem:

$$\max_{x_1, x_2 \geq 0} u(x_1, x_2), \quad \text{when } p_1x_1 + p_2x_2 \leq m$$

The Lagrange function (拉格朗日函数) is

$$\mathcal{L} = u(x_1, x_2) + \lambda(m - p_1x_1 - p_2x_2)$$

where λ is the Lagrange multiplier (拉格朗日乘数), 其实 λ 是影子价格, 衡量的是金钱的边际效用, 指的是预算总额变动的一单位 (约束放松一

点点), 那么目标函数 L 最大值的改变量。(其实也是增加一单位 x_1 所得到的效用和在市场上消耗的金钱之比)

If (x_1^*, x_2^*) is the optimal choice, it must satisfy:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{\partial u(x_1^*, x_2^*)}{\partial x_1} - \lambda p_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{\partial u(x_1^*, x_2^*)}{\partial x_2} - \lambda p_2 = 0$$

$$\frac{\partial u(x_1^*, x_2^*)}{\partial x_1} = \lambda p_1$$

$$\frac{\partial u(x_1^*, x_2^*)}{\partial x_2} = \lambda p_2$$

$$\Rightarrow \frac{p_1}{p_2} = MRS_{12}(x_1^*, x_2^*)$$

One obvious advantage of the method of Lagrange multipliers is that it is general enough to deal with choices with more than two goods

Consider

$$\max_{x_i > 0} u(x_1, x_2, \dots, x_n), \quad \text{when} \quad \sum_{i=1}^n p_i x_i \leq m$$

If $(x_1^*, \dots, x_n^*)$ is the optimal choice, then it satisfies:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u(x_1^*, \dots, x_n^*)}{\partial x_i} - \lambda p_i = 0$$

Consumer's Choice & Comparative Statics

姜峻赫

2022.10.8

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1 Individual Demand Function

- This section studies how people change their optimal consumption choices when income or the prices of the good changes
 - such analysis is known as comparative statics
 - “comparative”: compare situations before and after changes
 - “statics”: only examine the equilibrium (均衡) choice rather than the adjustment process

- The goal is to analyze the behavior of demand functions

Suppose the consumer's preference is monotone and strictly convex.

- the consumer's optimal consumption choice given any prices $p = (p_1, p_2)$ and income m will be unique and on his budget line

The consumer's demand functions give the optimal amounts of each of the goods as a function of the prices and income:

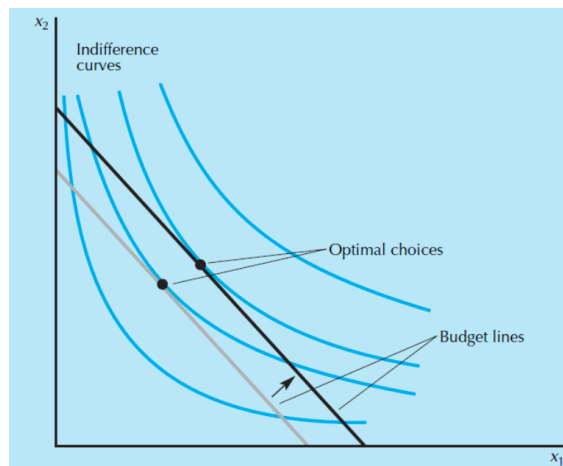
$$x_1(p_1, p_2, m), x_2(p_1, p_2, m)$$

or simply

$$x_1(p, m), x_2(p, m)$$

2 Income Effect

- Income effect refers to the change in optimal consumption due to a change in income, holding prices fixed
- also called wealth effect



2.1 Normal Good

- Normally, we think that the demand for each good would increase when income increases
 - normal goods
 - $\frac{\partial x_1}{\partial m} > 0$

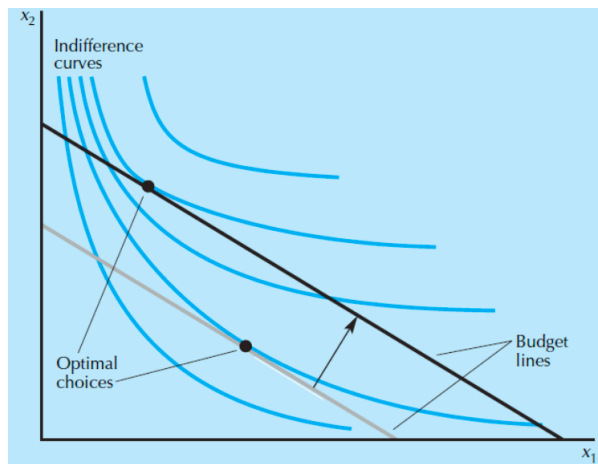
2.2 Inferior Good

- It is also possible that an increase of income results in a reduction in the consumption of one of the goods
 - inferior good (低档品)
 - $\frac{\partial x_1}{\partial m} < 0$
- Low quality goods are usually inferior goods.(但是 Inferior goods 不一定是低档品, 比如便宜的车、住的酒店的级别、买的廉价衣服等等, 不一定是质量差, 可能只是因为不在你的偏好上而不得不消费的东西。)
- Whether a good is inferior depends on the income level that we are examining
- very poor people might consume more low-quality goods as their income increases
- after a point, the consumption of low quality goods would probably decline as income continued to increase
- 也是有阶段性的, 在一定的收入增加的时候, 可能消费不会减少, 这个阶段是低档品, 但是钱更多了可能会减少。所以是不是劣等品有阶段性。
- 可不可能存在财富同时消费两种商品 (花完), 两种商品都是劣等品?

—

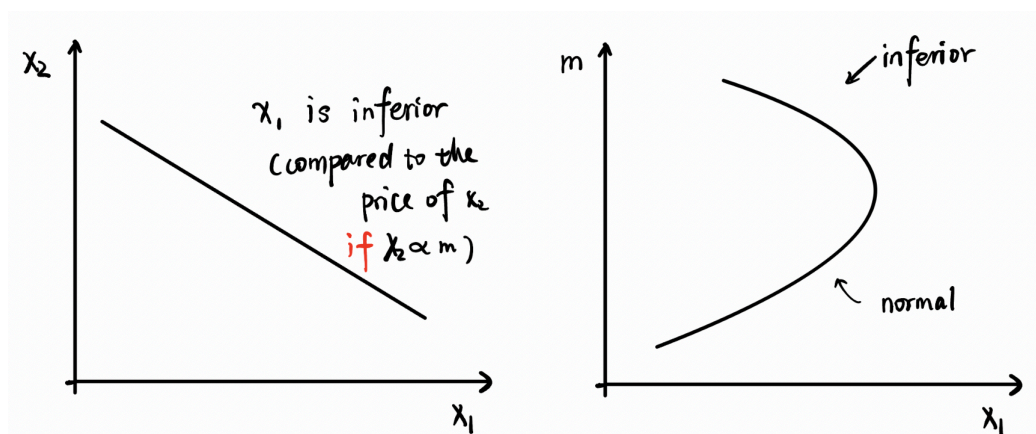
$$p_1 \frac{dx_1}{dm} + p_2 \frac{dx_2}{dm} = 1$$

- 两种商品都是劣等品的话, 预算约束线对 m 求导之后不满足等于 1, 事实上, 如果两种都是劣等品, 那么最优解都是收入增加的时候买的东西都减少, 那么钱一定会剩下
- 如果钱没花完, 去储蓄了, 那就增加一个商品“储蓄”, 同样, 这三种商品不可能都是劣等品。
- If $u(x, y) = u_1(x) + u_2(y)$ where $u'_i > 0, u''_i < 0$, Q: Is it possible that x or y is inferior?
- Assume x is inferior, (x_1, y_1) is the optimal choice, then $\frac{MU_x|_{x=x_1}}{MU_y|_{y=y_1}} = \frac{p_x}{p_y}$ if $m \uparrow$, $x \downarrow$, the other optimal choice will be (x_2, y_2) , where $x_2 < x_1, y_2 > y_1$, $\frac{MU_x|_{x=x_2}}{MU_y|_{y=y_2}} = \frac{p_x}{p_y}$, but we have $u''_i < 0$, then MU_i is monotonous decreasing, so $MU_x|_{x=x_2} > MU_x|_{x=x_1}, MU_y|_{y=y_1} < MU_y|_{y=y_2}, \frac{MU_x|_{x=x_2}}{MU_y|_{y=y_2}} > \frac{p_x}{p_y}$, this is contradictory to the equation.
- 劣等品的例子: x_1

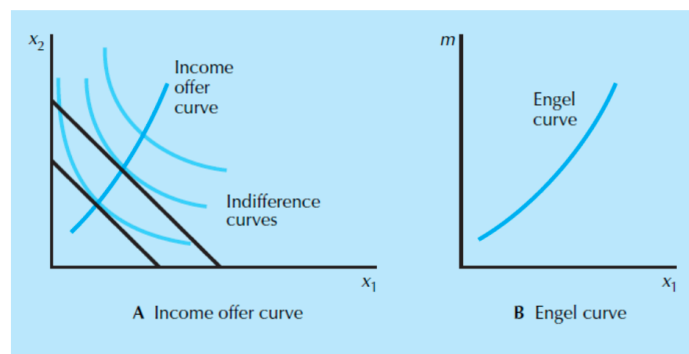


2.3 Income consumption curve and Engle curve

- 把最优解连起来，就是收入消费线（Income consumption curve）或收入提供曲线（Income offer curve）或收入扩张曲线（income expansion path）
- 几种特殊的线：下降的收入消费线和弯折的收入消费线（对应着劣等品和特定时段改变的劣等品）



- 单纯给出第一个图，只能说明 x_1, x_2 中有一个劣等品，但是不知道是哪个（假设 x_2 是正常品，则 x_2 和 m 呈正比关系，而我们知道劣等品和 m 的关系是随着 m 增长而减少消费的，从而 x_1 是劣等品。如果假设 x_1 正常品，则得到相反的结论）
- 第一个图不知道具体的斜率关系，如果有了无差异曲线图，那么就能确定出谁是真正的劣等品。
- Engel curve: 对特定收入和特定商品的关系曲线（graph of quantity demanded and income）
- Engel law: 随着收入增加，食品的消费占比会下降（不代表食品是劣等品，因为食品消费没减少，只是占比减少）



2.4 Income Elasticity of Demand

- We can further divide normal goods into different sets based on the relative consumption of different goods changes as income changes
 - Income elasticity of demand (需求的收入弹性) is defined as

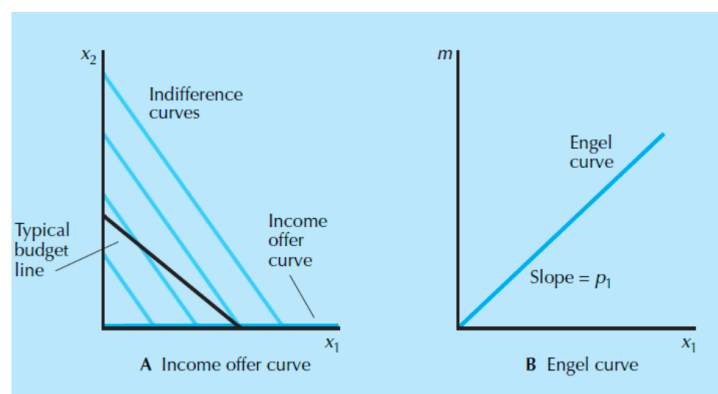
$$e_m = \frac{\frac{dQ_d}{Q_d}}{\frac{dm}{m}} = \frac{m}{Q_d} \frac{dQ}{dm}$$

- Luxury goods(奢侈品): $e_m > 1$
- Necessities: $0 < e_m < 1$
- 可以通过画一条 45 度线来判断:

2.5 Examples

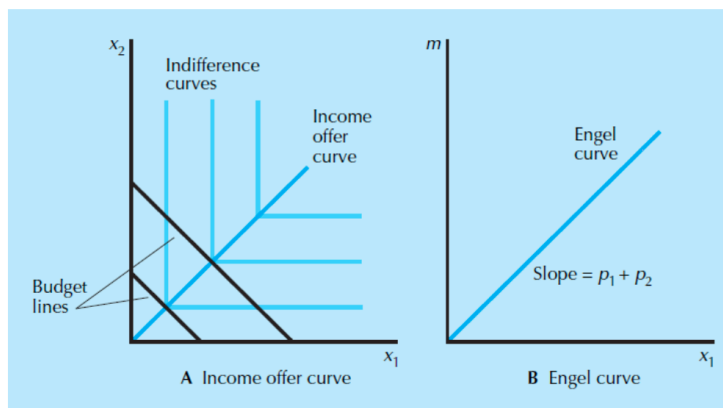
2.5.1 Perfect substitutes

$$u(x_1, x_2) = x_1 + x_2 \quad \text{and} \quad p_1 < p_2 : x_1 = m/p_1 \quad \text{or} \quad m = p_1 x_1$$



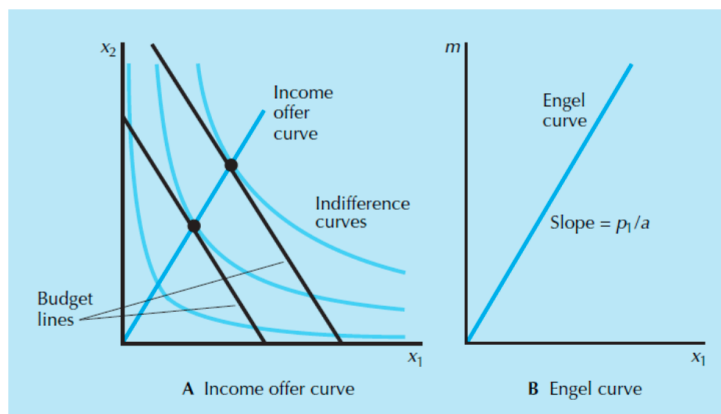
2.5.2 Perfect complements

$$u(x_1, x_2) = \min\{x_1, x_2\}: x_1 = m/(p_1 + p_2) \text{ or } m = x_1(p_1 + p_2)$$



2.5.3 Cobb-Douglas Preferences

$$u(x_1, x_2) = x_1^a x_2^{1-a}: x_1 = am/p_1 \text{ or } m = p_1 x_1/a$$

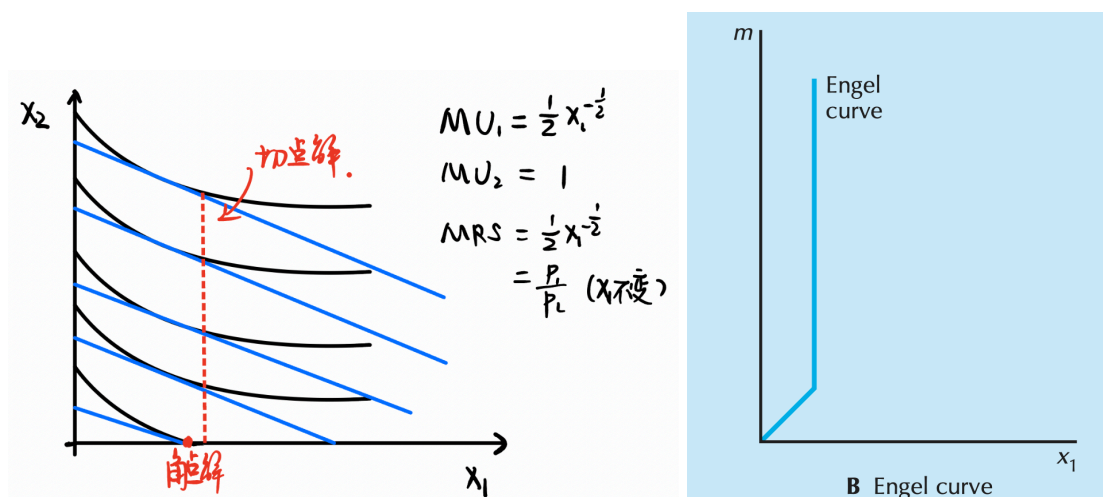


2.5.4 Homothetic Preferences and Linear Engel Curves

- Engel curves are straight lines if consumer's preferences are homothetic (同位偏好/相似偏好)
- A consumer's preferences are homothetic if and only if

$$(x_1, x_2) \succ (y_1, y_2) \Leftrightarrow (tx_1, tx_2) \succ (ty_1, ty_2), \forall t > 0$$

- That is, the consumer's MRS is the same anywhere on a straight line drawn from the origin(也就是 MRS 只取决于两者的相对数量之比)
- Often assumed in dynamic optimization problems
- Nonhomothetic example: $u(x_1, x_2) = \sqrt{x_1} + x_2$ (quasilinear (拟线性); preference)
- 绘图表示则如下



- 在角点解时，随着 m 升高会相应的外拓角点解，直到到达满足 $0.5x_1^{-0.5} = p_1/p_2$ 的时刻才保持不变，有图二所示。

3 Price Effect

- Price effect refers to the change in optimal consumption due to a change in price, holding income fixed.
- Focusing on the consumption of one particular good, e.g., good 1, there are two kinds of price changes.
- change in its own price or change in the price of the other good
- The effects of these two kinds of changes respectively are

$$\frac{\partial x_I}{\partial p_I} \quad \frac{\partial x_I}{\partial p_K} \quad (k \neq I)$$

3.1 Ordinary Good and Giffen Good

- Intuitively, when the price of a good increases, the demand for this good will decrease

$$\frac{\partial x_I}{\partial p_I} < 0$$

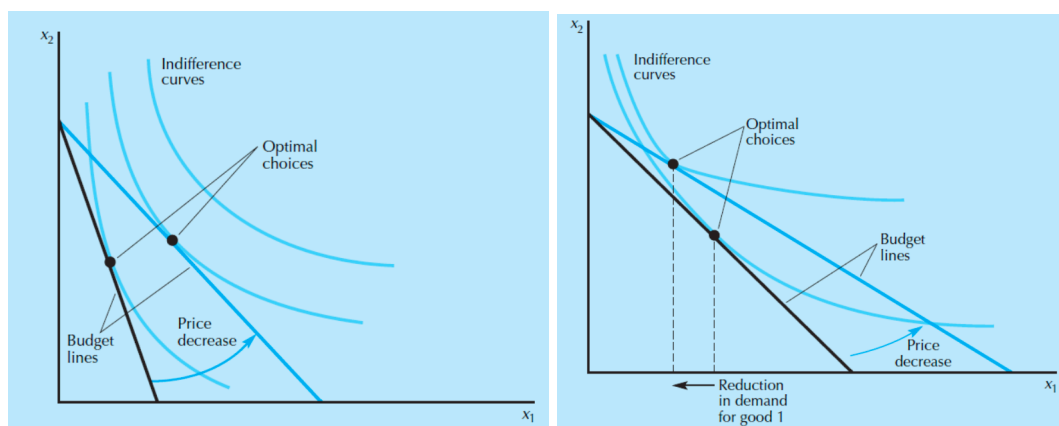
- such a good is an ordinary good (普通商品)
- It is logically possible to find well behaved preferences for which a decrease in the price leads to a reduction in the demand

$$\frac{\partial x_I}{\partial p_I} > 0$$

- such a good is called a Giffen good (吉芬商品)
- giffen good 很难找，比如有钱人豪车越贵越买，其实是附加的身份地位的提高导致你想买，而不是商品本身的价值带来的，这不符合吉芬商品的定义

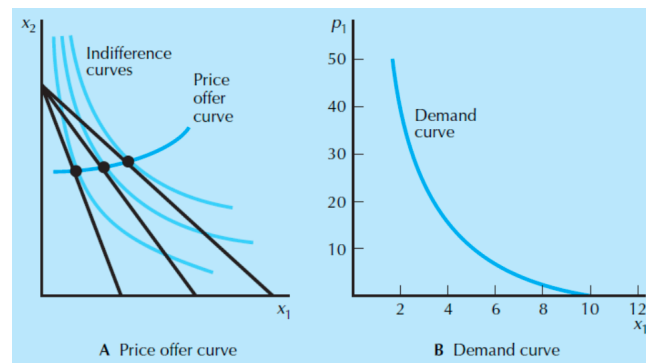
- 闲暇是吉芬商品嘛？我收入越高，闲暇的相对的机会成本越高，但是我好像越喜欢闲暇，但是这也不是吉芬商品，因为闲暇的价格发生改变的时候，收入也是在改变的，而吉芬商品衡量的是收入不变的时候，单纯商品价格改变导致商品消费数量的改变。
- 果农在苹果商品上涨的时候，他消费的苹果会增加。（这也是由于苹果价格上涨让收入上涨了）
- 所以真的很难找啊喂！
- 极端的例子：只有土豆和牛肉，土豆占比很大，我需要充饥，土豆价格上涨的时候会导致我继续削减牛肉的预算（因为不充饥！），然后增加土豆的购买。

graphic illustration



3.2 Price consumption curve(P.C.C) and Demand curve

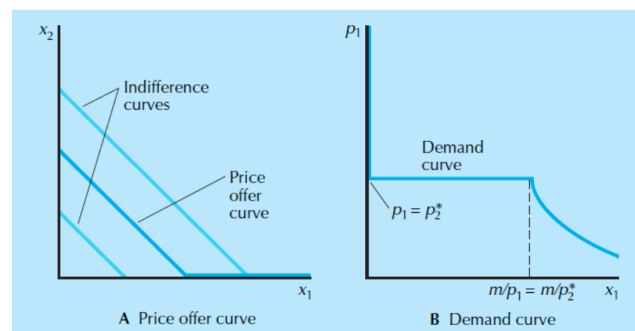
- Change the price of one good, holding income and the price of the other good constant
- Price consumption curve(P.C.C)(价格消费线)/ Price offer curve (价格提供曲线):
- contains all utility-maximizing bundles traced out as one price changes
- Demand curve: graph of quantity demanded (optimal consumption) and price
- limit the study to the relationship between the quantity demanded and changes in the own price of the good
- demand curve 上的每一个点都是效用最大化求出来的一个最优解
- 另外一种理解方式：需求函数 $x_1 = f(p_1)$, 反需求函数 $p_1 = g(x_1) = f^{-1}(x_1)$, 最优点都满足 $p_1 = MRS(x_1)p_2$, MRS 是每增加一单位 x_1 愿意放弃的 x_2 的数量, 乘上 p_2 就是愿意放弃的 x_2 的价值, 在最优需求点上的时候, 我愿意为多持有 x_1 而愿意支付的价格刚好就是 x_1 的价格, 代表了边际支付意愿



3.3 Examples

3.3.1 Perfect substitutes

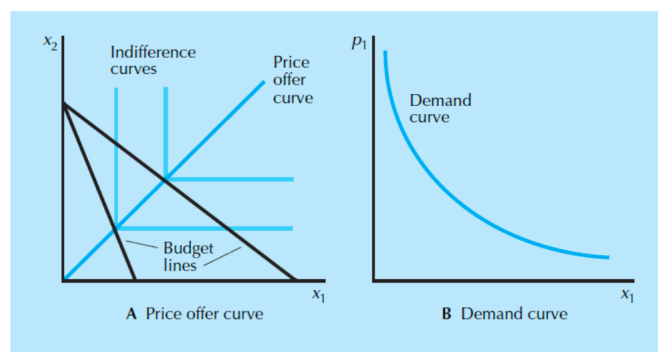
$$u(x_1, x_2) = x_1 + x_2$$



- 在 x_1, x_2 价格相等的时候，最优解 (x_1, x_2) 会任意沿着一条无差异曲线移动，从而 demand curve 中也会保持水平。
- 在 x_1 价格变动的时候，若低于 p_2 ，则会全部都购买 x_1 ，从而满足 $p_1 x_1 = m$ ，也就是呈反比例关系；同时在图 1 中，最优解已经变成右下角的角点解，随着 p_1 减少，角点解也会外拓。
- 在 p_1 价格高于 p_2 时，不会购买任何 x_1 ，则表现为 demand curve 中沿着 y 轴攀升的一段。

3.3.2 Perfect complements

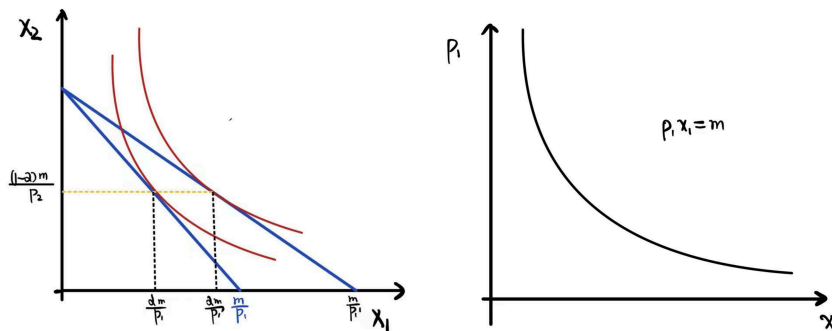
$$u(x_1, x_2) = \min\{x_1, x_2\}: x_1 = m/(p_1 + p_2)$$



- x_1, x_2 始终保持相等的量

3.3.3 Cobb-Douglas Utility

- $u(x_1, x_2) = x_1^a x_2^{1-a}$, $x_1 = \frac{am}{p_1}$, $x_2 = \frac{(1-a)m}{p_2}$
- x_2^* is constant (flat): not $f(p_1)$
- x_1^* is rectangular hyperbola (直角双曲线)



3.4 substitutes and complements

- Now let's consider $\frac{\partial x_m}{\partial p_k}$ for $k \neq l$
- If $\frac{\partial x_m}{\partial p_k} > 0$ demand for good l goes up when the price of good k goes up
- called: good l is a substitute for good k (替代品, 如咖啡 or 茶, 一种商品价格上涨会导致替代品的需求增加)
- If $\frac{\partial x_m}{\partial p_k} < 0$, demand for good l goes down when the price of good k goes up
- called: good l is a complement for good k (互补品, 如汽油 and 车, 一种商品价格上涨会导致其互补品的需求减少)

Consumer's Choice & Comparative Statics 2

姜峻赫

2022.10.15

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1 Price Effect: A Decomposition

(PS: 十分抱歉上次此部分笔记有诸多笔误)

现实情境: 食堂的菜变贵了, 那么我们会两个反应: 我的钱的购买力变少了 (m 没变, p 变了, 相对的我变穷了), 同时面包的价格没变, 可能我去吃更多的面包了 (有一个替代的关系), 那么想要分解这两种心理感受 (变穷: 收入效应, 转而买别的: 替代效应)

价格效应 = 收入效应 (单纯变穷) + 替代效应 (价格贵了我去买别的了)

- We want to understand why Giffen goods can arise in our theoretical analysis and know under what conditions goods are ordinary.
- When the price of a good changes, the consumer's budget line rotates accordingly.
- In this process, two effects arise
 1. change in **relative price**
 2. change in the **total purchasing power**
- To understand the total effect, we can analyze these two effects one by one: We do the following thought experiment
 - step 1: consider only relative price change, **without altering the total purchasing power** (substitution effect, 替代效应)
 - step 2: consider **only real purchasing power (实际购买力)** change, without changing the relative price from the first stage (income effect, 收入效应)
 - 简单来说, 替代效应就是保持**实际价格不变**的情况下, 因为相对价格变动而引起的最优解的变动 (也就是有人给了你钱或者拿走了你的钱的假象情况)

【PS: 斯勒茨基替代效应 会认为实际价格不变是购买力不变 (可以买相同量的商品), 也就是想象中的预算约束线 (辅助线) 要平行于旋转后的预算约束线, **并经过原先的最优点**, 从而满足在可以购买原先数量的商品的同时, 因为相对价格发生改变而选择了另外的解。

希克斯替代效应 会认为实际价格不变是购买到的**总效用不变**, 从而想象中的预算约束线 (辅助线) 要平行于旋转后的预算约束线, 并与原先的无差异曲线**相切**, 从而满足在可以达到原先效用的同时, 因为相对价格发生改变而选择了另外的解。

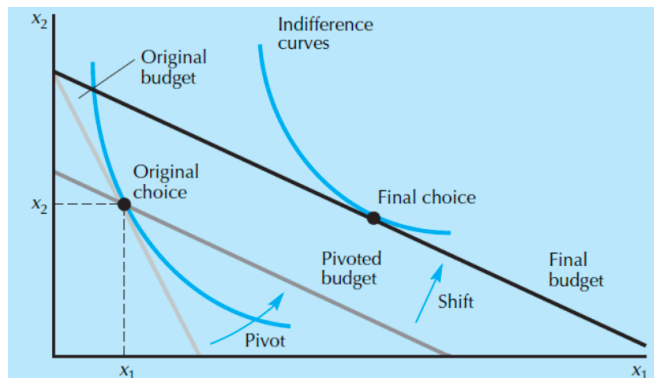
在老师课件上提到的是**斯勒茨基替代效应**】
 - 收入效应就是保持相对价格不变的同时, 因为总收入变动而导致的两种商品同比例增长的情况。

1.1 Substitution Effect

- In step 1, we imagine a change in price from p_1 to p'_1
- At the same time, we imagine that the consumer's income also changes so that the original consumption
- $(x_1(p, m), x_2(p, m))$ is still just affordable under $p = (p'_1, p_2)$
- Therefore, we have

$$m' = p'_1 x_1(p, m) + p_2 x_2(p, m)$$
- Under this new budget constraint, the consumer's demand becomes $(x_1(p', m'), x_2(p', m'))$ (compensated demand (补偿需求))
- The change from $(x_1(p, m), x_2(p, m))$ to $(x_1(p', m'), x_2(p', m'))$ is called the substitution effect

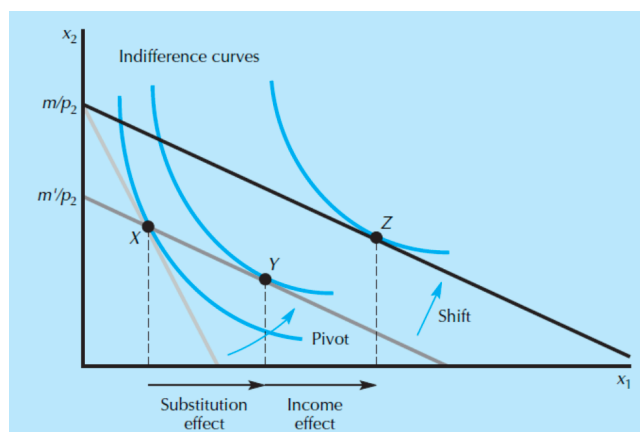
- $\Delta x_1^s = x_1(p'_1, m') - x_1(p_1, m)$



- 实际应用中就是：将旋转后的预算约束线平移到与原先最优点相交（如果是希克斯替代效应，将平移到与原先无差异曲线相切），此时得到的最优解和原来的最优解之差即为替代效用（与新的无差异曲线相切）

1.2 Income Effect

- In step 2, relative price is kept fixed as in step 1, but we alter the consumer's income to his real (and original) income m .
- This to examine the effect of only purchasing power change.
- In this stage, the consumer's demand changes from $(x_1(p', m'), x_2(p', m'))$ to $(x_1(p', m), x_2(p', m))$
- the income effect: $\Delta x_1^n = x_1(p'_1, m) - x_1(p'_1, m')$
- 实际应用中就是：价格效应-替代效应 = 收入效应。也可以通过再次平移回去看最优点的变动。



1.3 A Numerical Example

- Suppose that the consumer has a demand function for milk of the form

$$x_1 = 10 + \frac{m}{10p_1}$$

- Originally his income is \$120 per week and the price of milk is \$3 per quart.
- Now suppose that the price of milk falls to \$2 per quart.
- How much income would have to change in order to make the original consumption of milk just affordable when the price of milk is \$2 a quart?

$$x'_1 = 10 + \frac{120}{10 \times 3} = 14, \Delta m = (p'_1 - p_1) \cdot x_1 = -1 \times 14 = -14$$

$$m' = 120 - 14 = 106$$

- What is the consumer's compensated demand for milk at the new price?

$$x'_1 = 10 + \frac{m'}{10p'_1} = 10 + \frac{106}{10 \times 2} = 15.3$$

- What is the substitution effect? How about the income effect?

$$SE = x'_1 - x_1 = 15.3 - 14 = 1.3$$

$$x''_1 = 10 + \frac{m}{10p'_1} = 10 + \frac{120}{10 \times 2} = 16$$

$$IE = x''_1 - x'_1 = 16 - 15.3 = 0.7$$

事实上，可以通过 $TotalEffect = SE + IE$ 求出一个算即可

1.4 Slutsky Identity

- The price effect can be decomposed into the substitution effect and the income effect.
- In particular,

$$\begin{aligned} \Delta x_1 &= x_1(p', m) - x_1(p, m) \\ &= [x_1(p', m) - x_1(p', m')] + [x_1(p', m') - x_1(p, m)] \\ &= \Delta x_1^n + \Delta x_1^s \end{aligned}$$

- This equation is called the Slutsky identity

1.5 Sign of Substitution Effect

- An important property of substitution effect is the following **compensated law of demand**:

$$(p'_1 - p_1)[x_1(p', m') - x_1(p, m)] \leq 0$$

- It states that the compensated demand decreases if the prices increases(需求随着价格上升**单调递减** (负相关), 倘若正相关就变成吉芬商品了)
- For notational simplicity, let $(x_1, x_2) = (x_1(p, m), x_2(p, m))$ be the original demand and $(x'_1, x'_2) = (x_1(p', m'), x_2(p', m'))$ be the compensated demand.

- Because (x'_1, x'_2) is the compensated demand, we have

$$m' = p'_1 x'_1 + p_2 x'_2 = p'_1 x_1 + p_2 x_2$$

(其实就是说两者花掉了同样的钱)

- Because both (x_1, x_2) and (x'_1, x'_2) are affordable under (p', m') and the consumer chooses (x'_1, x'_2) , we know that the consumer prefers (x'_1, x'_2) to (x_1, x_2) (花同样的钱, 选了哪个喜欢哪个)
- But under (p, m) , we know the consumer chooses (x_1, x_2) instead of (x'_1, x'_2) (之前为什么不买呢, 只能猜测买不起, 毕竟你更喜欢另一个)
- This implies that (x'_1, x'_2) is not affordable at (p, m) :

$$p_1 x'_1 + p_2 x'_2 > m = p_1 x_1 + p_2 x_2$$

- 由上述一个等式一个不等式, 可以推知

$$m' = p'_1 x'_1 + p_2 x'_2 = p'_1 x_1 + p_2 x_2$$

$$p_1 x'_1(p', m') + p_2 x'_2(p', m') > m = p_1 x_1(p, m) + p_2 x_2(p, m)$$

$$(p'_1 - p_1)x'_1 < (p'_1 - p_1)x_1 \Rightarrow (p'_1 - p_1)(x'_1(p', m') - x_1(p, m)) < 0$$

- 需求反映了在满足预算约束线的情况下效用最大化的选择。

1.5.1 Revealed preference

- The most important step in this proof is the conclusion that (x'_1, x'_2) is preferred to (x_1, x_2)
- The argument draws conclusion about preferences from behavior.
- This kind of arguments is called revealed preference (显示偏好) argument
- let (x_1, x_2) be the bundle purchased at prices (p_1, p_2) when the consumer has income m
- if $p_1 x_1 + p_2 x_2 \geq p_1 y_1 + p_2 y_2$ and (y_1, y_2) is actually a different bundle from (x_1, x_2) , we say (x_1, x_2) is directly revealed preferred to (y_1, y_2)
- 简单说就是, 有钱买的时候一定买自己最想买的, 如果没买说明钱不够。
- From the previous analysis, we have

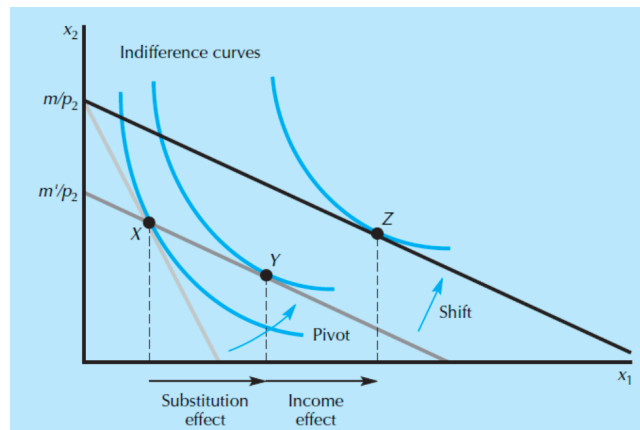
$$(p'_1 - p_1)[x_1(p', m) - x_1(p, m)] = (p'_1 - p_1)[x_1(p', m) - x_1(p', m')] + (p'_1 - p_1)[x_1(p', m') - x_1(p, m)]$$

- We've proved that $(p'_1 - p_1)[x_1(p', m') - x_1(p, m)] \leq 0$
- The remaining question is: the sign of $(p'_1 - p_1)[x_1(p', m) - x_1(p', m')]$
- If $p'_1 < p_1$, $(p'_1 - p_1)[x_1(p', m) - x_1(p', m')] \leq 0$ requires $x_1(p', m) - x_1(p', m') \geq 0$

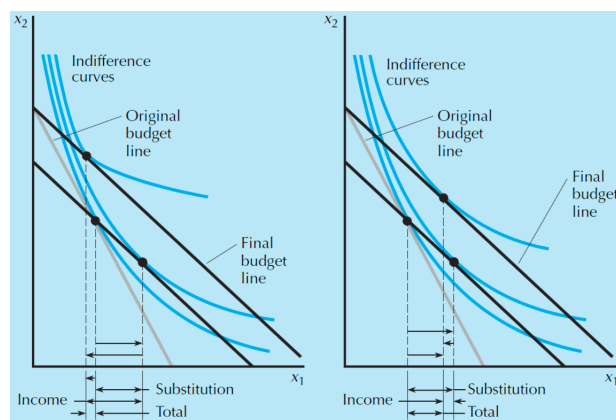
- we know $m' = p'_1 x_1(p, m) + p_2 x_2(p, m) < m$
- $x_1(p', m) - x_1(p', m') \geq 0 \Rightarrow$ good 1 is a normal good, i.e., its demand increases as income increases, holding prices fixed (at p')
- Thus, if good 1 is a normal good, we must have

$$(p'_1 - p_1)[x_1(p', m) - x_1(p, m)] \leq 0$$

- Since both the substitution and income effects increase demand when own-price decreases, the demand curve for a normal good must slope downwards
- income and substitution effects reinforce each other
- The Law of Downward-Sloping Demand therefore always applies to normal goods(normal $\Rightarrow$ ordinary)



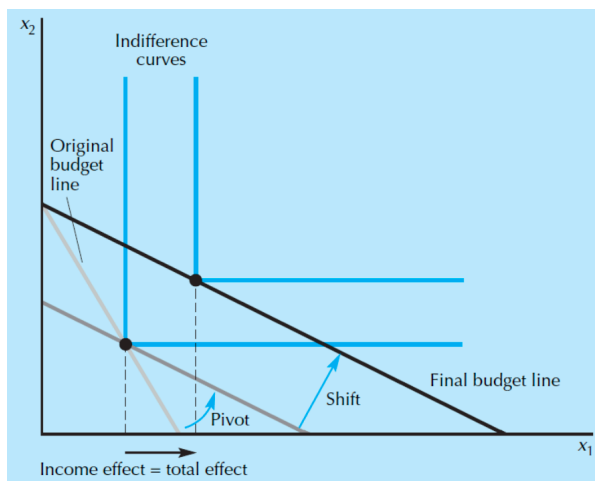
- Some goods are income-inferior; that is, demand is reduced by an increase in income
- the substitution and income effects oppose each other when a good's own price changes
- In the rare case of extreme income-inferiority, the income effect may be larger in size than the substitution effect, causing quantity demanded to decrease as own-price decreases.
- Such goods are called Giffen goods



2 Examples of IE and SE

2.1 Perfect Complements

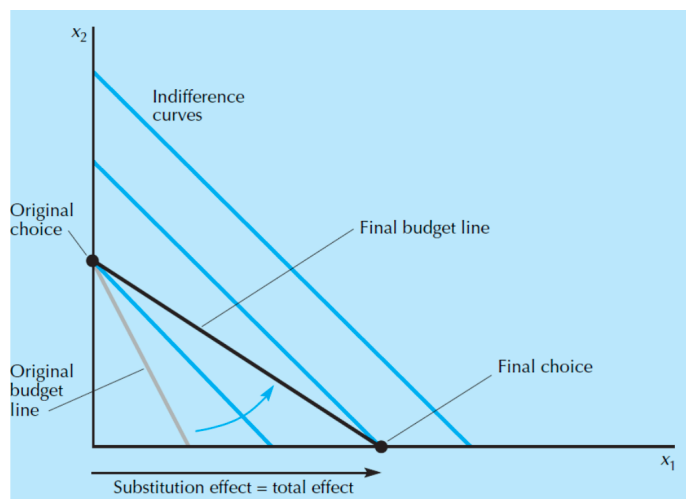
从直观上来讲，已经是完全互补品了，相应的，应该没有替代效应，两者没有相互之间的替代作用才能做到互补（正如你的左鞋不能替代右鞋）（下图之中没有 rotate 之后的最优点变化）



2.2 Perfect Substitutes

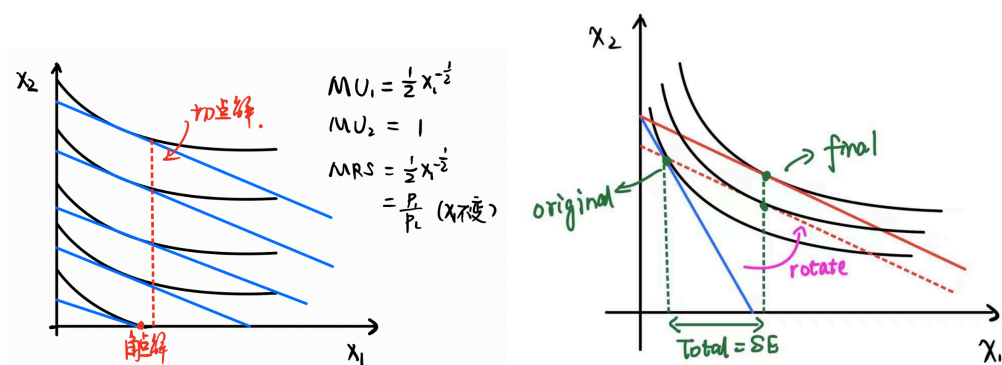
从直观上来讲，已经是完全替代品了，替代效应为全部价格作用（下图之中没有收入效应的平移）

特殊情况下，如果本来就是全消费 x_1 了，然后如果继续 x_1 价格减少，那么只会有 x_1 继续增长，此时没有替代效应，只有收入效应

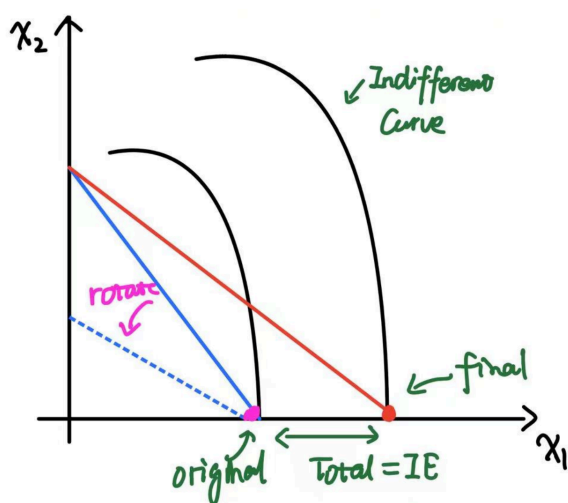


2.3 Special examples

没有收入效应的，不一定是完全替代，还有可能是拟线性情况 (Quasilinear preference)。上节课的例子 $u = \sqrt{x_1} + x_2$ 满足切点解的某一种商品数量完全不变。（如下图 1）



没有替代效应的，也不一定完全互补，如下面的例子



2.4 Numerical example

We have

$$u = x_1 + x_2, p_1 = 2, p_2 = 1, m = 100$$

what if $p'_1 = 0.5$, IE, SE, TE?

$$\frac{p_1}{p_2} = 2 > 1, x_1 = 0, x_2 = \frac{100}{p_2} = 100$$

$$m' = m + \Delta m = m + (p'_1 - p_1) \cdot x_1 = m = 100$$

$$p'_1 = 0.5, \frac{p'_1}{p_2} = 0.5 < 1, x'_1 = 200, x'_2 = 0$$

$$\text{Final point } m = 100, \frac{p'_1}{p_2} = 0.5 < 1, x''_1 = 200, x''_2 = 0$$

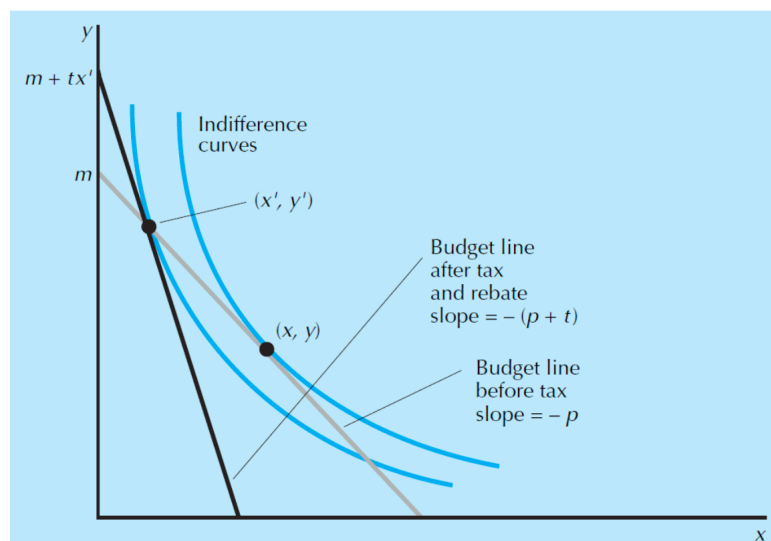
$$SE = x'_1 - x_1 = 200, IE = 0, TE = 200$$

3 Applications

3.1 Rebating a Tax

- There were many plans proposed to reduce the United States's dependence on foreign oil in the 1970s(不想过度依赖国外的石油，一种方法是降低你对汽油的消费，也就是收消费税)
- one such plan involved increasing the gasoline tax
- but a straight increase in the tax on gasoline would hit consumers (可能会激起民愤!)
- it was suggested that the revenues raised from consumers by this tax would be returned to the consumers in the form of direct money payments
- 政府征收石油的消费税，然后将这个税收按照补贴的方式发给消费者，这是不是相对于什么都没做呢？（政府的钱完全没变）
- What would be their impacts on consumers?

3.1.1 Analysis



- 让 y 代表除了汽油以外的其他商品， x 代表是汽油。标准化， y 的价格为 1， x 的相对价格为 p ，则斜率为 $-p$ ，收税之后 x 的价格变成 $p+t$ ，反映到预算约束线上就是斜率变陡为 $-p-t$ ，从而消费者会消费更少的汽油。
- 如果没有税收返还，那么灰色的线会沿着 m 点向内旋转让斜率变陡，得到旋转后的线 l （图中没有标出）
- 如果有税收返还，我们考察一般的消费者的收入消费曲线，此时一个奇妙的点是最优解 (x', y') 也在原先的预算约束线上。解释如下

$$(p + t)x' + y' = m + \hat{m}$$

- 政府的税收返还量取决于所有消费者的总消费量，所以平均到每一个消费者身上就是 $\hat{m}$ ，而对于特殊的个人来说通常 $\hat{m} \neq tx'$ ，其中 x' 是消费者最终的汽油消费量
- 所以上面这个等式的右边是常数，而左边是消费者可以选择的组合（只要保持等式不变）
- 我们如果考虑较为理想的情况，特定的消费者对于汽油的消费处于平均水平（也就是所谓的典型消费者），那么我们可以认为她/他最终的消费汽油所缴纳的税就是政府的补贴量，也就是 $tx' = \hat{m}$ ，于是我们就有如下等式成立

$$p' = p + t, R = tx'$$

$$(p + t)x' + y' = m + tx' \Rightarrow px' + y' = m$$

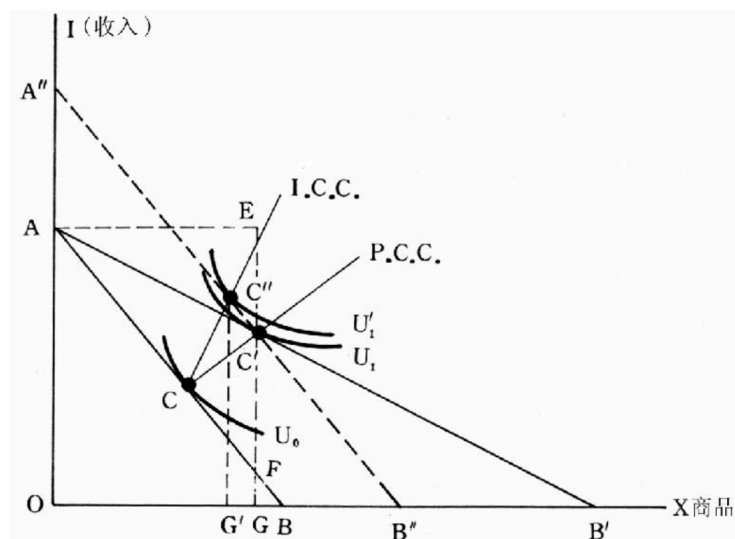
- 这也说明了对于我们所考虑的消费者来说， (x', y') 是最优解（所以和无差异曲线相切），同时也落在原先的预算约束线上（这是由于，对于我们所考虑的“典型”会满足其缴纳的汽油税刚刚好等于政府的补贴，由上面推导可以得到）
- 这种情况下，政府没有改变消费者的总收入，但是汽油的消费量少了
- 但是消费者的效用无差异曲线变低了，用显示偏好理论解释，消费者在相同的收入下，本来选了 (x, y) ，但是现在不得不选了 (x', y') ，这说明此时原先的 (x, y) 买不起了。消费者的福利确实是被损伤的。（但是政府达到了目标，税收返还一定程度上补偿了消费者的损失）

3.2 Direct or Indirect Subsidy

- In order to encourage the public to consume some items, the government generally adopts two methods
 - directly subsidy: give cash to consumers
 - indirectly: use coupons so that consumers can buy things at a discount rate
- Please compare the advantages and disadvantages of these two policies

3.2.1 Analysis

- 规律：扭曲你的行为越少，往往越好。消费者往往跟喜欢直接发钱。



- 纵轴是收入水平 (I)，也就是用于除了 X 的其他商品的 money
- 间接补贴：从 AB 到 AB'，X 的价格降低了， $C \rightarrow C'$
- 直接补贴：虚线！ $C \rightarrow C''$ ，AA'' 就是补贴的金额大小。
- 一个奇特的现象是 C' 和 C'' 在同一条预算约束线上。这是因为大前提是，两种补贴方式的补贴额度是一样的。
- 最优解应该是 C'' ，而 C' 的选择则是因为政府扭曲了消费者的行为（用商品价格的变化）（显示性偏好的解释：在同一条预算约束线上，最好选 C'' ，说明 C'' 效用更大）

3.3 Income or Consumption Tax

- The substitution effect is the name that economists give to the change in demand when prices change but a consumer's purchasing power is held constant
- The definition we have studied above is called the Slutsky substitution effect
 - hold the consumer's purchasing power constant the original bundle remains affordable
- Another definition is called the Hicks substitution effect
 - hold the consumer's purchasing power constant the consumer can purchase a bundle that is indifferent to his original bundle

Consumer Surplus & Market Demand

姜峻赫

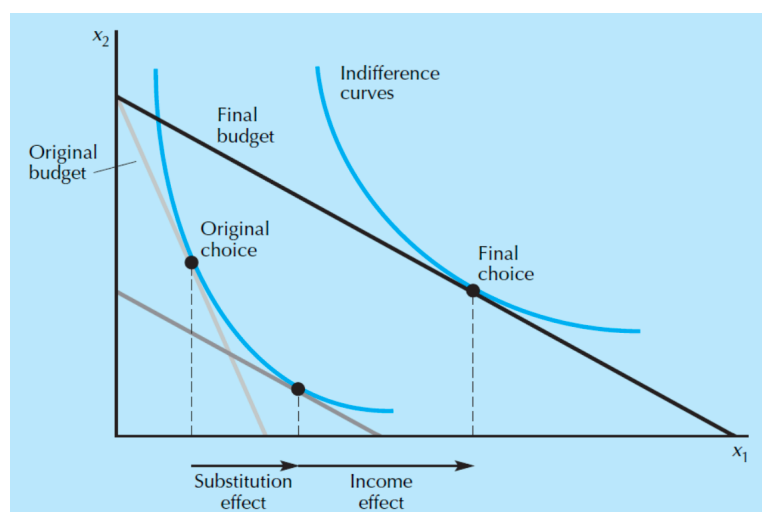
2022 年 10 月 22 日

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1 Another Type of Substitution Effect

- The substitution effect is the name that economists give to the change in demand when prices change but a consumer's purchasing power is held constant
- The definition we have studied above is called the Slutsky substitution effect (hold the consumer's purchasing power constant the original bundle remains affordable)
- Another definition is called the Hicks substitution effect (hold the consumer's purchasing power constant the consumer can purchase a bundle that is indifferent to his original bundle)



- hicks 分解的替代效应也是负的
- p_1, q_1 为初始价格, p_2, q_2 为最终价格, 在 p_1, q_1 下没有选择 y_1, y_2 只能是因为买不起, 第二个不等式也同理。

$$p_1 x_1 + p_2 x_2 \leq p_1 y_1 + p_2 y_2$$

$$q_1 y_1 + q_2 y_2 \leq q_1 x_1 + q_2 x_2$$

$$(q_1 - p_1)(y_1 - x_1) + (q_2 - p_2)(y_2 - x_2) \leq 0$$

$$p_2 = q_2 \Rightarrow (q_1 - p_1)(y_2 - x_2) \leq 0$$

2 Compensated Demand Curves

- We have seen how the quantity demanded changes as a price changes in three different contexts
 - holding income fixed $\Rightarrow$ the standard demand curve

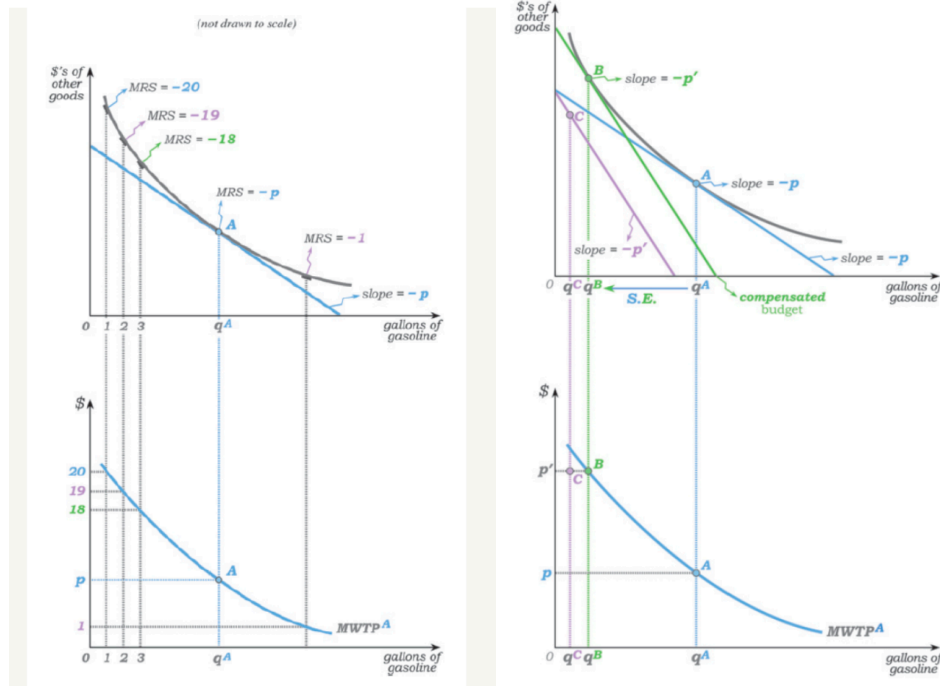
- holding purchasing power fixed $\Rightarrow$ Slutsky demand curve
 - holding utility fixed $\Rightarrow$ Hicks demand curve
- The Slutsky and Hicks demand curves are always downward sloping curves
- The ordinary (Marshallian/Walrasian) demand curve is a downward sloping curve for normal goods
- The Hicksian demand curve is sometimes called the compensated demand curve (useful in benefit-cost analysis)
- hicks 分解的替代效应得到的最优点会比 slutsky 分解替代效应的最优点低 and 左一些, hicks 分解是在同一条无差异曲线上的, 而 slutsky 是经过原先的最优点, 根据显性偏好原则, hicks 分解时原先的点和补偿需求的点在同一条无差异曲线上, 但是却没有选择那个点, 只可能是因为买不起, 而那个点跟 slutsky 分解的最优点在同一条预算约束线上, 从而说明 slutsky 的预算约束线更高 (价格更贵)。由于商品都是正常品, 从而 slutsky 替代效应会偏右

2.1 Consumers gain from trading

- How to measure consumers gain (or welfare) from trading at single price or policy change?
- Let consider an example of a consumer's choices over gasoline and a composite good denominated in dollars
- Now let's ask the following question:
 - how much better off is the consumer for being able to purchase gasoline at its current price rather than being excluded from the market for gasoline?
 - Or how much would the consumer be willing to pay for the opportunity to participate in the current market for gasoline?

2.2 MWTP: Marginal Willingness to Pay

- 边际支付意愿
- the MWTP curve is simply plotted by plotting the dollar values of the MRS at each gallon of gasoline



- Unfortunately, the MWTP curves are not observable. A simpler procedure that has seen extensively is to use the demand curve instead
- The previous graph illustrate the difference between the MWTP curves and the demand curve
- the two curves are generally not the same, would be the same when there are no income effects
- true only for tastes that are quasilinear (拟线性) in that good

3 Quasilinear Utility

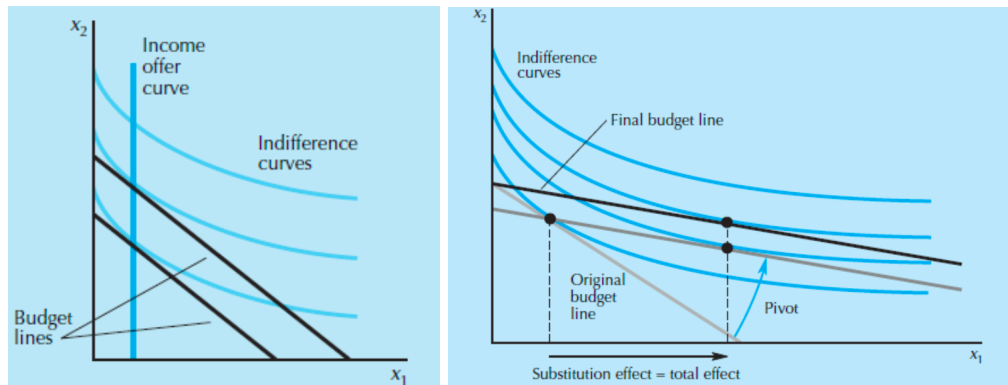
- A quasilinear utility function takes the form of

$$u(x_1, x_2) = v(x_1) + x_2$$

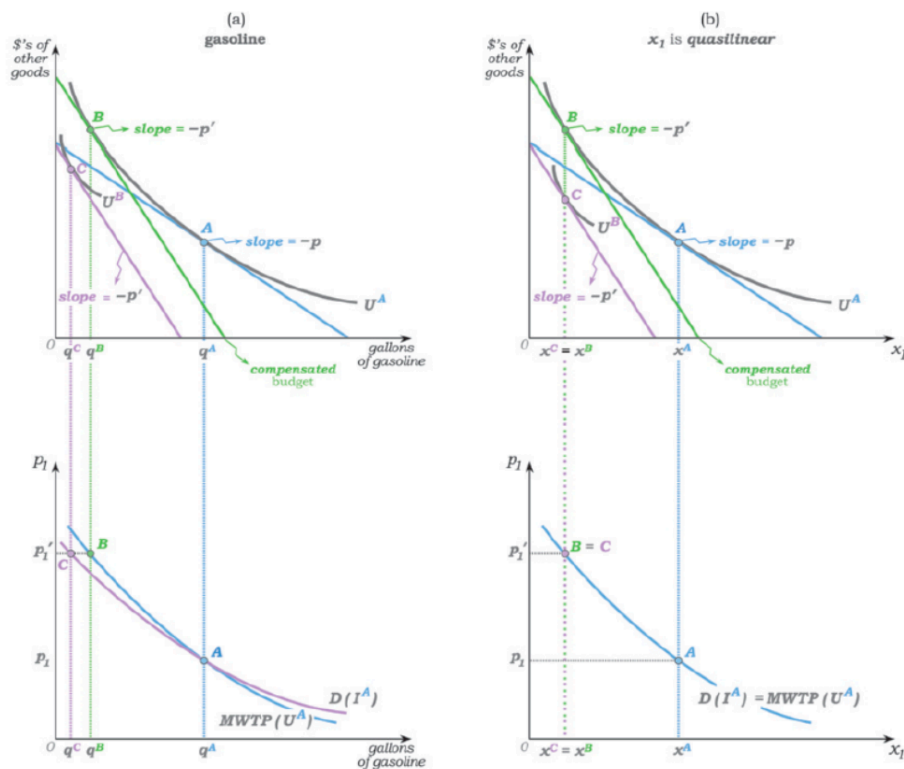
- Because x_2 enters into the utility function linearly and x_1 does not, u is quasilinear
- Assume $p_2 = 1$
- Demand for x_1 ($MRS = v'(x_1)/1 = p_1/1$) can be solved from

$$v'(x_1) = p_1$$

- 事实上这就是需求函数，且没有收入效应，因为收入甚至没有出现在这个表达式中，是可以根据这个函数解出来 x_1 的。



Relationship of demand and MWTP Curves of x_1 :



- What would be a real-life situation where this kind of thing might occur?
- Suppose x_1 is pencils and x_2 is money to spend on other goods: one extra unit of income will spread around so the income effect on x_1 would be small
- When we are examining a choice between all other goods and some single good that isn't a very large part of the consumer's budget, the quasilinear assumption may well be plausible

比如买笔，你在收入增加的时候不会全都买笔，或者说过了一个临界点后，收入的增加不会导致你买笔的增加，因为花在别的东西占大头，从而可以认为曲线的大部分不会有收入效应，可以用需求曲线替代边际支付意愿曲线。

- A popular application of the quasilinear utility is

$$u(x, m) = v(x) + m$$

- x : the amount of a consumption good of concern
- m : all money the consumer spends on other goods
- In reality, a consumer consumes a large variety of goods and has a very complicated preference over all these goods. In economic analysis, we sometimes only want to focus on the consumption of one particular good
- So we simplify the problem by summarizing the amount of money that the agent spends on all other goods by m and focus on the consumption x
- The optimality condition

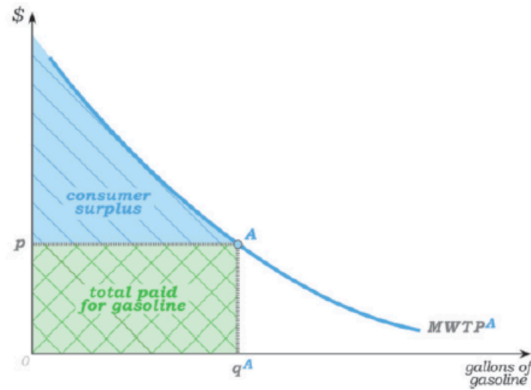
$$v'(x) = p$$

- defines the consumer's demand curve.
- This is why we sometimes say that a consumer's demand curve is just his marginal utility curve. You have to understand that this relies on quasilinear utility function

4 Total Willingness to Pay and Consumer Surplus

- 消费者剩余就是买东西获得的满足感，如果这个价格高于你的心理价格，不会很快乐，如果低于，那么会开心。那么可以用心理愿意支付的钱和实际支付的钱的差作为一个衡量：消费者剩余
- The consumer's total willingness to pay for all the gasoline purchased is equal to: marginal willingness to pay for the first gallon + his marginal willingness to pay for the second gallon + ...
- roughly equal to the area below the MWTP curve up to the quantity that he consumes
- The amount he actually had to pay is simply equal to p times the number of gallons he chose to consume
- The gains from trade is called the consumer surplus (消费者剩余)

- the difference between what the consumer was willing to pay and what he actually paid for his gasoline consumption



4.1 Compensating and Equivalent Variation

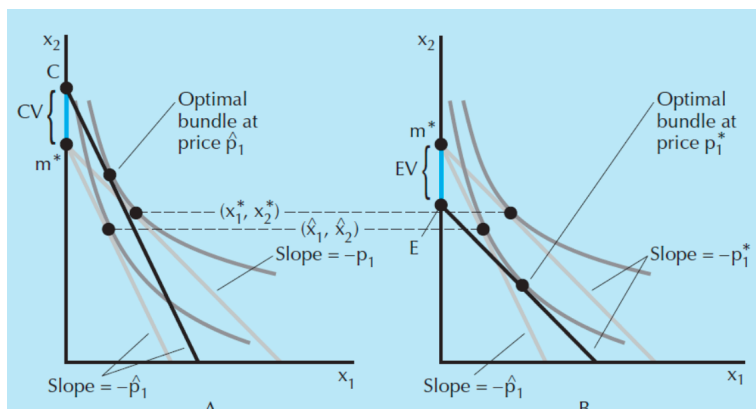
- In some applications it may be convenient to use certain monetary measures of utility
- How much money we would have to give a consumer to compensate him for a change in his consumption patterns?

4.1.1 compensating variation

- is to ask how much money we would have to give the consumer after the price change to make him just as well off as he was before the price change
- 补偿的额度：事后补偿你多少才能让你达到原先同样的开心程度（达到同样的效用）

4.1.2 equivalent variation

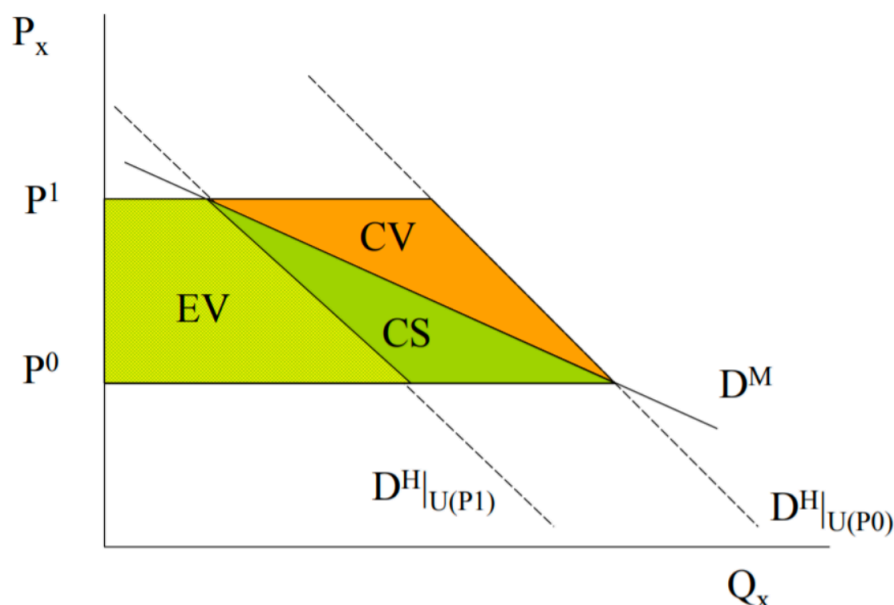
- Another way to ask how much money would have to be taken away from the consumer before the price change to leave him as well off as he would be after the price change
- 补偿的额度：当时拿走了你多少才让你如此的不开心（事前拿走你多少钱才能达到和现在收完税后你的效用水平）



- 图 A 中原有的线是较为平缓的灰色的线，收税后最终的线为较为陡峭的灰色的线。黑色的线代表在收税后补偿之后使得消费者效用（开心程度）相同
- 图 B 中灰色线同图 A，黑色线代表直接拿走你多少钱才能让你达到收税后的效用（让你如此不开心!）

4.1.3 An example

- $u(x_1, x_2) = \sqrt{x_1 x_2}$ $p_1 = p_2 = 1$, $m = 100$ $p'_1 = 2$
- $x_1 = \frac{m}{2p_1} = 50$, $x_2 = \frac{m}{2p_2} = 50$, $x'_1 = 25, x'_2 = 50$
- CV: 在 $(2, 1)$ 下的 $m' = ?$ $u = 50$, $\sqrt{(\frac{m'}{2 \times 2})(\frac{m'}{2 \times 1})} = 50 \Rightarrow m' = 100\sqrt{2}$, $CV \approx 41$
- EV: 在 $(1, 1)$ 下 $m'' = ?$ $u = \sqrt{25 \times 50}$, $\sqrt{(\frac{m''}{2})(\frac{m''}{2})} = \sqrt{25 \times 50} \Rightarrow m'' = 50\sqrt{2}$, $EV \approx 100 - 70 = 30$
- CS: $x_1 = \frac{100}{2p_1} = \frac{50}{p_1}$ $CS = \int_1^2 \frac{50}{p} dp = 50 \ln 2 \approx 35$

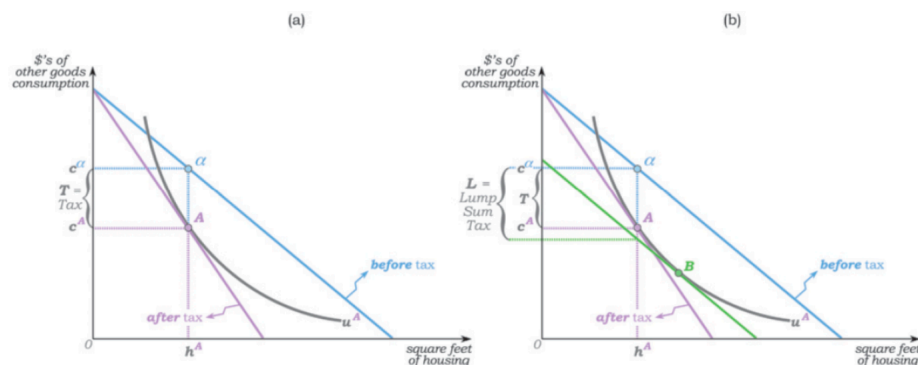


- 图中 $D^H|_{U(P_1)}$ 和 $D^H|_{U(P_0)}$ 都是 hicks 需求曲线，分别代表在价格变成 P_1 后的边际意愿曲线，和价格为 P_0 时的边际意愿曲线。可以看到 EV 在上面的图 B 中就是预算约束线沿着后来的无差异曲线滑动得到，CV 则是沿着原来的。 D^M 代表 Marshallian 需求曲线，也就是通常意义上的需求曲线。为了方便讨论都变成了直线
- 图中 CV 是两块图形之和，CS 是下面两块图形之和，EV 就是标出部分。所以有 $CV > CS > EV$
- 直观的理解这个结论，CS 部分是好理解的，就是我们通常意义上消费者愿意支付价格和实际支付价格之间关于购买量的积分
- EV 部分直观理解：EV 从上面的图 B 中来看，反映了在两个不同预算下达到相同快乐程度所要弥补的 x_2 数量，如果不弥补这部分 x_2 ，那么 x_1 消费是会减少的，从而减少消费者的快乐程度，减少的这部分也就是在 x_1 在价格 P_1 下的在两个价格之间边际支付意愿曲线的积分
- 也就是用了：需要增加的 x_2 价值 = 减少消费的 x_1 的价值 = 在某个价格下 x_1 的边际支付意愿曲线在两个价格下的积分
- 可以从数学上严格证明这一点。CV 也是同理。

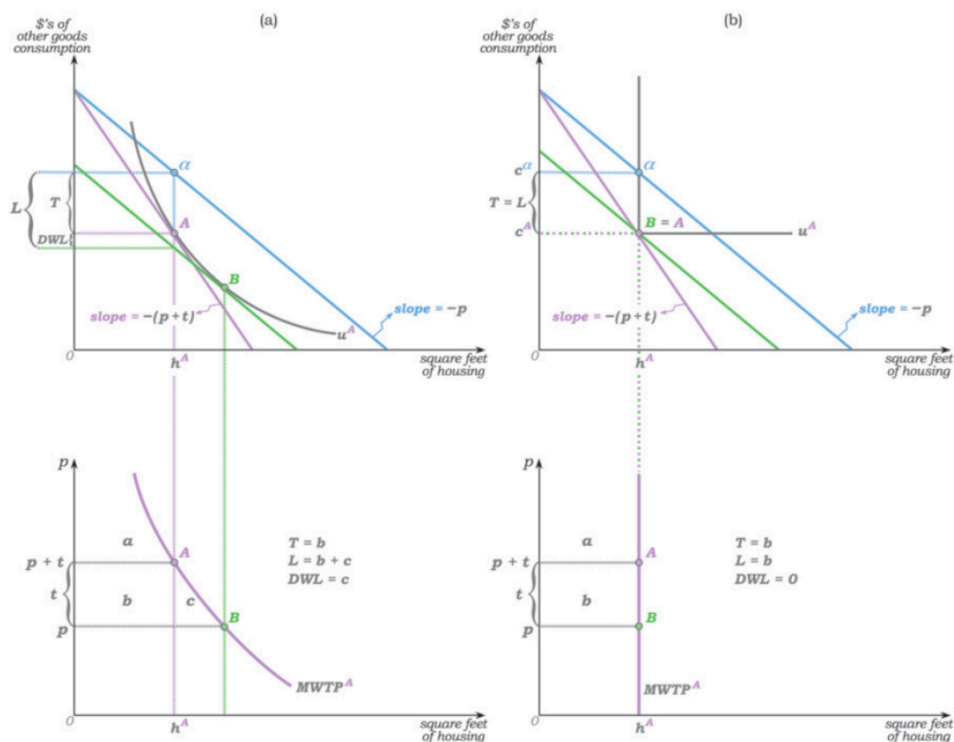
5 Deadweight Loss and Inefficiency of Taxation

- To see what makes a particular tax costly, we will try to answer the following question:
 - How much would a taxed individual be willing to bribe the government to get the tax rescinded?

- We will then compare this amount with the amount that the individual is actually paying in tax.
- if larger, then there exists a way that the government could have raised more revenue without making anyone worse off
- the difference between these two is our measure of deadweight loss



- 假设消费者向政府缴纳 L 可以达到征税后同样的效用水平，这个图中 $L-T$ 部分就是 Dead-weight loss，无效损失。(b) 中绿色的线就是相当于直接拿走了消费者一笔钱后消费者的预算约束线，所以分析和 EV 基本是一样的。



- 仿照上面的三色分析图，现在有直接拿走的消费者的钱 $L = b+c$, 收税 $T=b$, 那么 DWL 就是 c
- 为完全互补品的时候，没有替代效应，那么总额税和单个商品税收是没有差别的，此时只有收入效应，从而没有 deadweight loss，如上面图 (b) 所示

无谓损失的含义是：为了避免产生这个政策的钱和收税收到的钱的差。（只有在消费者愿意用更高的价格去贿赂政府不收税的情况下存在）

现实中为什么不直接征收消费税？很难不产生其他的影响：比如改变当前消费和储蓄（未来消费）的比例关系

经济学家不喜欢税收是因为替代效应（无效率的），大众不喜欢税收是因为收入效应（收税感觉自己变穷了啊!）

Market Demand & Choice Under Uncertainty

姜峻赫

2022 年 10 月 29 日

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1 Market Demand

- We have understood how each individual consumer makes purchasing decision. Now turn to adding up individual demand to form the market demand.
- Suppose there are n consumers in the economy, Individual i has a demand function good k as $x_i^k = D_i^k(p, m_i)$
- Suppose all consumers are price-takers
- Then market demand for good k is

$$X^k(p, m_1, m_2, \dots, m_n) = \sum_{i=1}^n D_i^k(p, m_i)$$

- The aggregate demand will generally depend on prices and the distribution of incomes
- If all consumers have identical preference that are homothetic, then

$$X^k(p, m_1, m_2, \dots, m_n) = X^k(p, M) \quad \text{where } M = \sum_i m_i \quad (\text{the aggregate income})$$

- 财富分配不重要时候（也就是没有收入效应的时候），所有人才会像一个人行动
- It is sometimes convenient to think of this as the demand of some “representative consumer” who has an income that is just the sum of all individual incomes.
- The conditions under which this can be done are rather restrictive. A complete discussion of this issue is beyond the scope of this course.

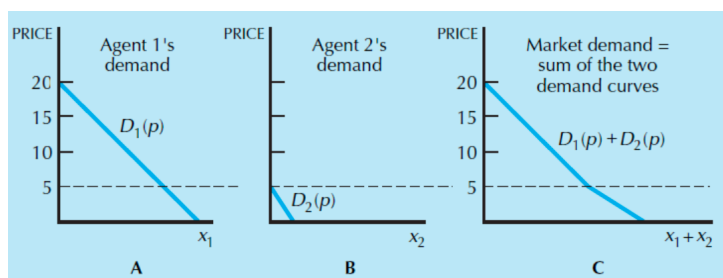
1.1 An Example

- Graphically speaking, market demand curve is “horizontal sum”(水平加总) of individual consumers' curves
- 水平加总！在同样的价格下进行水平加和，见下例！

$$D_1 = 20 - p, \quad D_2 = 10 - 2p$$

- if $0 \leq p \leq 5$

$$D(p) = 30 - 3p$$



2 Elasticity

- Often want to measure responsiveness of one variable in response to changes in another
- Elasticity measures this “sensitivity”(敏感性)
- Elasticity of $x = f(p)$ is defined as

$$\epsilon_{x,p} = \frac{\Delta x/x}{\Delta p/p}$$

- It is unit free
- 是变化率比上变化率!

2.1 Types of Demand Elasticity

- Own-price elasticity of demand (商品的自身价格需求弹性): quantity demanded of good i and price of good i
- Cross-price elasticity of demand (需求的交叉弹性): demand for good i and price of good j
- Income elasticity of demand (需求的收入弹性): demand for good i and income m

2.2 Own-Price Elasticity of Demand

- Own-Price elasticity of demand (q, p) is defined to be the percent change in quantity divided by the percent change in its own price
- Let $D(p)$ be the demand function. Suppose the current price is p and it increases to $p + \Delta p$
- the demand changes from $D(p)$ to $D(p + \Delta p)$
- The percent change in quantity for 1% percent change in price is measured by

$$\epsilon_{q,p} = \frac{\frac{D(p + \Delta p) - D(p)}{D(p)}}{\frac{\Delta p}{p}} = \frac{p}{D(p)} \frac{D(p + \Delta p) - D(p)}{\Delta p}$$

- If the demand is $D(p) = a - bp$ linear, then

$$\epsilon_{q,p} = \frac{-bp}{a - bp}$$

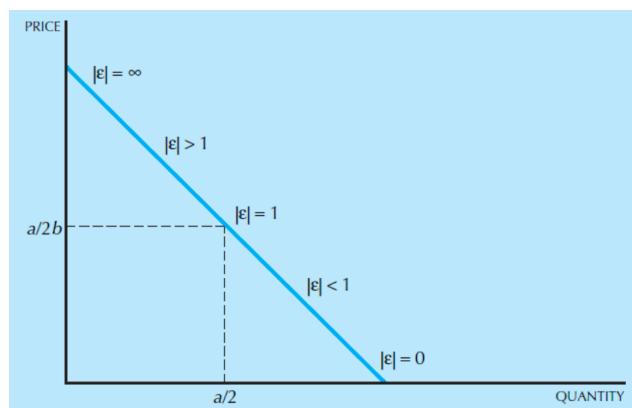
- In general, this is not the case. To avoid this ambiguity, we define the price elasticity of demand to be the limit when $\Delta p \rightarrow 0$:

$$\epsilon_{q,p} = \frac{p}{D(p)} D'(p)$$

- sometimes called the Point elasticity (点弹性)(想要接近线性, 取消 Δp 的影响)
- Price elasticity of demand is generally non-positive, i.e. $\epsilon_{p,q} \leq 0$
- However, when we think about price elasticity or verbally talk about it, we generally refer to its absolute value $|\epsilon_{p,q}|$ even if we do not explicitly mention “absolute value”
- In this way, large elasticity (that means large absolute value) implies that the demand is very sensitive to price changes
 - if $|\epsilon_{p,q}| > 1$: an elastic demand (富有弹性)
 - if $|\epsilon_{p,q}| < 1$: an inelastic demand (缺乏弹性)
 - if $|\epsilon_{p,q}| = 1$: unit elastic demand (单位弹性)
- 弹性小意味着不敏感, 价格变动导致的需求变动小
- 往往价格越高, 弹性越大: 符合直觉的!

$$\epsilon_{q,p} = \frac{-bp}{a - bp}$$

- when $p = 0$, $\epsilon_{q,p} = 0$
- when $p = \frac{a}{b}$, (i.e. $D(p) = 0$), $|\epsilon_{q,p}| \rightarrow \infty$
- when $p = \frac{a}{2b}$, $|\epsilon_{q,p}| = 1$



2.3 An Example: Constant Elasticity Demand

- Does there exist a demand curve that gives us a constant elasticity of demand?
- Consider the demand curve:

$$D(p) = Ap^{\epsilon}$$

where $A > 0$ and $\epsilon < 0$ are constant.

$$\epsilon_{q,p} = \frac{p}{Ap^{\epsilon}}(\epsilon Ap^{\epsilon-1}) = \epsilon$$

- So the elasticity is just $\epsilon_{Q,P}$, a constant. Therefore, the demand is called the constant elasticity demand (固定弹性需求)

3 Revenue and $\epsilon_{q,p}$

- Now consider how $\epsilon_{q,p}$ affects the response of revenue to price changes.
- Seller's revenue is $R(p) = pD(p)$

$$\frac{dR}{dp} = D(p) + p \frac{dD(p)}{dp} = D(p) \left(1 + \frac{p}{D(p)} \frac{dD(p)}{dp} \right) = D(p) (1 - |\epsilon_{q,p}|)$$

- Unit elastic demand: a change to price doesn't alter revenue
- Inelastic demand: a price increase raises sellers' revenue
- Elastic demand: a price increase reduces sellers' revenue
- 不是什么行业都可以用罢工来导致结果的! 罢工希望的是通过减少产出量让老板的收入下降, 但是如果产出下降导致价格涨的很厉害, 可能老板会赚的更多
- 政府是否真的可以提高税率增加政府的收入? 税率提高增加了商品价格, 但是征收的税不一定多了, 这也是取决于市场弹性的大小

3.1 Factors that Influence $\epsilon_{q,p}$

- 影响弹性的大小是通过影响两种效应来影响弹性的
- Closeness of substitutes: The closer the substitutes, the more elastic the demand
- Proportion of income spent on the good: The greater the proportion of income spent on a good, the more elastic the demand. (譬如奢侈品, 弹性很大, 没钱就不买了!)
- Time elapsed since price change: the longer the time, the more elastic the demand. (时间越长, 越有可能产生竞争商品, 可替代性更强)

4 Random Variables

- A proposition that takes the value True with probability π and False with probability $1 - \pi$ is a random variable with distribution $(\pi, 1 - \pi)$
- The (probability) distribution (概率分布) of a random variable X with n values $x_1, x_2, \dots, x_n$ is:

$$(\pi_1, \pi_2, \dots, \pi_n)$$

- with $\pi(X = x_i) = \pi_i$ and $\sum_{i=1}^n \pi_i = 1$
- The expected value of X after doing A is

$$E[X] = \sum \pi_i x_i$$

5 Lotteries

- When we consider consumers' choices under certainty, the basic element is consumption bundles.
- When we think about consumers' choices under uncertainty, the basic element becomes lotteries.
- For simplicity, we assume now that there is only one consumption good. You can even think of this consumption good as money

- ▶ Let $X = \mathbb{R}$ be the set of all outcomes.
- ▶ A lottery over X can be written as

$$p_1 \circ c_1 + p_2 \circ c_2 + \dots + p_n \circ c_n$$

where $n \geq 1$ is a finite integer, $c_1, \dots, c_n \in X$, and $(p_1, \dots, p_n)$ is a probability distribution.

- ▶ So a lottery specifies
 - ▶ the number of mutually exclusive states: n ;
 - ▶ the probability of each state: $(p_1, \dots, p_n)$;
 - ▶ the consumption in each state: $(c_1, \dots, c_n)$.

- Just imagine a lottery in our real life
- there is also a probability of winning for each prize, e.g., 0.001 for the first prize, 0.01 for the second prize, 0.1 for the third prize, and all remaining probabilities for no prize
- This real lottery can be written as

$$0.001 \circ 10000 + 0.01 \circ 1000 + 0.1 \circ 100 + 0.889 \circ 0$$

5.1 Expected Utility

- 尽管期望相同，但是在不同风险偏好的人的心中这些选择是不相同的。譬如一个古老的游戏：第一次抛到正面给 2 元，第二次才抛到正面给 4 元，第 3 次才抛到给 8 元……
- 这个预期的价格是个无穷大的值，但是还是有人愿意玩这个唉，，

- In fact, there are many reasonable assumptions we can make on the consumers' preferences over lotteries
- A most widely adopted assumption is that consumers have expected utility functions
- Formally, we say a consumer has expected utility function over lotteries if there exists a function $v : X \rightarrow \mathbb{R}$ such that the utility that the consumer gets from the lottery $p_1 \circ c_1 + p_2 \circ c_2 + \dots + p_n \circ c_n$ is

$$u(p_1 \circ c_1 + p_2 \circ c_2 + \dots + p_n \circ c_n) = p_1 v(c_1) + \dots + p_n v(c_n)$$

- 可以认为这些之间是不相关的，彼此独立的，所以可以写成上面的形式
- The utility function u is sometimes called a von Neumann-Morgenstern utility function (冯诺伊曼效用函数) and the function v is called a Bernoulli utility function.
- Think about v as the consumer's utility function for non-random outcomes. Then the consumer's expected utility from a lottery is simply the expected value of his utility from the realized outcomes.
- In words, the psychic value of a lottery is given by the expected utility of the dollar payoffs

$$u(L) = E(v(c))$$

5.2 Attitudes Towards Risk

- Based on expected utility, we can talk about consumers' risk attitude.
- 其实效用函数随着 x 的增长的增长不一定快还是慢，所以产生了对于风险态度的差异。
- Risk aversion (风险规避): the consumer prefers certainty to any lottery that yields the same expected wealth
- Risk loving (风险偏好): the consumer prefers random lottery that yields the same expected wealth to certainty
- Risk neutral (风险中性): indifferent between any gambles (random or nonrandom) that yield the same expected wealth
- Suppose the consumer has an expected utility function. The Bernoulli utility function is $v : R \rightarrow R$
- risk averse if v is concave
- risk loving if v is convex (我是赌狗，虽然有风险，但是他们实在给的太多辣!!) (干这一票的! 一辈子不用干活了!)

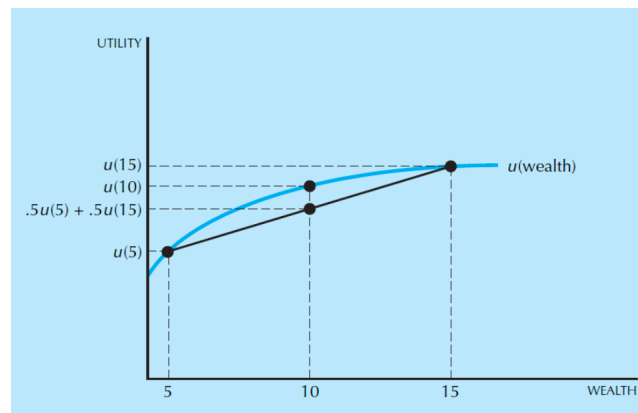
- risk neutral if v is linear
- 比较风险偏好的时候一定要这个人在不同的风险下的财富期望是相同的，否则不能得出任何结论

5.3 An Example

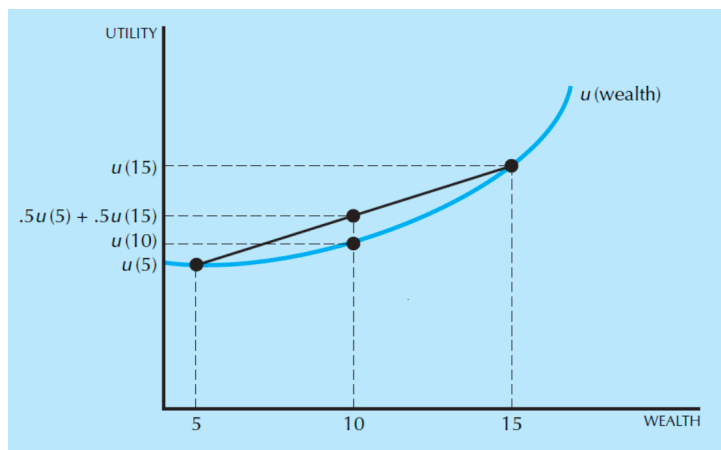
- To understand the idea, suppose the consumer currently has c units of money
- He is contemplating a lottery that gives him a 50 percent probability of winning $c' > 0$ units of money and a 50 percent probability of losing c' units of money.
- If he does not take this lottery, his expected utility is $v(c)$.
- If he takes this lottery, his wealth will be $c + c'$ with 0.5 probability and $c - c'$ with 0.5 probability
- his expected utility becomes

$$0.5v(c + c') + 0.5v(c - c')$$

- If v is concave, then the consumer prefers not taking the lottery



- If v is concave, then the expected utility of wealth is greater than the utility of the expected value of wealth



6 Certainty Equivalent and Risk Premium

6.1 Certainty Equivalent

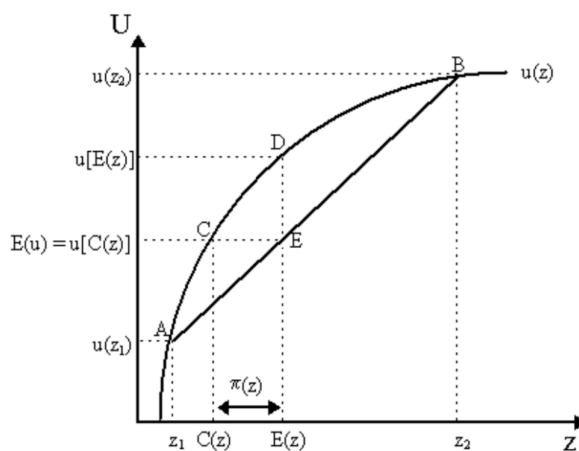
- Certainty Equivalent (CE, 确定性等值)
- the amount of money (wealth) obtained with certainty, which gives the same utility as the expected utility of an uncertain outcome

$$v(CE) = p_1 v(c_1) + (1 - p_1) v(c_2)$$

6.2 Risk Premium(风险溢价)

- 为了规避风险，宁愿丢掉一些财富来不参与带有风险性的行为（这样的期望效用还是会不低于参加风险行为！）
- the difference between the Expected Wealth of an action and the Certainty Equivalent

$$RP = [p_1 c_1 + (1 - p_1) c_2] - CE$$



- Risk averse individual: the horizontal EU line reaches the concave utility curve before it reaches the vertical line that corresponds to the expected value
- so risk premium is positive
- EU and risk premium depends on: decision maker's utility function; probability distribution for the payoffs

7 Insurance

- 虽然可以风险对冲来减少自己的风险，尽可能的多选择收益负相关的产品，多样化产品篮子，但同时也会降低自己的期望收益。另一种方法是买保险。
- Consider a risk averse consumer who has an initial wealth of w but who runs a risk of a loss of D
- The probability of the loss is $\pi \in (0, 1)$
- It is possible, however, for the consumer to buy insurance. One unit of insurance costs p dollars and pays 1 dollar if the loss occurs.
- One unit of insurance costs p dollars and pays 1 dollar if the loss occurs.
- Thus, if α units of insurance are bought, the wealth of the individual will be
 - $w - \alpha p$ if there is no loss
 - $w - \alpha p - D + \alpha$ if the loss occurs
- In this case, the consumer's expected wealth is

$$w - \pi D + \alpha(\pi - p)$$

- and the expected utility is

$$(1 - \pi)v(w - \alpha p) + \pi v(w - \alpha p - D + \alpha)$$

- 公平保险：一份保单保险公司的收益是： $\pi(p - 1) + (1 - \pi)p = p - \pi$ ，在完全竞争市场的情况下，这个收益会不断趋于 0，从而保险公司收取的一单位费用就是发生事件的概率（这种情况下没有收益）
- it is equal to the expected cost of insurance $p = \pi$
- How much insurance should the consumer buy?
- The consumer's problem is

$$\max (1 - \pi)v(w - \alpha\pi) + \pi v(w - \alpha\pi - D + \alpha)$$

- One way to solve this problem is to consider its first order condition.
- we take a more “economic” way. First, note that when $p = \pi$, the consumer’s expected wealth if he buys insurance is

$$w - \pi D + \alpha(\pi - p) = w - \pi D$$

- Second, because the consumer is risk averse, he prefers a certain outcome to any lottery whose expected wealth is equal to that outcome:

$$(1 - \pi)v(w - \alpha\pi) + \pi v(w - \alpha\pi - D + \alpha) \leq v(w - \pi D)$$

- This inequality states that the consumer’s expected utility can not exceed $v(w - \pi D)$ regardless of his choice of insurance α
- Third, by purchasing $\alpha = D$, the consumer gets expected utility
- Therefore, $\alpha = D$ is optimal for the consumer
- 两个东西相等意味着不管损失发生或者不发生，我的财富都是相同的，也就是完全没有风险！最佳的购买保险的情况（完全保险），也就是完全没有风险的状态：certain 的状态！
- 保险公司如果可以自由定价，能够收取的最高金额就是消费者的风险溢价！
- 大规模的传染病是很少有保险公司愿意赔付的，传染病不是一个独立事件！可能会有连锁反应，一个人生病可能要赔一个小区的，这个概率就变得很大了！一旦暴雷就没钱付了（类似于挤兑）
- 但是保险公司如果规模够大，可以分散风险，这个是 ok 的当然保险公司存在的前提是不存在道德风险，就是骗保，，比如谎报身体状况报人寿保险等等。

上述结论都是在假设消费者会按照效用期望来行事，但是实际上有 Allais paradox 的挑战。

Technology

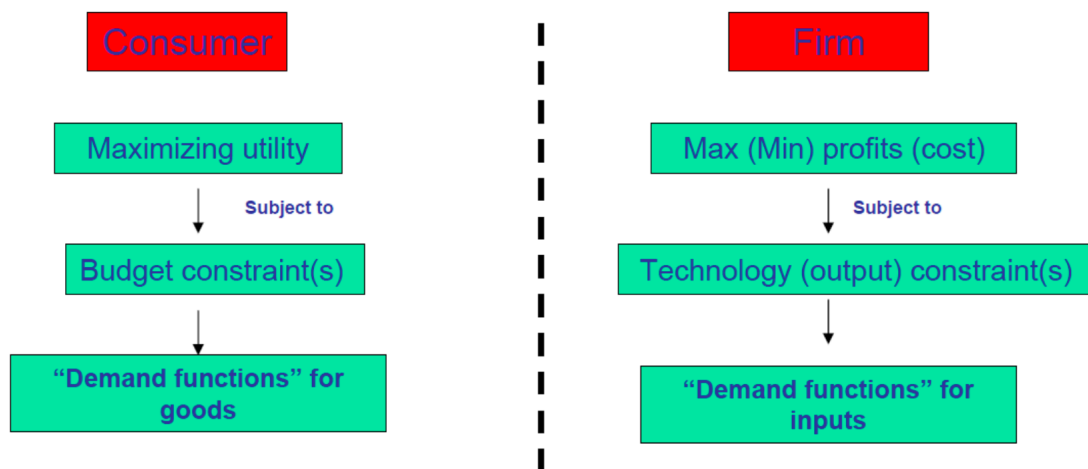
姜峻赫

2022.11.05 & 11.12

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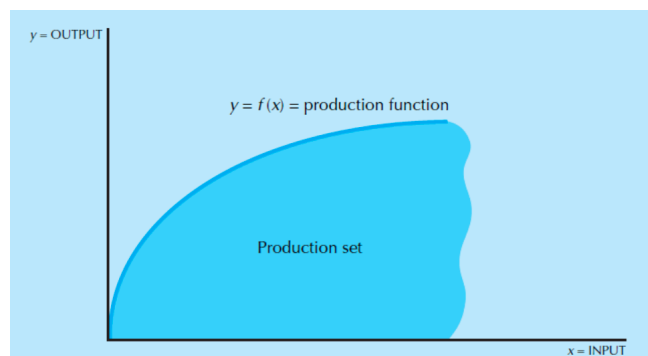
1 Symmetry Between Consumer and Firm Theory



- 厂商也像消费者有预算约束一样，他们的生产也受到技术条件的约束！

2 Technology

- We now turn to our study of firms' behavior.
- A firm uses some inputs to produce outputs. inputs are also called “factors of production” (生产要素)
- examples: labor (L) (skilled, unskilled); physical capital (K) (有形资本) (equipment, structures, inventories, land); intangibles(无形资本)
- Only certain combinations of inputs are feasible ways to produce a given amount of output
- Firm must limit itself to technologically feasible production plans
- 最传统的说法：劳动，土地，资本。现在数据（data）也被作为生产资料来处理！
- Suppose a firm uses n inputs to produce an output.
- A technology (生产技术) is represented by a production function (生产函数) $f : \mathbb{R}^n \rightarrow \mathbb{R}$, that maps every possible combination of inputs $(x_1, x_2, \dots, x_n)$ to its maximum possible output $f(x_1, \dots, x_n)$
- Here x_i is the amount of the i th input.
- When there is only one input, we can easily plot production function in the input-output plane



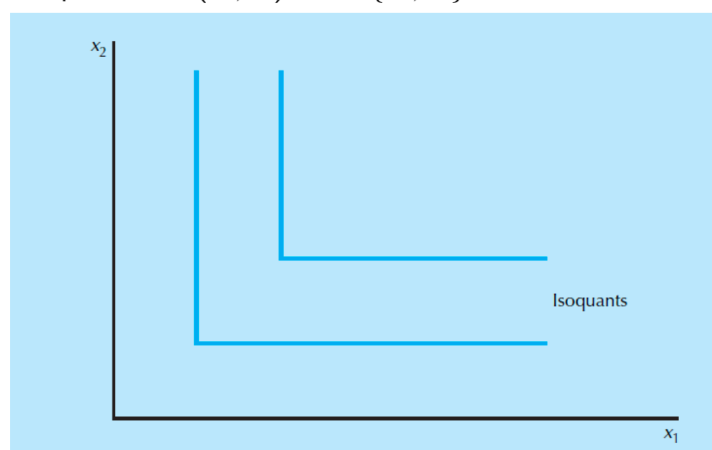
3 Isoquant Curves

- 边界线就是生产函数，假定所有人没有在摸鱼都在好好工作的最大产量！
- What about production functions with two inputs?
- isoquant curves (等产量线)

- An isoquant is the set of all possible combinations of inputs 1 and 2 that are just sufficient to produce a given amount of output
- Analogous to indifference curve. Implies substitution across inputs possible
- 和无差异曲线的差别在于，无差异曲线描述的是消费者的一个目标函数，而等产量线描述的是生产者在做选择的时候的约束条件，而描述消费者的选择时的约束条件，用的是消费约束线。
- Indifference curve 我们只关注高低，不在乎实际的数值，这是因为我们只关心效用的比较关系，也就是序数关系。而等产量线上的数字是有意义的，是实际的东西。对于无差异曲线我们可以做单调变换改变线和线之间的距离，但是等产量线却不可以，他们之间的距离是很重要的。

4 Examples

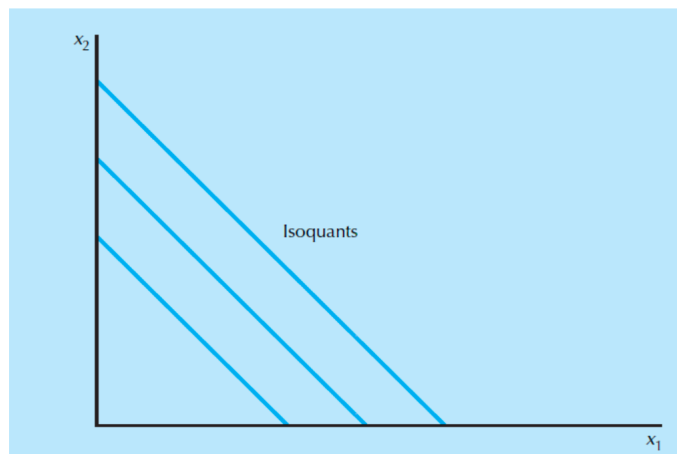
Fixed portions: $f(x_1, x_2) = \min\{x_1, x_2\}$



- 在生活中很常见，比如汽车的轮子，给我两个的效用和三个是差不多的，比如杯子和杯盖，标配的理念，构成完全互补！

Perfect complement

Perfect substitute: $f(x_1, x_2) = x_1 + x_2$



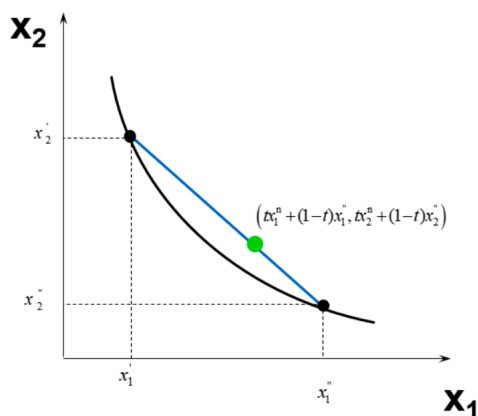
Perfect substitutes

- 举个例子，比如一个沙拉的不同产地的蔬菜，就是完全替代的。（或许家三在饭里面配的菜是油麦菜还是豆芽其实是等价的？）

Cobb-Douglas production function

- $f(x_1, x_2) = Ax_1^a x_2^b$
- 这个地方不能做单调变换 (不能!!! 不能!!!!)，每一个字母都是有确切含义的!
- $A > 0$: the scale of production, i.e., how much output we would get if we used one unit of each input.
- $a > 0$ and $b > 0$: two inputs' respective share of output
- A measures the portion of output not explained by x_1 and x_2 .
- It is often called total factor productivity (TFP) (全要素生产力)
- When we studied Cobb-Douglas utility function, we assumed $A = 1$ and argued that we can assume without loss of generality that $a + b = 1$. This is because utility is ordinal, only relative magnitude matters and utility levels can be arbitrary
- This is not true for Cobb-Douglas production function, output level is fixed for a given inputs combination for a fixed technology
- Similarly as preferences, we usually impose some natural properties on technologies/production functions.

- We say a technology is monotonic (单调) if its production function is increasing. If you increase the amount of at least one of the inputs, it should be possible to produce at least as much output as you were producing originally.
- 也就是说生产函数不存在类似于厌恶品的东西!
- We say a technology is convex (凸的) if for any output level y , the set $\{(x_1, x_2) | f(x_1, x_2) \geq y\}$ is convex. Suppose input bundles x and x' both provide q units, then the mixture $tx + (1-t)x'$ provides at least q units of output for all $t \in (0, 1)$. For example, if f is concave, then the technology is convex.
- 正如下图所示, 中间这个点要高于等产量线所代表的产量。相对于绝对也更偏好于平均!



5 Marginal Product

- Recall that the marginal utility of a good measures the increase in utility is one more unit of this good is consumed
- Similarly, the marginal product (边际产出) of an input measures how many more output can be produced if one more unit of this input is used
- The marginal product of input 1 is

$$MP_i(x_1, x_2) = \frac{\partial f(x_1, x_2)}{\partial x_i}$$

- As more of a variable input (labor) is added to a fixed input (plant), additions to output get smaller and smaller

$$\frac{\partial MP_1}{\partial x_1} \leq f_{11} \leq 0$$

$$\frac{\partial MP_2}{\partial x_2} \leq f_{22} \leq 0$$

- 给定其他条件都不变的情况下，一个生产要素增加的时候，边际生产率是递减的!!
- Note that adding workers increases output but the increases become smaller and smaller as more workers are hired. Possible reason: teamwork (团队协作) and specialization (专业化)
- 人口红利：农村人口到城市，城市工业化开始，人口到工业区可以增加城市工业化的边际产出！而不减少农业的边际产出！（会出现拐点）
- 耕地红线：给定耕地的情况下还有边际产出递减，再减少耕地又有更多的农业人口转出，产出粮食更少了。
- 马尔萨斯陷阱：新产出的人口带来的边际产出是递减的，但是人口的消耗量不变，也就相当于越来越多的人只吃饭不干活，从而随着人的增长，一定会发生大饥荒。（但是条件是不满足的：保持边际产出递减，其他的投入应当是不变的，当代农业的“资本”的数量是在不断增加的，技术也在进步）（一定要检查定理的前提条件是否存在!!）

Technical Rate of Substitution

- The technical rate of substitution (技术替代率, sometimes MRTS) of input 1 for input 2 at input level (x_1, x_2) is

$$TRS_{12}(x_1, x_2) = \frac{MP_1(x_1, x_2)}{MP_2(x_1, x_2)}$$

- The idea is the following: suppose the firm is producing $f(x_1, x_2)$ units of output if we decrease one unit of input 1, how many units of input 2 shall we increase to keep the same output level (减少一单位 1, 需要多少单位 2 来弥补?)
- Mathematically, $TRS_{12}(x_1, x_2)$ is the “slope” of the isoquant curve at (x_1, x_2)
- 边际替代率递减：斜率越来越平。解释：伴随着 x_1 增加， MU_1 在减少，而 x_2 在同步减少， MU_2 会增加，整体来看就是 TRS_{12} 在递减。
- 边际产出递减指的是其他的要素不变，考察其中一种要素的变化，边际产出是递减的；技术替代率指的是给定产出水平不变，同时调整 x_1 和 x_2 ，调整比例不变，边际替代率是递减的。
- 例如

$$f(x_1, x_2) = x_1^a x_2^b, \quad MP_1(x_1, x_2) = ax_1^{a-1} x_2^b, \quad MP_2(x_1, x_2) = bx_1^a x_2^{b-1}$$

$$TRS_{12}(x_1, x_2) = \frac{ax_2}{bx_1}$$

- 技术替代率还和技术移民有关，要让自己不下岗，就要让自己特殊一些，替代率才会越来越小。（越机械化、越稳定的生产越容易被机械化智能化）

6 Returns to Scale

- 规模报酬
- Unlike marginal productivity, which considers the output change due to exactly one input level changes while holding all other inputs fixed, we now consider the output changes because of all inputs change.

- CRS: A production function exhibits constant returns to scale (规模报酬不变) if

$$f(tx_1, tx_2) = tf(x_1, x_2) \quad \forall (x_1, x_2), t > 1$$

- IRS: A production function exhibits increasing returns to scale (规模报酬递增) if

$$f(tx_1, tx_2) > tf(x_1, x_2) \quad \forall (x_1, x_2), t > 1$$

(初期投入 or 专业化分工) (此处并购会带来更好的产出)

- DRS: A production function exhibits decreasing returns to scale (规模报酬递减) if

$$f(tx_1, tx_2) < tf(x_1, x_2) \quad \forall (x_1, x_2), t > 1$$

(后期投入) (与其有一个大厂, 不如拆成很多小厂子)

- Note that a technology can exhibit different kinds of returns to scale at different levels of production.

7 Long Run and Short Run

- For a firm that uses several inputs to produce its output, usually the levels of some inputs can be easily changed in a short-period of time, while the levels of other inputs can only be changed after a long while
- Intuitively, it is usually easy for a firm to hire some workers in a short-period of time, while it takes much longer for a firm to build a new factory
- Thus, although the number of factories is a variable that can be chosen by the firm and determines the firm's output, the firm can only take it as given if we consider its production only within a short period of time.
- Economists, therefore, like to divide the length of time into two kinds based on whether all inputs can be arbitrarily changed.
- 通俗来讲, 你的厂子和土地短期处理不了, 但是长期来看可以当作被卖掉了。短期这笔投入收不回来, 是不能懂的, 长期任何要素都可以变动

- Short run means a certain period of time during which at least one input can not be changed
 - fixed factor: a factor of production that is in a fixed amount for the firm
 - variable factor: a factor can be used in different amounts
- Long run means a long period of time during which all inputs can be changed.

Profit Maximization & Cost Minimization Problem

姜峻赫

2022.11.12

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1 Profit Maximization

- There are two ways to investigate firm's profit maximization problem. One is a direct approach.
- The other is an indirect approach.we divide a firm's profit maximization problem into two stages:one is cost minimization and the other is profit maximization.

1.1 Competitive Markets and Price Takers

- For now, we assume that firms are price takers (价格接受者) in both market for the factors of production it uses and market for the output goods it produces
- 也就是说市场很大很稳定很有竞争性，从而单个企业的变化对整个市场影响不大（也就是，不是 price maker 而是 price taker）
- Thus, from firms'point of view, prices for both factors and outputs are fixed, regardless of their demands in the factor markets and supplies in the output markets.

- Markets in which firms take prices as given are called competitive markets (竞争性市场). these participating firms are called competitive firms.
- 差异化不明显产品（比如白 T 恤）的厂商，基本上做不到一家独大，从而可以认为是竞争性市场。
- The justification for price taking behavior is large markets. because there are many many participants in the same market, each individual participant only have negligible effect on the market price

1.2 Profit

- Suppose a firm sells y units of output at price p
- To produce these outputs, the firm uses inputs $(x_1, \dots, x_n)$ at price $(w_1, \dots, w_n)$.
- The profits (利润) the firm receives can be expressed as

$$\pi = py - \sum_{i=1}^n w_i x_i$$

- 这里的 p 和 w_i 都是给定不变的（因为竞争性市场）
- py is the firm's revenue. $\sum_{i=1}^n w_i x_i$ is total cost.
- In the expression for cost, we should be sure to include all of the factors of production used by the firm, valued at their market price.
- if an individual works in his own firm, then his labor is an input and it should be counted as part of the costs
- his wage rate is simply the market price of his labor – what he would be getting if he sold his labor on the open market
- Economic costs (经济成本) like these are often referred to as opportunity costs (机会成本).
- 假定你已经有土地了，而不是真的要去借你的土地，机会成本就是你的土地被用来干别的所能达到的最大收益。

1.3 Short-Run Profit Maximization

- Let's consider the short-run profit-maximization problem when input 2 is fixed at some level x_2 .
- Let $f(x_1, x_2)$ be the production function for the firm. Then the profit maximization problem facing the firm can be written as

$$\max p f(x_1, x_2) - w_1 x_1 - w_2 x_2$$

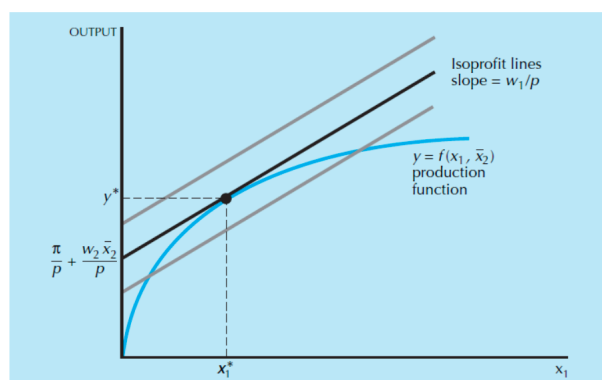
- The first order condition is

$$pMP_1(x_1^*, x_2) = w_1$$

- the value of the MP of a factor should equal its price (边际成本等于价格)
- Recall MP_1 is interpreted as the additional output that can be produced if one more unit of input 1 is used.
- So $pMP_1(x_1, x_2)$ is the benefit of using one more unit of input 1, while w_1 is the cost
- if $pMP_1(x_1, x_2) > w_1$, the firm should use more input 1 and produce more
- if $pMP_1(x_1, x_2) < w_1$, the firm should use less input 1 and produce less

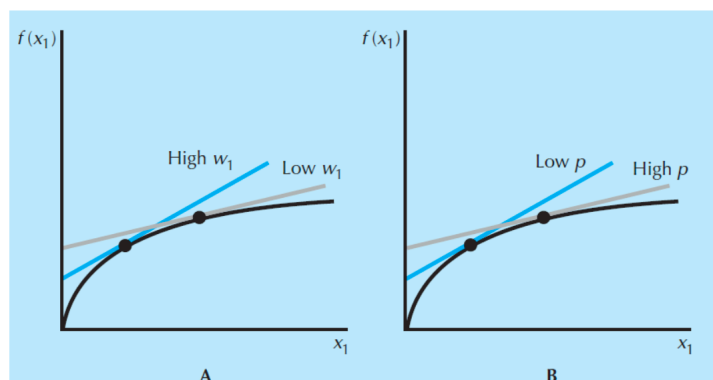
$$\pi = py - w_1x_1 - w_2\bar{x}_2$$

Isoprofit lines (等利润线): $y = \frac{\pi}{p} + \frac{w_2}{p}\bar{x}_2 + \frac{w_1}{p}x_1$



- 给定约束: production function set, 找等利润线的最高位置: 还是相切!
- 短期内, 负利润是会存在的!

How does the optimal choice of factor 1 vary as we vary its factor price w_1 ? What if the output price decreases?



$$w_1 \uparrow, x_1^* \downarrow, x_2^* \downarrow$$

$$p \downarrow, x_1^* \downarrow, x_2^* \downarrow$$

- 容易理解，卖的价格下降或者成本价格提高，都会带来产量下降

$$w_2 \uparrow, \Delta x_1^* = 0, \pi \downarrow$$

- 规模报酬递增的情况会和完全竞争市场矛盾的！这两者不能同时存在（或者只在某个阶段可以共存），否则达成均衡的时候只会有有限数量的大企业，此时的 p 是会随着市场变动而变动的。

2 Cost minimization problem v.s.Profit Maximization

- An important implication of the firm choosing a profit-maximization production plan is that there is no way to produce the same amounts of outputs at a lower total input costs
- cost minimization is a necessary condition for profit maximization
- Reasons to study firm's cost minimization problem
 - leads us to a number of results that are technically useful
 - when the firm is not a price taker, we can no longer use the profit function for analysis
 - if the production set exhibits nondecreasing returns to scales, the profit maximization problem does not work well
- Firm's profit maximization problem can be divided into two steps
- first, for various levels of output, the firm chooses the optimal factors that minimize production costs
- second, the firm decides how much to produce to maximize its profit
- We now consider the cost minimization problem (成本最小化问题)

$$\min w_1 x_1 + w_2 x_2 \quad s.t. \bar{y} = f(x_1, x_2)$$

- Assume f is differentiable
- Suppose $x_1^* > 0$ and $x_2^* > 0$ solves this problem. Using Lagrange method, we know

$$\frac{w_1}{w_2} = \frac{MP_1}{MP_2} = TRS_{12}(x_1^*, x_2^*)$$

- At optimal, TRS_{12} must equal factor price ratio

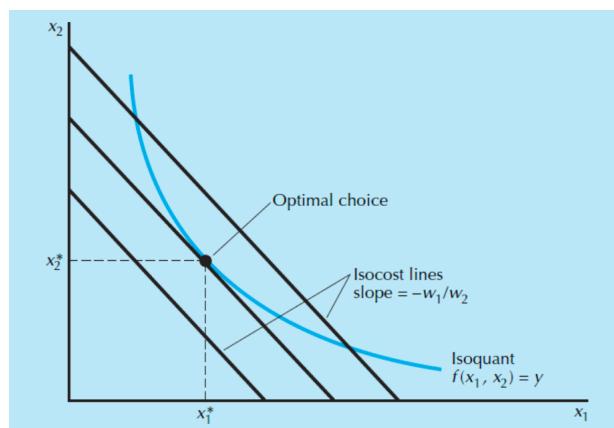
- if $w_1/w_2 > TRS_{12}(x_1, x_2)$, imagine we lower x_1 by one unit, we can increase x_2 by $TRS_{12}(x_1, x_2)$ units and get the same output level, the difference in costs is $w_1 + TRS_{12}(x_1, x_2)w_2 < 0$, so (x_1, x_2) is not cost minimizing
- Suppose that we want to plot all the combinations of inputs that have some given level of cost, C . We can write this as

$$w_1x_1 + w_2x_2 = C$$

- which can be rearranged to give

$$x_2 = \frac{C}{w_2} - \frac{w_1}{w_2}x_1$$

- As we let the number C vary we get a whole family of isocost lines. The choice of factors that minimize production costs can be determined by finding the point on the isoquant that has the lowest associated isocost curve.



- 注意此处的约束和目标跟效用处对偶的无差异曲线位置对换了，此处约束是曲线，目标函数是直线！
- The solution can be written as $x_1(w_1, w_2, y)$ and $x_2(w_1, w_2, y)$
- if $f(x_1, x_2) = x_1^a x_2^b$, $x_1 = ((\frac{b}{a} \frac{w_1}{w_2})^{-b} y)^{\frac{1}{a+b}}$
- called the conditional factor demand functions (有条件的要素需求函数), or derived factor demands (派生的要素需求)
- The corresponding cost function is $c(w_1, w_2, y)$

2.1 Cobb-Douglas Technology

- Consider the Cobb-Douglas technology $f(x_1, x_2) = x_1^a x_2^b$ for some $a, b > 0$

$$\frac{w_1}{w_2} = \frac{ax_2}{bx_1}$$

- Combined with

$$x_1^a x_2^b = y$$

- we get

$$x_1 = \left(\left(\frac{b}{a} \frac{w_1}{w_2} \right)^{-b} y \right)^{\frac{1}{a+b}}$$

- So the minimum cost is

$$\begin{aligned} c(y) &= w_1 x_1 + w_2 x_2 \\ &= \frac{a+b}{a} w_1 x_1 \\ &= \frac{a+b}{a^{\frac{a}{a+b}} b^{\frac{b}{a+b}}} w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}} \end{aligned}$$

- if $a + b = 1$ we have $c(y) = \frac{1}{a^a b^b} w_1^a w_2^b y$

2.2 Perfect Complements

- Consider $f(x_1, x_2) = \min\{x_1, x_2\}$
- factors are perfect complements
- To produce y units of output, we need at least y units of x_1 and y units of x_2
- So the minimal cost is

$$c(y) = (w_1 + w_2)y$$

2.3 Perfect Substitutes

- Consider $f(x_1, x_2) = x_1 + x_2$
- To produce y units of output, we can use y units of x_1 and 0 unit of x_2 , or 0 unit of x_1 and y units of x_2
- So the minimal cost is

$$c(y) = \min\{w_1 y + w_2 \times 0, w_1 \times 0 + w_2 y\} = \min\{w_1, w_2\}y$$

Cost Curves

姜峻赫

2022 年 11 月 19 日

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- In producers' theory, the difference between short-run and long-run is whether all factors in production are free to vary. In the short-run, some factors are fixed
- Suppose x_2 is fixed in the short-run. Then the firm's cost minimization problem becomes

$$\min_{x_1} wx_1 + w\bar{x}_2 \quad y = f(x_1, \bar{x}_2)$$

- 长期成本是所有短期成本最小的一个值。长期成本和短期成本有些时候是相同的，此时恰好是选到了长期最优的规模 x_2 ，那么长期和短期最优是一样的
- 长期求解：固定 x_2 然后让 x_1 使得最优，从而再求解 x_2 的最优化问题

► Now let's consider the following minimization problem

$$\min_{x_2} c_s(w_1, w_2, y; x_2)$$

► Plugging in $c_s(w_1, w_2, y; x_2)$, we can write

$$\min_{x_2} w_1 \left(\frac{y}{x_2^b} \right)^{\frac{1}{a}} + w_2 x_2$$

► We can show that the minimized cost is

$$\frac{a+b}{a^{\frac{a}{a+b}} b^{\frac{b}{a+b}}} w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}}$$

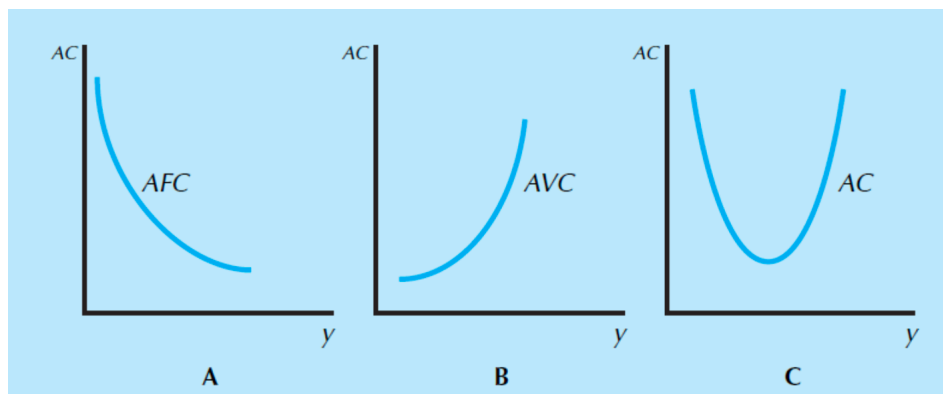
which is exactly the $c(w_1, w_2, y)$ we have calculated before

1 Average Costs

- fixed costs: costs that must be paid regardless of the level of production 【不变成本 (fixed): 就算你产量为 0 也收不回的投入】;
- $F \equiv c(0)$
- variable costs: costs that change when output changes 【可变成本 (variable): 随时停产随时不用的成本。】
- $c_v(y) \equiv c(y) - c(0)$
- Average cost function $AC(y)$ (平均成本函数), measures the costs per unit of output

$$AC(y) = \frac{c(y)}{y}$$

-
- 初期可能实现边际产出递增，因为专业化。后期递减，是因为承载能力有限。
-
- 平均可变成本函数是和边际产出相反的，所以相应的会有一开始减少，后来增多。或者一直增多（两种可能性）



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2 Marginal Costs

- The marginal cost curve (边际成本曲线) measures the change in costs for a given change in output

$$MC(y) = c'(y) = c'_v(y)$$

- 不变成本是常数，求完导就没了

3 MC Curves v.s. AC Curves

- MC curve and AVC curve coincide at $y = 0$

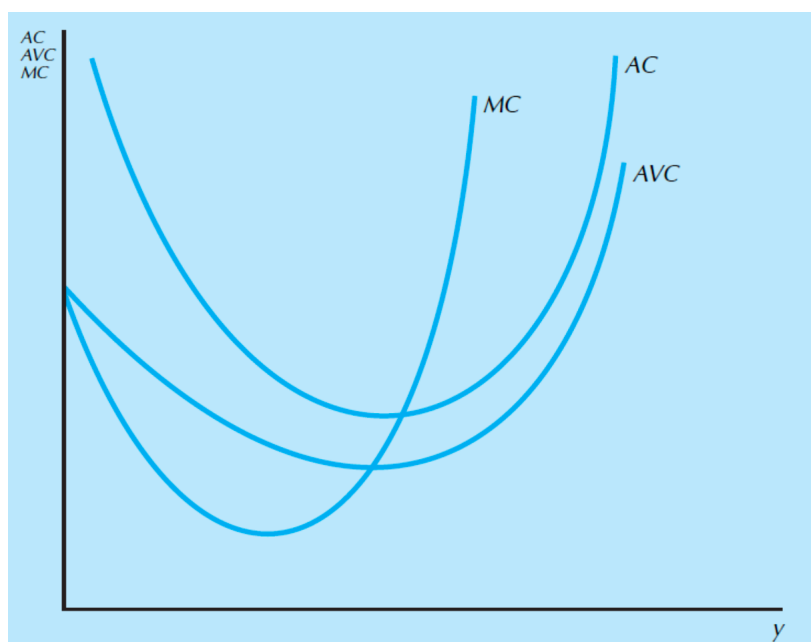
$$MC(0) = \lim_{\Delta y \rightarrow 0} \frac{c(\Delta y) - c(0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{c(\Delta y)}{\Delta y} = AVC(0)$$

- 离散情况下理解，也就是生产第一单位的时候的边际产出刚好就是平均产出（因为只生产了一单位）
- If $AVC(y)$ decreases in the interval (y_1, y_2) , then

$$MC(y) \leq AVC(y) \quad \forall y \in (y_1, y_2)$$

- in a range of output where AVC curve is decreasing, MC curve must lie below AVC curve
- $AVC(y)$ 这一段递减，说明此时的成本（边际成本）一定会低于平均（边际小于平均，那么平均递减；大于平均，平均递增）
- 像老师算平均分的时候，加一个低于平均分的同学加入其中会让平均分减少
- 另一种理解方式，用导数来衡量递增递减

$$\frac{\partial AVC}{\partial y} = \frac{\partial \frac{C}{y}}{\partial y} = \frac{MC}{y} - \frac{C}{y^2} = \frac{1}{y}(MC - AVC)$$



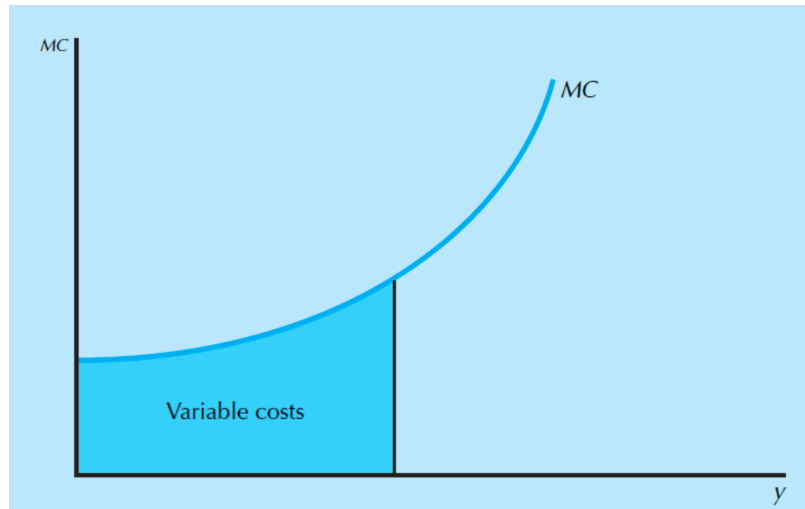
- 要注意几个特殊点：MC 和 AVC 的交点在 y 轴（已经讨论），还有 AVC 的最低点（这由于只要 AVC 大于 MC，就会保持递减，从而一直到两者相等），MC 也和 AC 交在最低点，讨论同上。
- AVC 和 AC 的间距会越来越小（这是由于平均固定成本占比越来越小，从而间隔先大后小）

4 Marginal Cost v.s. Variable Costs

- If the marginal cost is continuous on $[0, +1)$, the area below a marginal cost curve is the associated variable costs
- Consider the case where the output good is produced in discrete amounts

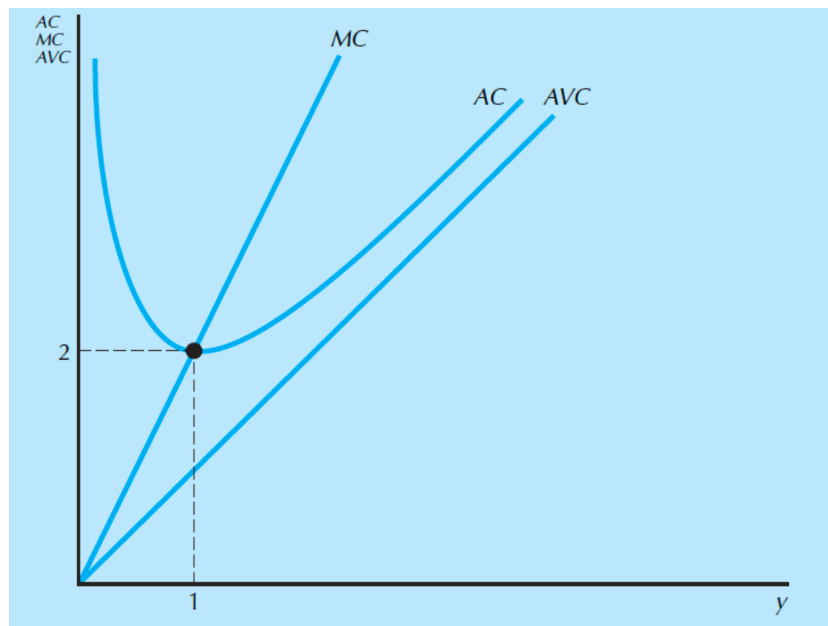
$$c_v(y) = (c_v(y) - c_v(y-1)) + (c_v(y-1) - c_v(y-2)) + \cdots + (c_v(1) - c_v(0))$$

$$c_v(y) = MC(y-1) + MC(y-2) + MC(y-3) + \cdots + MC(0)$$



5 An Example

- Let's consider the cost function $c(y) = y^2 + 1$
- $c_v(y) = y^2$ $c_f(y) = 1$ $AVC(y) = \frac{y^2}{y} = y$ $AFC(y) = \frac{1}{y}$ $AC(y) = y + \frac{1}{y}$ $MC(y) = 2y$



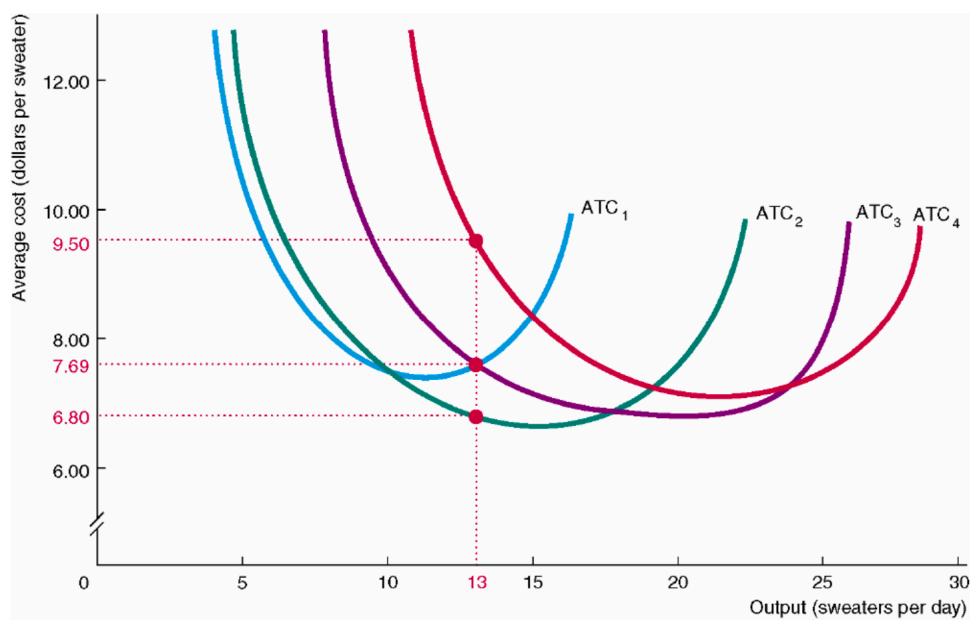
6 Long-run v.s. Short-run Cost Curves

- Previous analysis applies to both long-run and short-run cost functions. We only need to assume $F = 0$ when we think $c(y)$ as a long-run cost function

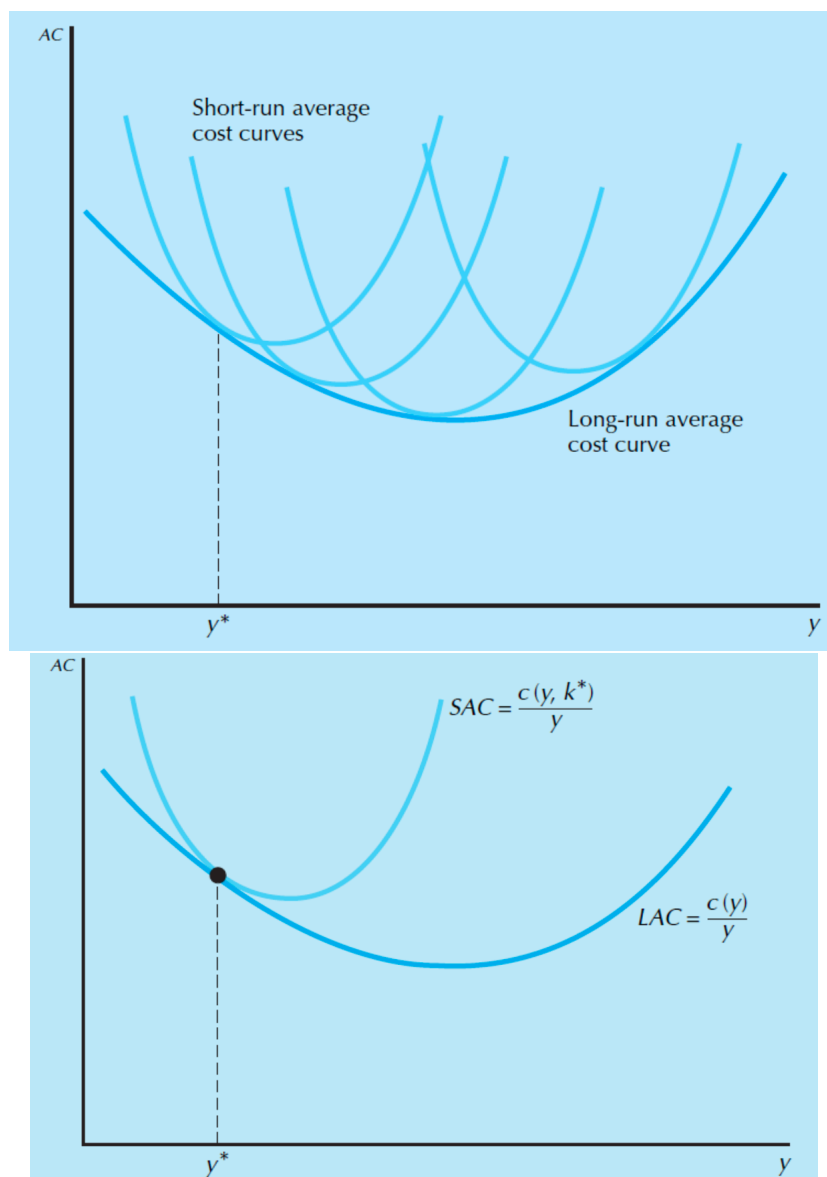
- What are the relationships between long-run and short-run cost curves?
- For simplicity, assume that in the short-run capital k is the only fixed input. Long-run cost function then can be expressed as

$$c(y) = \min_{k \geq 0} c_s(y; k)$$

- If for each y , $k(y)$ solves this minimization problem, we can also write $c(y) = c_s(y; k(y))$
- 取到长期最优成本下的成本函数如上，长期成本会比短期成本高！
- long-run cost function is the lower envelop (下包络线) of all short-run cost functions



- 每一条不同颜色的线都是不同的 k 决定的线。生产者在 13 的这个时刻，肯定选择的是最低的 6.8，选择了绿色线。也就是说，在可选的时候，选择的值一定是最低的。从而长期来看，如果时刻更为密集，选择的曲线就更加平滑，我的选择就是这些曲线的下包络线。



- the long-run average cost function is also the lower envelop of all short-run average cost functions
- 长期成本线和短期成本线事实上是有一个切点的。LAC 只是 SAC 的某一种情况，在取定 k 后得到的那一个点就是他们两个的切点。
- 切点不一定是平均成本曲线的最低点，只是在固定的 k 的时候，所有曲线之中最低的一条线 1，而不一定是这条线 1 的最低点。（下包络线和曲线的切点不一定是最低点）
- Moreover, we also have

$$LAC(y) = \frac{c(y)}{y} = \frac{\min_{k \geq 0} c_s(y; k)}{y} = \min_k \frac{c_s(y; k)}{y} = \min_k SAC(y; k)$$

- where LAC:long-run average cost and SAC:short-run average cost
- i.e. $LAC(y) = SAC(y; k(y))$

LMC vs SMC

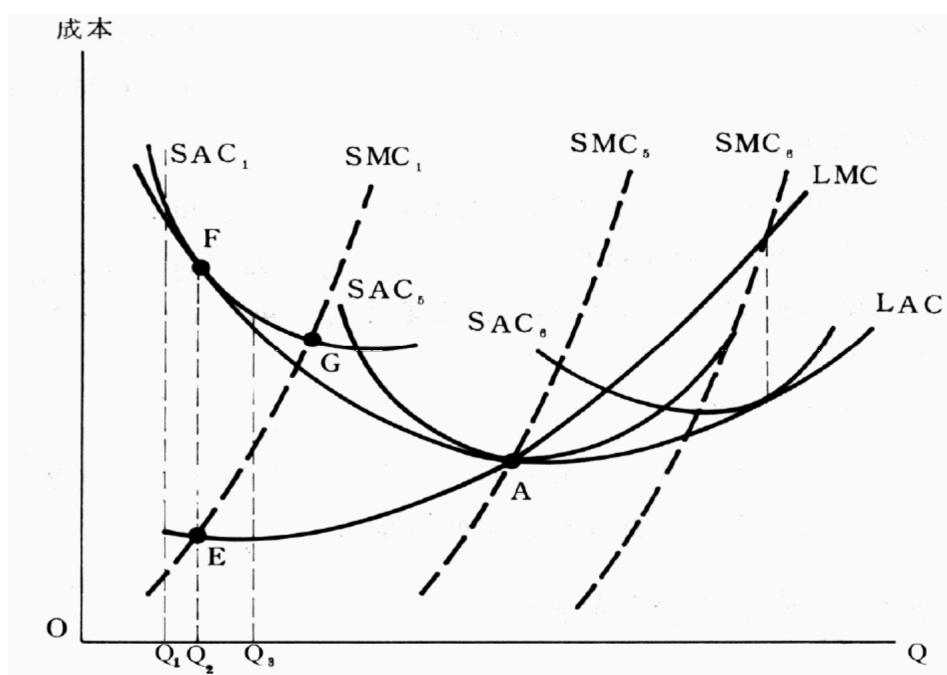
- Because $c(y) = c_s(y; k(y))$, by chain rule, we have

$$MC(y) = \frac{\partial c_s(y; k(y))}{\partial y} + \frac{\partial c_s(y; k(y))}{\partial k} k'(y)$$

- 如果 k 刚好是最优化下的 k ，那么一定满足一阶条件

$$\frac{\partial c_s(y; k(y))}{\partial k} = 0$$

- 短期边际成本和长期边际成本也是有一个交点的，就是长期选定的 k 刚好是短期的 k (此时的交点横坐标对应的产量，刚好是包络线和曲线的切点的横坐标，但是不是边际曲线的切点，而是交点)



- 这个图中还有一个要点：边际小于平均的时候，平均是递减的，大于平均的时候，平均递增。边际和平均的交点就是最低点。（这对于短期和长期都是成立的）
- 长期边际成本线和每条短期成本线都有交点，事实上，长期成本线就是这么画出来的。因为每条短期成本线和长期成本线的交点的横坐标刚好是切点的横坐标，根据已知的短期成本线此时的纵坐标就可以得到长期成本线对应的函数在这个点的取值)

- 从 Q_1 到 Q_2 , SAC 是在不断接近 LAC 的, 这时候 SAC 下降的比 LAC 快很多! 从 Q_2 到 Q_3 , 同一条 SAC 在远离 LAC, SAC 下降的要慢于 LAC。

Supply & Partial Equilibrium

姜峻赫

2022 年 11 月 19 日

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1 Market environments

- How does a firm decide how much product to supply?
- goals, e.g. profit maximization; technology, e.g. cost function; market environments (市场环境)
- Market environments describe the ways that firms respond to each other when they make their pricing and output decisions, also referred to as market structure (市场结构)
- Four examples of market environments: monopoly (垄断), oligopoly (寡头), monopolistic competition (垄断性竞争), pure competition (完全竞争).

1.1 Monopoly

- only one firm in this market
- the only seller determines the quantity supplied/the market-clearing price

1.2 Oligopoly

- a few firms that sell identical products
- the decisions of each influence the payoffs of the other firms
- 企业之间不同的行为会很大程度上影响对方，所以涉及到博弈环节

1.3 Monopolistic competition

- many firms that sell slightly different products
- each firm's output level is small relative to the total
- 在足够细分的厂商里面，每一个厂商在自己的范围内有垄断力量，但是这些细微的差异不会让人觉得是完全不同的东西。比如苹果手机和安卓手机。

1.4 Pure competition

- many many firms that sell identical products

2 Pure competition

2.1 Assumptions

- There are many buyers and sellers (大到每个人都很渺小，自己对于价格和需求没有明显的影响力)
- homogeneous product (同质产品) (垄断竞争是差异化，而完全竞争是每个人卖的东西都是完全相同的，只要卖家卖的比别人高，就没人买它。像手机市场就不能认为完全不同，但是农产品领域就容易出现同质产品，近似于完全竞争市场，但不完全等同)
- freedom of entry and exit (进入与出自由) (有正的利润我就加入，一旦出现亏损就马上退出，事实上进出都有壁垒，还有浸没投入等，完全自由化是很难做到的；很多行业的进出壁垒明显低于别的，比如农产品低于高科技)
- perfect information (完全信息) (我知道所有人卖什么，卖多少价格，不管是买方还是卖方都是这样)

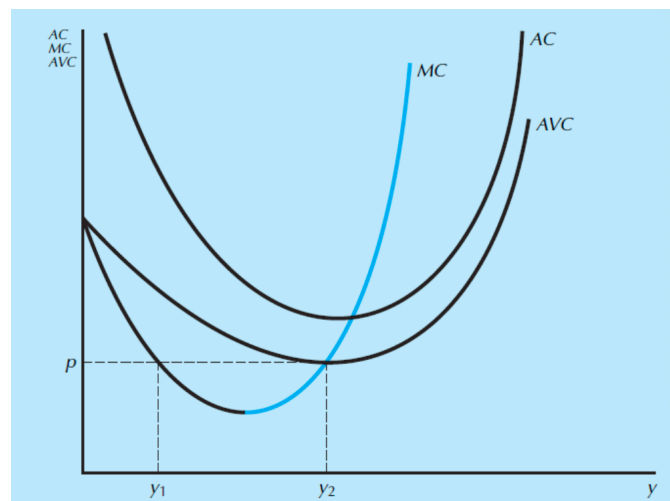
2.2 Price taker

- Because there are many firms selling identical products, each individual firm is small relative to the whole market
- Each individual firm has no influence over the market price.
- In principle, each firm could decide how much to sell at what price, but if its price is higher than the market price, then no one would buy from it; if its price is lower than the market price, then every buyer would like to buy from it, but then the firm would like to raise its price.
- implies that each firm would like to sell its product exactly at the market price
- firms are price-takers.
- More specifically, from each firm's point of view, price of its product is fixed at the market price and all it has to worry about is how much to produce.
- Firms which are price takers are usually referred to as competitive firms
- 所以此时的最优化问题，变量之间相互独立不会影响彼此，可以使用一阶条件

$$p = MC(y) \quad \text{gives us the inverse supply function}$$

- price as a function of output

3 Firm's supply curve



- 有两个交点 y_1 和 y_2

- 很明显不是 y_1 ，如果是 y_1 ，那么此时的边际成本等于价格，而此时的边际成本曲线是下降的，也就意味着多生产一些的时候，可以以低于价格的情况生产，从而继续盈利。从而向下倾斜的点都不是供给曲线的点。

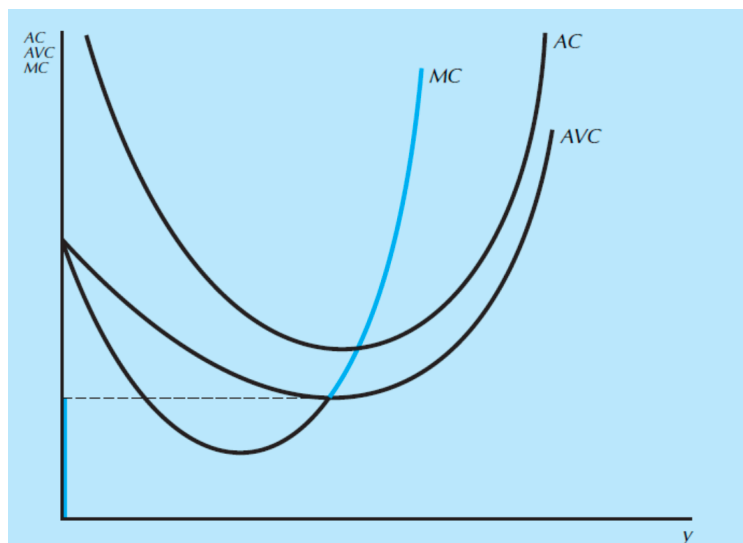
- 数学上理解

$$MR = MC \quad MR' < MC' \Rightarrow 0 < MC'$$

- The supply curve of a competitive firm must lie along the upwardsloping part of the MC curve.
- the “Giffen good” phenomenon cannot arise for supply curves (供给曲线都是向上倾斜的!)
- The previous discussion is assuming that it is profitable to produce something
- It could be that the best thing for a firm to do is to shut down (停工)
- The firm is better off going out of business when

$$-F > py - c_v(y) - F \Leftrightarrow AVC(y) > p \quad \text{the shutdown condition}$$

- The supply curve is the upward-sloping part of the MC curve that lies above the AVC curve



An Example

- Assume a firm's short-run cost function is

$$c_s(y) = 0.1y^3 - 2y^2 + 15y + 10$$

$$p = MC = 0.3y^2 - 4y + 15 \Rightarrow y = \frac{4 + \sqrt{1.2p - 2}}{0.6}$$

- shutdown point: 两种求法, 求 AVC 最低点 or AVC 和 MC 交点 (容易错的点: AVC 中的不变成本没有! 求出 y 要求的是 p!)

$$AVC = \frac{c_s(y)}{y} = 0.1y^2 - 2y + 15$$

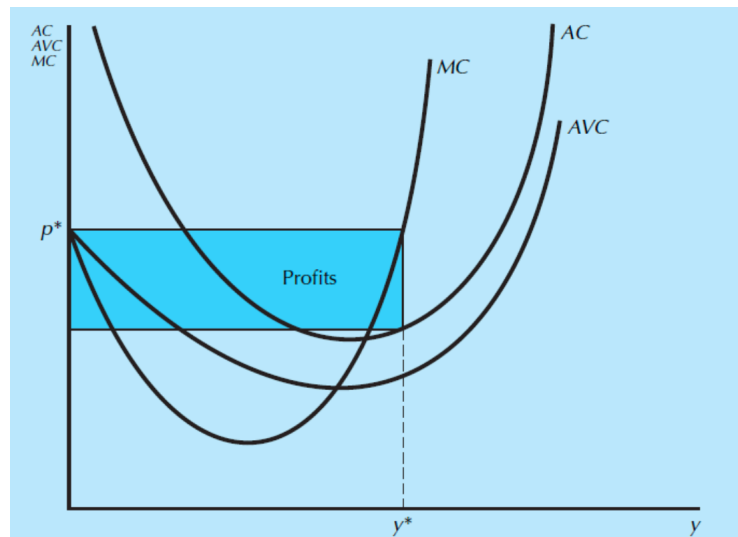
$$AVC = MC \Leftrightarrow 0.1y^2 - 2y + 15 = 0.3y^2 - 4y + 15 \Rightarrow y = 10 \Rightarrow p = 5$$

$$S_s = \begin{cases} \frac{4 + \sqrt{1.2p - 2}}{0.6} & p \geq 5 \\ 0 & p < 5 \end{cases}$$

4 Profits and Producer's surplus

4.1 Profit

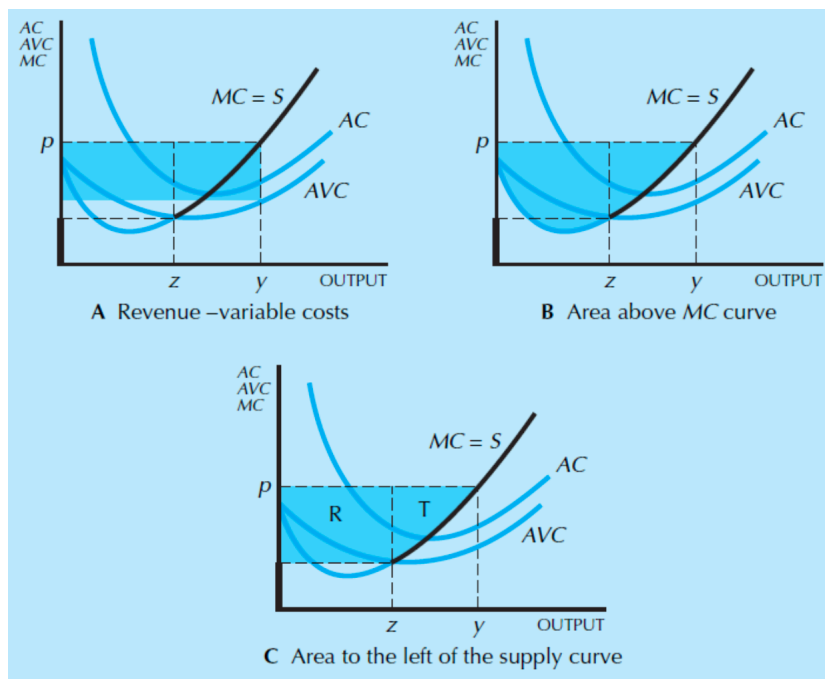
- If the market price is p and firm supplies y units of output, then the profit is $py - c(y)$ and this can be easily illustrated



- $\Pi = p^*y^* - ACy^*$

4.2 Producer's surplus

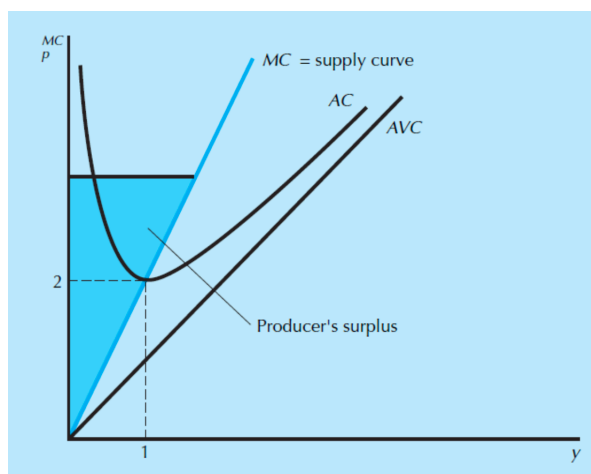
- The producer surplus is the sum over all units produced of the difference between the market price of the good and the marginal cost of production
- The producer surplus is the sum over all units produced of the difference between the market price of the good and the marginal cost of production
- 短期来说生产者剩余是利润和固定成本之和



- 第一张图是全都用 $py - c_v(y)$ ，第二张是用积分定义（被积部分），第三张是左半部分是用一的方法（不是一的部分）右半部分用二的方法，更像经原中学到的 supply curve 和 p 围起来的部分（停工点之前的部分）

4.3 An Example

- Assume $c(y) = y^2 + 1$
- Then $MC(y) = 2y \Rightarrow y^*(p) = p/2$
- In this case profit is $p^2/2 - p^2/4 - 1 = p^2/4 - 1$
- In addition producer's surplus is $(1/2) \times (p/2)p = p^2/4$



4.4 Long-run Supply

- Let $c_L(y)$ be the firm's long-run cost function
- If the market price is p , then the firm's long-run profit maximization problem is

$$\max_{y \geq 0} py - c_L(y)$$

- A necessary condition (from the method of Lagrange multipliers) is

$$p \leq MC_L(y) \text{ with equality if } y > 0$$

- So we only need to use the long-run cost function and then do the same exercises we did before.
- That is, the long-run supply curve will be given by

$$p = MC_L(y) = MC(y, k(y))$$

- The short-run supply curve is given by price equals marginal cost at some fixed level of k

$$p = MC(y, k)$$

- The long-run supply curve is more elastic than the short-run supply curve. When the price of output changes, the firm has more choices to adjust in the long run than in the short run
- 数量对价格越敏感就越有弹性。长期要比短期更不敏感，图像上来看，长期线更为躺平，同样的价格变化使得 y 变化量更大，从而更加有弹性。直觉上长期比短期做的调整是更多的，可以调整 k ，固定成本也是可以调的。
- 长期来看 shutdown point, F 都是 0，从而条件是 $AC > p$ ，这点上是由于长期来说 AC 和 AVC 是同样的。

Partial Equilibrium & Monopoly

姜峻赫

2022 年 11 月 26 日

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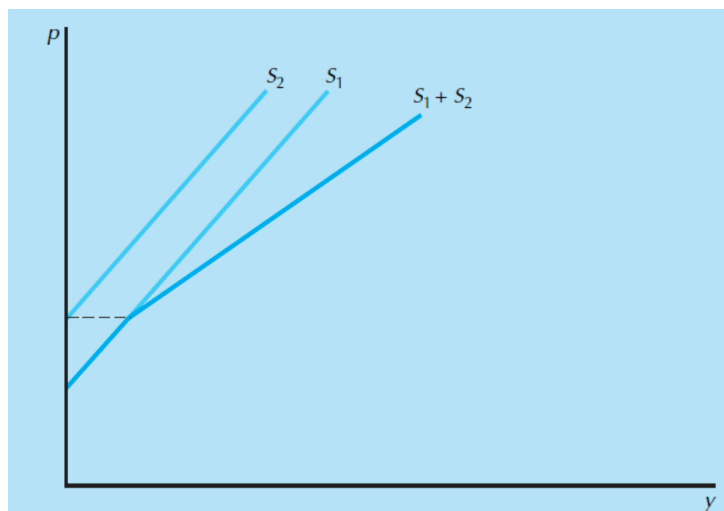
1 Partial Equilibrium

1.1 Short-run industry supply

- 短期的企业是不会退出市场的，数量是确定的。
- The industry supply curve will be the sum of the supplies of all the individual firms

- In the short-run, the number of firms which can produce a certain product is finite and fixed. These firms “already exist” and are in some sense able to be up and running simply by acquiring the necessary variable inputs.

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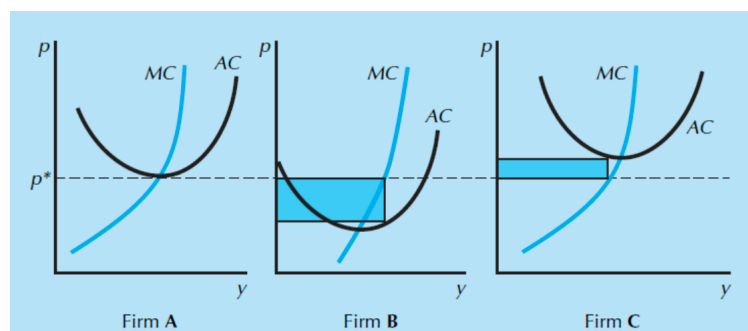


1.2 Short-run equilibrium

- 短期局部均衡!
- Suppose the aggregate (or market) demand curve is $D(p)$. Then the intersection of the demand curve and supply curve is the short-run equilibrium of this industry.
- Formally, a short-run partial competitive equilibrium of this market (or industry) is a price-quantity pair (p^*, y^*) such that

$$D(p^*) = S_s(p^*) = y^*$$

- 市场出清 (market clears): An equilibrium simply requires that demand be equal to supply. 适用于完全竞争市场，并且不考虑和其他的市场的互动，所有企业的数量是不动的（短期）
- This is partial equilibrium, only consider one market, ignoring the interconnections among different markets.
- 在这个情况下，单个企业无法决定均衡数量。A 厂商平均成本等于价格，是零利润的。B 有利润，C 是零利润。



- 在短期来说，由于固定成本的存在，所以停产条件在短期是在 AVC 条件下，不是看总利润为负，而是看是不是在 AVC 之上，所以短期内可以存在负利润，但是有正的会计利润

1.3 Long-run equilibrium

- In the long run, incumbent firms are free to choose optimal levels of all inputs
- they will move from their short-run to their long-run cost curves
- 厂商可以自由处置生产要素，并且进出自由，且如果存在很多有利润的企业，会有更多的企业进入。如果是负利润，企业会退出市场。
- They are also free to leave the industry (exit)
- Moreover, new firms may decide to begin producing the good (entry).
- In most competitive industries there are no restrictions against new firms entering the industry, in this case we say the industry exhibits free entry
- In a long-run equilibrium, we shall require that: the market clears, no firms has an incentive to enter or exit the industry.
- When do firms have incentives to enter or exit? If some firms are making negative profits, they want to and will exit. If some firms are making positive profits, firms outside the industry will adopt the technology of these firms and enter the industry.
- Therefore, condition 2 requires that all firms operating in this industry in the long-run earn zero profit.
- Now for simplicity, we assume that firms are homogeneous.
- They all have the same long-run supply curve $S_l(p)$ and corresponding profit function $\pi(p)$.
- As before, the aggregate demand function is $D(p)$. A long-run partial competitive equilibrium is a triple (p^*, y^*, n^*) such that $D(p^*) = n^* \times S_l(p^*) = y^*(condition1)$, $\pi(p^*) = 0(condition2)$
- p^* : equilibrium price. y^* : equilibrium quantity. n^* : equilibrium number of firms

1.4 An example

- Suppose the long-run cost function is

$$c(y) = \frac{1}{3}y^3 - y^2 + 2y$$

$$LMC(y) = y^2 - 2y + 2$$

$$p = LMC = y^2 - 2y + 2 \Rightarrow y = \frac{2 + \sqrt{4 - 4(2 - p)}}{2} = 1 + \sqrt{p - 1}$$

- Shutdown point

$$\min LAC \Rightarrow LAC = LMC \Rightarrow \frac{1}{3}y^2 - y + 2 = y^2 - 2y + 2 \Rightarrow y = 1.5, \quad LAC = 1.25$$

$$S_s = \begin{cases} \sqrt{p-1} + 1 & \text{if } p > 1.25 \\ \{0, 1.5\} & \text{if } p = 1.25 \\ 0 & \text{if } p < 1.25 \end{cases}$$

- Now suppose that the aggregate demand is

$$D(p) = 90 - 24p$$

- If $p^* < 1.25$, then we know $S_i(p^*) = 0$ but $D(p^*) > 0$
- If $p^* > 1.25$, then each firm must earn positive profit because $p > LAC(S_i(p^*))$
- If $p^* = 1.25$ and $S_i(p^*) = 0$, $D(p^*) > 0$
- Then we are left with the case that every firm produces 1.5 at $p^* = 1.25$
- Market clearing condition leads to $n^* = 40$

1.5 Equilibrium price

- In this example, the equilibrium price is equal to the lowest level of firm's long-run average cost. This is not accidental
- Assume that the firm's long-run average cost function is minimized at $\hat{y}$ and $LAC(\hat{y}) = \hat{c}$. Assume also that $D(\hat{c}) > 0$
- if every firm produces $y = 0$, then market does not clear
- if every firm produces $y > 0$, then $py - c(y) < y(\hat{c} - \frac{c(y)}{y}) \leq 0$
- Any $p > \hat{c}$ can not be an equilibrium price:
- if every firm produces $y = 0$, then market does not clear;(需求大于零, 但是生产为零)

- if every firm produces $y > 0$, then $py - c(y) \geq p\hat{y} - c(\hat{y}) = \hat{y}(p - c(\hat{y})/\hat{y}) = \hat{y}(p - LAC(\hat{y})) = \hat{y}(p - \hat{c})$
- 这个式子里面第一个不等式是因为左边的 y 是要求利润最大化的 y ，一定大于任何 y 。 $\hat{y}$ 是达到 LAC 最小的时候的 y ， $\hat{c}$ 也是 LAC 最小的时候的 c
- So the only possible equilibrium price is $p = \hat{c}$
- 短期的企业数量不变，只有市场出清这一个条件；长期来说除了市场出清，还要有零利润的条件（因为进出自由）

2 Monopoly

2.1 Perfect Competition

- Assumptions of pure competition: many buyers and sellers. homogeneous product. freedom of entry and exit. perfect information
- So firms are price takers
- The optimal output in the long run satisfies:

$$p = MC_L = AC_L = AC_S = MC_S$$

2.2 Examples of monopoly

- utility companies; toll highways; new technology

2.3 Causes of monopoly

- legal restrictions(譬如专利保护费)
- control of critical resources(比如 opec，管控石油)
- government-authorized franchises(政府许可证，某些行业只能由政府规定的企业才能经营，比如烟草)
- economies of size (natural monopoly)(单纯存在规模报酬的好处，自然发生的，比如电信的供应)
- brand loyalty and extensive advertising (high entry cost)(比如极高的产品忠诚度)

2.4 Monopoly pricing

- Unlike competitive firms which take market price as given, a monopolist has its power to determine what the price of its product is
- It is the freedom of setting price schemes that makes a monopolist's problem much more complicated than that of a competitive firm
- uniform pricing/ price discrimination: first, second, third degree price discrimination
- 价格不再是外生变量，而是内生的

2.5 Profit-Maximization with Uniform Pricing

- The simplest form of monopoly pricing is uniform pricing. The monopolist treats the market as a whole and simply adopts a linear pricing strategy.(i.e. sets a price p for each unit demanded)
- This is also known as simple monopoly pricing or standard monopoly pricing
- If the monopolist charges unit price p , then the market demand is $D(p)$
- total revenue is $pD(p)$, total cost is $c(D(p))$
- The monopolist's problem is to choose p to maximize its profits:

$$\max_p pD(p) - c(D(p))$$

- Because every price determines an output level and vice versa, we can reformulate the monopolist's problem as

$$\max_{y \geq 0} P(y)y - c(y)$$

- where $P(y) = D^{-1}(y)$ is the inverse demand function
- A necessary condition for the monopolist's problem is

$$P'(y)y + P(y) \leq MC(y) \text{ with equality if } y \geq 0$$

- When $P(0) > MC(0)$, which will be maintained in our course, this necessary condition boils down to

$$P'(y)y + P(y) = MC(y)$$

- The LHS $P'(y)y + P(y)$ is the monopolist's marginal revenue: $P'(y)y$ is marginal change in revenue due to price change; $P(y)$ is marginal change in revenue due to quantity change

- 在完全竞争市场中, $P'(y)$ 是 0, 也就是边际利润就是价格 (额外卖一单位产品的收益就是价格)
- The RHS $MC(y)$ is the monopolist's marginal cost.
- This necessary condition states that marginal revenue (MR) must equal MC at the monopolist's optimal output level.
- Notice $P'(y)\frac{y}{P(y)} = \frac{1}{\epsilon(y)}$, where $\epsilon(y)$ is the elasticity of the demand curve when quantity is y
- This implies the optimality condition can be rewritten as

$$P(y)(1 - \frac{1}{|\epsilon(y)|}) = MC(y)$$

- Notice this equality can not hold when $|\epsilon(y)| < 1$, a monopolist will never choose a quantity level at which the demand elasticity (absolute value) is less than 1
- 垄断厂商一定不会在弹性小于 1 的产品上取最优解, 一定会选择有弹性的产品。经济学理解: 如果缺乏弹性, 企业可以通过减产来提高利润。没有弹性的情况下, y 下降, p 会上升, p 的上升幅度要大于 y 的下降幅度, 从而通过减产来让总收益上升, 减产还带来了成本的下降, 从而此处的利润更高。
- This further implies that $P(y) > MC(y)$
- market price is higher than the monopolist's marginal cost

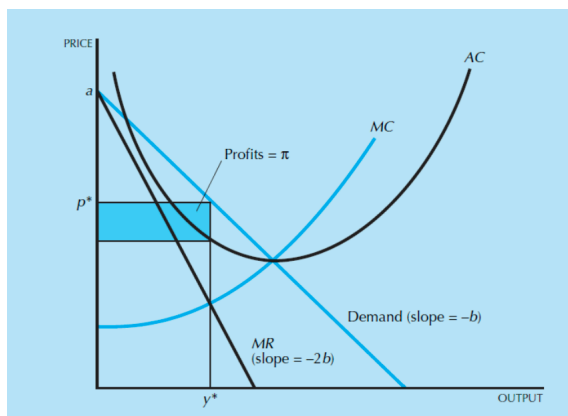
2.6 An Example: Linear Demand Curve

- Assume $P(y) = a - by$ and $c(y) = cy^2$
- Profit function $P(y)y - c(y)$
- $MR(y) = a - 2by$, $MC = 2cy$

$$MR(y^*) = MC(y^*)$$

$$a - 2by^* = 2cy^*$$

- $y^* = \frac{a}{2(b+c)}$



- Monopoly does not guarantee profit in the short run. (简单来说，只是有边际利润，但是由于固定成本的存在，可能只是赔的少了) However, monopolist typically earn normal profits in long run.
- Monopolists will try to maximize profits or minimize losses
- If AC is above AR (demand curve) the monopolist will suffer a loss

2.7 Markup Pricing and Monopoly Power

- Markup pricing (加成定价): output price is the marginal cost of production plus a “markup”
- Rearranging the previous equation we have

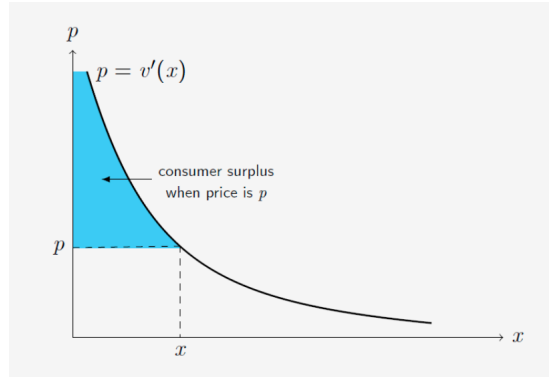
$$p(y) = \frac{MC(y)}{1 - \frac{1}{\epsilon(y)}}$$

- 定价和弹性负相关，弹性越大，价格越小。
- market power 理解方式：消费者对价格不敏感的时候，厂商可以定高价，而不敏感就是用弹性来刻画。越有弹性的市场，厂商越不能提高价格。

2.8 Inefficiency of Monopoly

- 效率就是要考虑总体的福利，不能只偏好一方
- A market is Pareto efficient (帕累托有效率) if it achieves the maximum possible total gains-to-trade (total welfare): there is no way to make someone better off without making anyone else worse off. Otherwise a market is Pareto inefficient
- When is the market efficient? Recall that a quasi-linear utility function has the form $u(y, m) = v(y) + m$ where y is the amount of a consumption good and m is money

- consumer's demand curve is $v'(y) = p$, consumer surplus is $v(y) - v(0) - py$ (since $v(y) - v(0) = \int_0^y v'(\hat{y}) d\hat{y}$)



- We've also learned that the producer surplus is

$$PS = \int_0^y (p - MC(\hat{y})) d\hat{y} = py - c(y) - c(0)$$

- In our discussion of “representative consumers” we noted the difficulty of treating market demand as if it had arisen from a single representative agent: income effects $\Rightarrow$ distribution of wealth matters
- There are no analogous income effects to cause any difficulty in the discussion of producers
- we can simply treat the market supply curve as if it was the supply curve of a single representative producer
- Assume there are n consumers.
- Consumer i has a quasilinear utility $v_i(y) + m$
- Then consumer i 's inverse demand curve is $p = v_i'(y)$, demand curve be $D_i(p) = v_i'^{-1}(p)$
- The market demand is $D(p) = \sum_i D_i(p)$
- Finally, assume the representative producer's cost function is $c(y)$

$$CS_i = v_i(y_i) - v_i(0) - py_i$$

$$CS = \sum CS_i = \sum v_i(y_i) - \sum v_i(0) - p \sum y_i$$

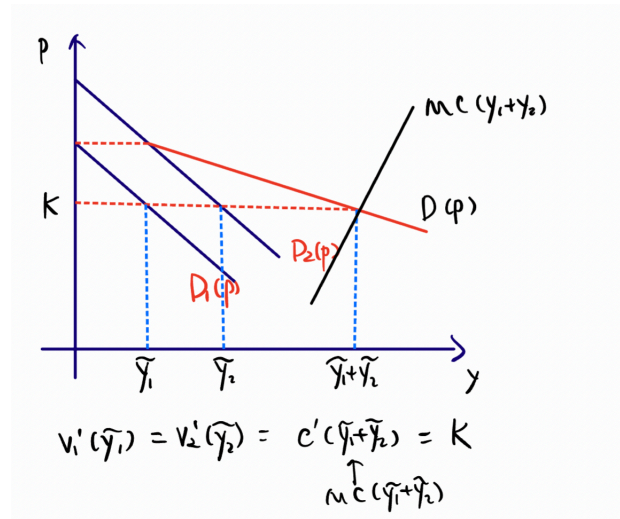
$$PS = p \sum y_i - c(\sum y_i) - c(0)$$

$$TS = \sum v_i(y_i) - \sum v_i(0) - (c(\sum y_i) - c(0))$$

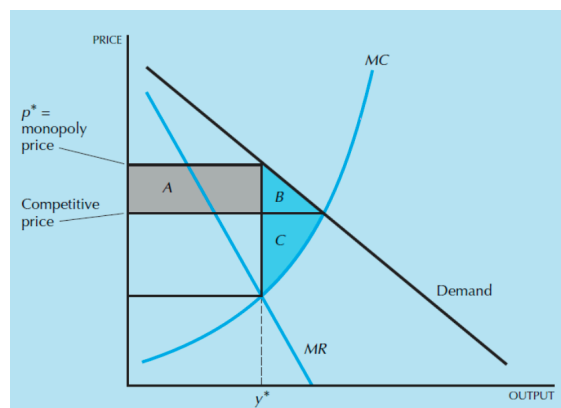
- This is maximized if

$$v'_k(y_k) = c'(\sum y_i)$$

- the optimal amount of consumption good is determined by the intersection of firm's MC and market demand



- We've already known that monopolist's price is higher than its marginal cost. This means the output is lower than the one determined by the intersection of demand and marginal curves.
- Thus a monopolist usually produced too little compared to socially optimal level. In terms of social surplus, this leads to a deadweight loss of welfare.
- changes in the producer's and consumers' surplus from a movement from monopolistic to competitive output.
- The deadweight loss due to the monopoly is given by the area B + C



- 此时存在帕累托改进：增产降价都可以让消费者和生产者的福利都提升。（因为此时的价格还是高于边际成本）

- 竞争厂商考虑的是边际，而对于垄断厂商来说，考虑的不单单是多生产的一单位，他还会因为多生产一单位产品，还会拉动所有之前产品的变化，所以不是在比较边际，所以会出现这种无效率。
- 此处的改进并不是以作为垄断厂商的身份达成的，垄断厂商定价是统一的，如果可以进行协商，以更低的价格再多卖一单位确实可以达到更好的效果，但那就违背了统一定价的原则。譬如按照价格歧视 or 谈判而非统一定价就可以增加利润。

2.9 Price Discrimination

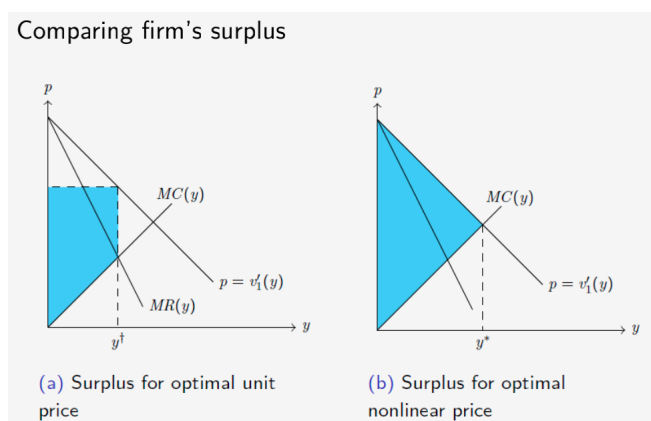
- So far a monopoly has been thought of as a firm which has to sell its product at the same price to every customer
- However, there is no a priori reason to restrict attention to trading mechanisms where the firm sets a unit price and all customers decide how much to buy
- In fact there are more clever pricing schemes that gives a monopolist higher profits than those obtained from uniform pricing, if the consumers are considered as heterogeneous(异质的)
- Price discrimination is the practice of selling different units of a good or service for different prices. Note that price differences that arise from cost differences are not price discrimination
- 1st-degree (一级价格歧视): each output unit is sold at a different price. Prices may differ across buyers. (理想化的，细致化的三级价格歧视)
- 2nd-degree (二级价格歧视): price paid by a buyer can vary with the quantity demanded by the buyer (but all customers face the same price schedule)(家庭装，小分装，批发，阶梯电费等)
- 3rd-degree (三级价格歧视): price paid by buyers in a given group is the same for all units purchased, but price may differ across buyer groups (学生票和非学生票)

2.9.1 First Degree Price Discrimination

- Assume there is only one consumer, i.e. $n = 1$.
- Facing this consumer, the monopolist adopts the following pricing strategy:

$$P(y) = \begin{cases} 0 & \text{if } y = 0 \\ v_1(y^*) - v_1(0) & \text{if } y = y^* \\ \infty & \text{otherwise} \end{cases}$$

- where y^* is the intersection of firm's marginal curve and this consumer's demand curve, i.e. $MC(y^*) = v'_1(y^*)$
- Essentially, the monopolist offers two choices to the consumer: either to buy y^* units for a price $v_1(y^*) - v_1(0)$ or not to buy anything
- Notice the price $v_1(y^*) - v_1(0)$ is just the price at which a consumer is just indifferent between purchasing or not purchasing a good for y^* units (reservation price, 保留价格) (we know the consumer will buy, 现实中可以少一个 ϵ , 那消费者还是会买)
- This results in profits $v_1(y^*) - v_1(0) - c(y^*)$, $CS = 0$, firm's surplus: $v_1(y^*) - v_1(0) - c(y^*) + c(0)$
- 生产者拿走了所有的消费者剩余! 在一级价格歧视的情况下, 厂商是可以达到帕累托最优的, 因为拿到了所有的剩余



- When there are more than one consumers in the market, if the monopolist is able to perfectly identify consumers and treat each consumer individually, then previous analysis still applies
- adopting a nonlinear price strategy for each consumer and extract all the surplus
- The monopolist's problem is

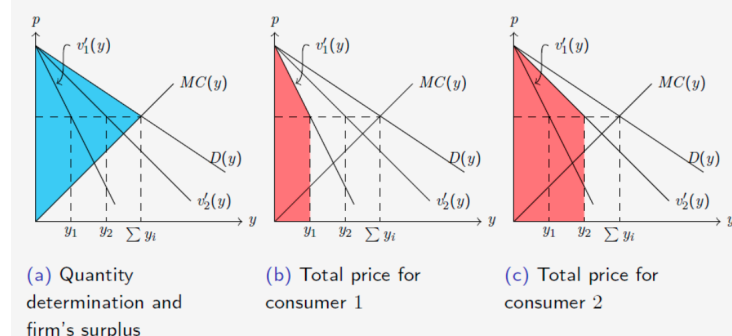
$$\max_{P_i, y_i} \sum v_i c(y_i) - \sum v_i(0) - c(\sum y_i)$$

- This is maximized if

$$v'_k(y_k) = c'(\sum y_i)$$

- 和之前的帕累托有效率的形式是完全相同的!
- Again, the optimal total amount of products is determined by the intersection of the monopolist's marginal curve and the market demand curve.

► Consider when $n = 2$



- 对消费者 1 说，就要红色区域的所有价格！对消费者 2，就要第二个红色区域的所有价格！要么买，要么不买！你自己选吧！
- 理解上来说，在完全竞争市场，每个消费者都可以以低于自己心理预期的价格买到自己想要数量的东西，然后这种市场福利用消费者剩余来刻画：这些钱是我最大愿意付出的钱，但是因为市场完全竞争，我得以不用那么高的价格，省下了很多钱
- 但是在垄断企业采取一级价格歧视的情况下，精准的刻画了我为了这部分数量的商品愿意付出的最大价值，事实上这个是连续变化的，操作上就采取要么买固定数量以你心里认为能够为之付出的最多的钱（保留价格），要么不买，从而把消费者剩余变成了生产者的利润。从而实现了帕累托最优，但是消费者没有得到任何福利。

Monopoly & Game Theory

姜峻赫

2022 年 12 月 3 日

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1 Monopoly

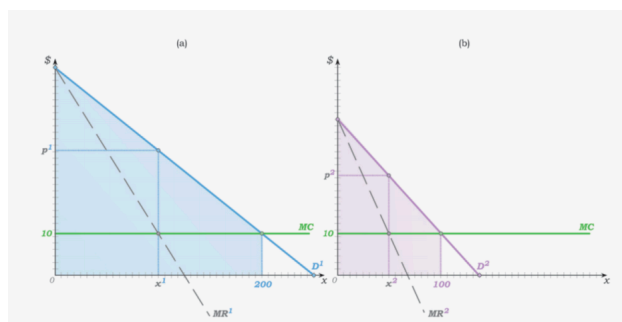
1.1 Third Degree Price Discrimination

- 一级价格歧视可以达成社会总剩余没有浪费的情况，达到了帕累托有效率，只不过此时消费者剩余已经变为 0 了。但是一级价格歧视是一个非常理想的状态，当下的科技发展使得一级价格歧视越发可能了，当下的大数据杀熟就是一个例子，或者不同数额的满减红包。
- So far, we assumed that the monopolist can fully identify each consumer and thus completely separate consumers into each individual. the monopolist adopts a nonlinear pricing scheme for each individual.

- However, in reality it is hard. for the firm to know exactly each consumer's preference, for the firm to know exactly each consumer's preference
- Another form of price discrimination is called third degree price discrimination. The monopolist first divide all consumers into groups and then charge uniform price within each group, price discriminate "imperfectly"
- For example, international companies charge different unit price for the same product in different countries or children/student/veteran/senior citizen discount
- Suppose the monopolist divide the n consumers into two groups, A and B. The inverse market demand within each group is $P_A(y)$ and $P_B(y)$

$$\max_{y_A, y_B} y_A P(y_A) + y_B P(y_B) - c(y_A + y_B) \Rightarrow MR_A(y_A) = MR_B(y_B) = c'(y_A + y_B)$$

- Comparing 1st degree and 3rd degree price discrimination with 2 consumers and constant MC



- 在每一个市场内部满足垄断价格的关系

$$P_A(y_A)(1 - \frac{1}{|\epsilon_A(y_A)|}) = P_B(y_B)(1 - \frac{1}{|\epsilon_B(y_B)|}) = c'(y_A + y_B)$$

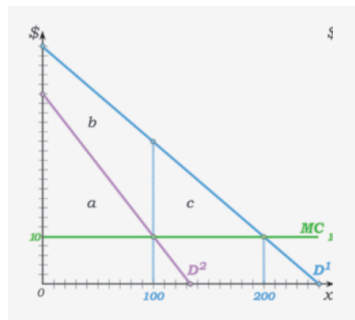
$$P_A(y_A) \geq P_B(y_B) \Leftrightarrow |\epsilon_A(y_A)| \leq |\epsilon_B(y_B)|$$

- higher price for price-insensitive market and lower price for price-sensitive market.
- A deadweight loss will arise under imperfect (or third degree) price discrimination

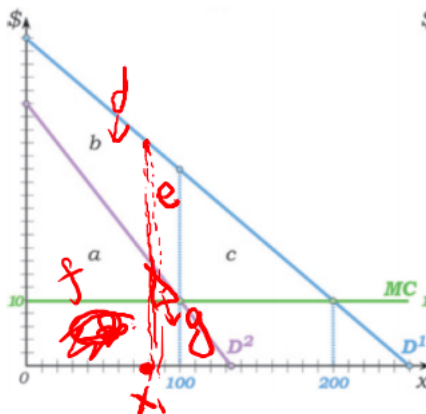
1.2 Second Degree Price Discrimination

- 虽然厂商知道有哪些类型，但是不知道每个人的类型，就可以 offer 多种类型让消费者选择
- We have said that the assumption that the monopolist can distinguish each consumer is too stringent and not realistic.

- A much more realistic assumption is that the monopolist knows that there are n consumers but does not know who is who.
- Then the monopolist can not separate the consumers and must treat the market as a whole. But then the pricing scheme in first degree price discrimination will fail immediately



- 第一个人可以买 200, 付出 $a+b+c+2000$, 第二个人可以买 100, 付出 $a+1000$, 但是第二个人事实上不会去买 200, 因为此时一点消费者剩余都没有, 而第一个 offer 还会有消费者剩余。他会假装成为 1。存在自选择偏差。
- When the monopolist cannot tell which consumers are type 1 and which are type 2, she cannot implement this perfect price discrimination. No one will pick the 200 unit package, leaving the firm with surplus of only $2a$.
- In order to induce type 1 consumers to behave differently from type 2 consumers, the monopolist must therefore come up with a different set of price/quantity packages. For instance, the monopolist might continue to offer 100 units at the price $1,000 + a$ while reducing the price of 200 units to $2,000 + a + c$
- would equalize the surplus a type 1 consumer will get under the two packages. The monopolist would be able to expect a surplus of $2a + c$, larger than the surplus of $2a$
- 自动对号入座, 看起来平等。事实上还有更优的选择, 譬如将 100 那条线左移



- 原有厂商的剩余是 $2a+c$ ，此时消费者 1 可以付出 $f+g+c+e$ 的剩余，消费者 2 可以付出 f 的剩余，所以厂商可以得到 $2f+g+c+e$ ，只要 $e>g$ ，厂商就可以有更好的二级价格歧视方法。
- 通过不同数量的差异性定价，实现了对号入座的自选择。
- Quantity: bulk discounts、charges for phone service/Quality: first/business/economy class, different storage capacities for cell phone.
- 满减、阶梯电价、数据流量包、家庭装和非家庭装也是二级价格歧视的例子。

1.3 Summary

- The four pricing schemes we have just discussed can be summarized in the following table.

	Integrated market	Segmented market
Linear pricing	uniform pricing	3rd degree
Nonlinear pricing	2nd degree	1st degree

There are many many other pricing strategies and what pricing scheme is profit maximization depends on the specific situation that a monopolist is facing.

2 Game Theory

- In a competitive market, consumers and firms are assumed to be price takers.
- In a monopoly market, there is only one firm in the market. But many markets/social activities involve only a small number of participants and economic agents can interact strategically in a variety of ways
- several firms compete in a certain market, Oligopolies, Cartels
- The outcome to a participant depends not only on his own choice but also on what others do. When participants make decisions, they strategically take into account their opponents' behavior.
- Game theory provides a rigorous mathematical tool to study these situations.

2.1 Elements of a Game

- Players (博弈参与人) :decision making entities firms, governments, individuals
- Strategies(策略) (or choices or actions) for each player:output, prices, advertising budget

- Payoffs (得益) to players as a function of strategy profiles: depend not only on own strategies, but on other's strategies as well

2.2 Examples of Simple Games

- Matching pennies (猜硬币博弈): player B gets \$1 and player A loses \$1 if pennies match. player B loses a dollar and player A gains \$1 if pennies don't match
- Prisoner's dilemma (囚徒困境): 0 if you confess while other denies; -1 if you both deny; -3 if you both confess; -6 if you deny and the other guy confess

2.3 Important Terminology

- Decision nodes: player make decisions at various points in a game
- Actions: the set of choices available each decision node in a game
- Pure Strategy (纯策略): a rule that tells the player what actions to take at each of his information set in the game
- Mixed strategy (混合策略): consists of a probability distribution on the set of pure strategies
- Common knowledge (公共知识): if every player knows it, every player knows that every other player knows it, every player knows that every other player knows that every other player knows it, and so on. we usually assume that the complete description of the game is common knowledge
- Simultaneous move (同时行动): if players in a game have to make their decision at the same times
- Sequential move (序贯行动): if players make their decisions in a particular sequence, one after another
- Perfect information: every player at every decision node knows the decision taken previously by every other player

2.4 Classification of Games

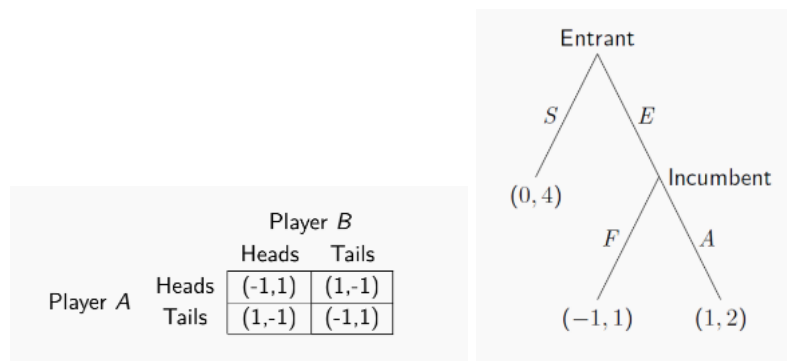
- These are several ways of classifying games
- by the nature of the states: stochastic v.s. deterministic game
- perfect information vs. imperfect information game: Perfect information (完美信息博弈): at each move in the game, the player with the move knows the full history of the play of the game thus far

- complete information v.s. incomplete information game: Incomplete information (非完全信息) : at least one player is uncertain about another player's payoff function (Asymmetric or private information)

	Complete	Incomplete
Static	Nash Equilibrium	Bayesian NE
Dynamic	Subgame Perfect NE	Perfect Bayesian NE

2.5 Ways to Describe a Game

- There are two ways to describe a game
- the "Normal" or matrix form (标准型或矩阵型): there are only 2 (sometimes 3) players; there are a finite number of strategies; actions approximately simultaneous
- the "Extensive" form (扩展型): if actions are sequential



2.6 Dominant Strategies

- We can describe the prisoners'dilemma in its normal form

		Prisoner B	
		Confess	Deny
Prisoner A	Confess	(-3,-3)	(0,-6)
	Deny	(-6,0)	(-1,-1)

- If B chooses to confess, then you should confess because $-3 > -6$
- If B chooses to deny, then you should confess because $0 > -1$
- In sum, regardless of B's choice, A should (will) choose to confess
- In games, if a strategy is always strictly optimal for a player regardless of other players' choices, we call this strategy a strictly dominant strategy (严格占优策略) for this player

- strategy s_i (strictly) dominates strategy s_i' if, for all possible strategy combinations of opponents, s_i yields a (strictly) higher payoff than s_i' to player i
- Confession is the strictly dominant strategy for both players in the prisoners'dilemma
- If a player has a strictly dominant strategy, this player will always play this strategy. If all players have a strictly dominant strategy, then the game is dominance-solvable, because we know every player will simply play their strictly dominant strategy.
- Strictly dominant strategy is natural and intuitive. But unfortunately, in many games, players do not have a strictly dominant strategy.

2.7 Nash Equilibrium

- Battle of the sexes:

		Wife	
		Soccer	Opera
Husband	Soccer	(2,1)	(0,0)
	Opera	(0,0)	(1,2)

- A couple wishes to go out for entertainment. The husband (row player) prefers soccer to opera while the wife (column player) prefers opera to soccer. They both prefer going together to going alone
- A Nash equilibrium of a N player normal form game is a strategy profile $(s_1, \dots, s_n)$ such that for each player i , s_i is optimal given other players' strategy $(s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$
- no player can improve her payoff by unilaterally deviating. 给定别人选择之后, 我单边偏离我的选择对于我来说没有好处。
- Technically, a Nash equilibrium is a strategy profile $(s_1^*, \dots, s_n^*)$ such that for every player i :

$$u_i(s_1^*, \dots, s_{i-1}^*, s_i^*, s_{i+1}^*, \dots, s_n^*) \geq u_i(s_1^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*) \quad \forall s_i \in S_i$$

- where u_i is the payoff function of player i
- For two player game, this boils down to a pair of strategies (s_1, s_2) such that given s_2 , s_1 is optimal for player 1. given s_1 , s_2 is optimal for player 2.
- In the battle of the sexes, there are four possible strategy profiles:
- (S, S), [or (O, O)] Nash equilibrium: given that the husband goes to soccer, it is optimal for the wife to go to soccer. given that the wife goes to soccer, it is optimal for the husband to go to soccer

- (S, O)[or (O,S)] not Nash equilibrium:given that the husband goes to soccer, it is not optimal for the wife to go to opera
- The definition of Nash equilibrium does not say anything about which one should/will be played.
- 而囚徒困境就只有一个纳什均衡解。但是这不是最好的情况，且是无效率的情况，需要第三方协调机构进行串供。
- 引申：军备竞赛就像囚徒困境，坦白对应增加军事支出，否定对应减少军事装备。
- 抛硬币问题就没有 nash 均衡解，不管选什么，我都有动力去偏离原有选择，所以 nash 均衡不一定存在。

		Player B	
		Heads	Tails
Player A	Heads	$(-1, 1)$	$(1, -1)$
	Tails	$(1, -1)$	$(-1, 1)$

- Underlying assumptions of Nash equilibrium:Players have predictions about other players' play and they optimally respond to their predictions.Players' predictions are correct.
- Dominant Strategies:"I'm doing the best I can no matter what you do." "You're doing the best you can no matter what I do."
- Nash Equilibrium:"I'm doing the best I can given what you are doing" "You're doing the best you can given what I am doing."
- 另一个有意思的例子：智猪博弈 $(\cdot(\infty)\cdot)$

Boxed pigs game(智猪博弈)

- 只需要一只猪去按按钮，就有吃的，按按钮花费 2
- 大猪先吃，分配比例为 9:1，小猪先吃，分配比例为 6:4

		大猪	大猪
		Press	Wait
小猪	Press	$(1, 5)$	$(-1, 9)$
小猪	Wait	$(4, 4)$	$(0, 0)$

- 有点像真实生活中的搭便车行为，能力强的人获得多，反而就会多做；没什么能力的人反而只坐享其成，什么都不做。

- 如果两者可以达成合作，不管谁按，一起吃，此时存在两个 nash 均衡解，团队搭便车的现象可能有所减少。

		大猪	大猪
		Press	Wait
小猪	Press	(1, 5)	(1, 7)
小猪	Wait	(3, 5)	(0, 0)

- 如果有一个策略被其他的策略“占优”了，那么这个策略就一定不是 nash 均衡解。

		Player II	
		Left	Right
Player I	Top	1,2	4,1
	Middle	3,2	2,1
	Bottom	2,1	1,3

- Bottom is dominated by Middle (for Player I): 首先排除最后一行
- Right is dominated by Left (for Player II): 排除最右一列
- Top is dominated by Middle (for Player I): 排除掉剩下的 (1, 2)
- 重复剔除严格被占优策略: Iterated elimination of strictly dominated strategies: eliminate all strategies which are dominated, relative to opponent's strategies which have not yet been eliminated
- If iterated elimination of strictly dominated strategies yields a unique strategy profile, then this strategy profile is the unique Nash equilibrium

2.8 Mixed Strategies

- 怎么做到概率选项呢? 长期来说不让别人看出来到底怎么做，而是让人知道这件事情发生的概率。抛硬币的时候不让人看出想怎么抛。
- 此时的最优就是得到最高的期望收益。
- To deal with non-existence of Nash equilibrium, we extend players' strategy spaces. We assume players in the matching pennies games can randomize between H and T.
- We use $p \circ H + (1 - p) \circ T$ to denote the strategy which plays H with probability p and T with probability $1 - p$
- This kind of strategies is called mixed strategies (混合策略)

- If player i have K pure strategies available. Then a mixed strategy for player i is a probability distribution over those K strategies
- The strategies we studied previously, e.g. H and T in the matching pennies game, are called pure strategies (纯策略).
- just special cases of mixed strategies, e.g. H is equivalent to $1 \circ H + 0 \circ T$
- There are many interpretations/justifications for mixed strategies: Literally, randomization in players' brain. Large population: fraction p of people play H and fraction $1-p$ play
- Use expected payoffs to evaluate payoffs from mixed strategies. Player 1 adopts a mixed strategy σ_1 : plays s_1 with probability p and s'_1 with probability $1-p$
- Player 2 adopts a mixed strategy σ_2 , Hence the expected utilities are given by

$$u_1(\sigma_1, \sigma_2) = pq u_1(s_1, s_2) + p(1-q) u_1(s_1, s'_2) + (1-p)q u_1(s'_1, s_2) + (1-p)(1-q) u_1(s'_1, s'_2)$$

$$u_2(\sigma_1, \sigma_2) = pq u_2(s_1, s_2) + p(1-q) u_2(s_1, s'_2) + (1-p)q u_2(s'_1, s_2) + (1-p)(1-q) u_2(s'_1, s'_2)$$

- For example, in matching pennies game, assume

$$\sigma_1 = 0.5 \circ H + 0.5 \circ T \text{ and } \sigma_2 = H$$

$$u_1(\sigma_1, \sigma_2) = 0.5 u_1(H, H) + 0.5 u_1(T, H) = 0.5 \times (-1) + 0.5 \times 1 = 0$$

2.9 The Best Response Correspondence

- A Nash equilibrium in mixed strategies is similarly defined
- (σ_1, σ_2) is a Nash equilibrium if given σ_2 , σ_1 is optimal for player 1, given σ_1 , σ_2 is optimal for player 2
- In general, it is more difficult to find equilibria in mixed strategies than in pure strategies.
- For 2×2 games (2 players, 2 pure strategies for each) the best response correspondence (最优反应曲线) is a powerful tool to help us analyze
- Let's consider the matching pennies game again.
- Assume player 1 plays $p \circ H + (1-p) \circ T$ and player 2 plays $q \circ H + (1-q) \circ T$, Then the payoff to player 2 is:

$$u_2(p, q) = pq + p(1-q)(-1) + (1-p)q(-1) + (1-p)(1-q) = 2(2p-1)q + (1-2p)$$

- 考虑的是第二个人，所以要把 q 给拎出来， q 是选择空间，选择一个 q 让效用最大。

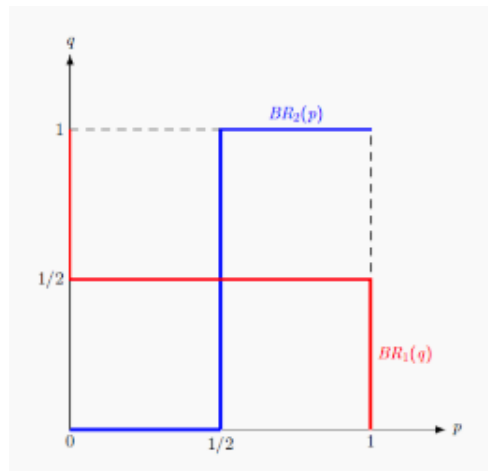
- Player 2's optimal choice can be expressed as

$$BR_2(p) = \begin{cases} \{0\} & p < 0.5 \\ [0, 1] & p = 0.5 \\ \{1\} & p > 0.5 \end{cases}$$

- Similarly, we can calculate player 1's optimal choice

$$BR_1(q) = \begin{cases} \{1\} & q < 0.5 \\ [0, 1] & q = 0.5 \\ \{0\} & q > 0.5 \end{cases}$$

- We can plot it in the p-q plane, The intersections of the two best response correspondences are Nash equilibria



Game Theory & Oligopoly

姜峻赫

2022.12.10

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1 Game Theory

1.1 Mixed Strategies

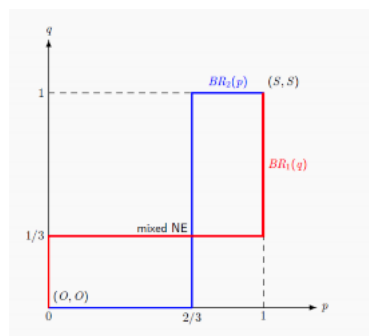
- Battle of the sexes:

		Wife	
		Soccer	Opera
Husband	Soccer	(2,1)	(0,0)
	Opera	(0,0)	(1,2)

- $husband : p \circ S + (1 - p) \circ O$

- $wife : q \circ S + (1 - q) \circ O \quad u_w(p, q) = pq + 2(1 - q)(1 - p) = (3p - 2)q + 2(1 - p)$

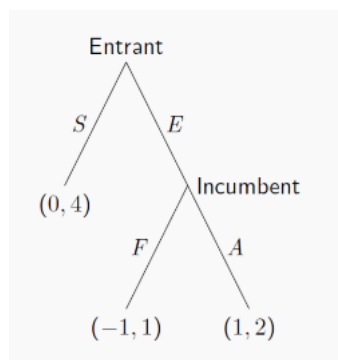
$$BR_w(p) = \begin{cases} 0 & \text{if } p < \frac{2}{3} \\ [0, 1] & \text{if } p = \frac{2}{3} \\ 1 & \text{if } p > \frac{2}{3} \end{cases}$$



- 最优点 $(p, q) = (2/3, 1/3)$
- 此时验证不存在有机可乘的现象：给定丈夫的选择的时候，妻子选哪个的期望都一样。给定妻子的选择的时候，丈夫选哪个的期望也都一样。
- 混合策略 nash 均衡。妻子的期望收益是 $2/3$ ，丈夫的期望收益也是 $2/3$ 。但是两个纯策略 nash 均衡的收益都更高，混合策略更低。
- 理解：不想协作，想的是不希望对方有机可乘，反而会达成更坏的收益。协商或许更好！考虑那张图的时候，其实有三个混合策略 nash 均衡解。

1.2 Games with Sequential Moves

- Consider the following chain store game. There is an entrant who chooses to either enter a market (E) or stay out (S). There is an incumbent who chooses to either fight (F) or accommodate (A) if the entrant enters
- Coca-Cola is an entrant while Pepsi is the incumbent in the Russian market around 1990.



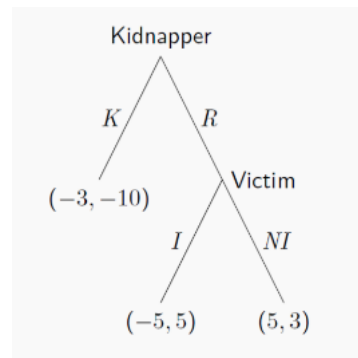
- (E, F) 是否是 nash 均衡? 不是, 因为给定 E, 回击不是在位者的更好的选择, (E, A) 是更可能偏向的选择。
- (S, A) 也不是, 给定 A, 进入者会偏离到 (E, A)
- (S, F) 是 nash 均衡, 给定 F, 会选择不进入。给定 S, 不管选什么都一样。(仿佛像是一个威胁, 这是一个不太可信的威胁, 对于在位者来说, 如果给定了 Enter, 选择 A 会更好, 这就称为不可置信的威胁)
- (E, A) 也是 nash 均衡解。
- 需要新的概念排除!

1.3 Subgame Perfect Equilibrium

- A Nash equilibrium is said to be a subgame perfect Nash equilibrium (子博弈精炼纳什均衡) if and only if it is a Nash equilibrium in every subgame of the game
- A subgame(子博弈) in an extensive-form game has the following properties:
- it begins at a node of the tree corresponding to an information set reduced to a singleton (the set contains only one set)(单结)
- it encompasses all parts of the tree following the starting node
- it never divides an information set

Backward Induction

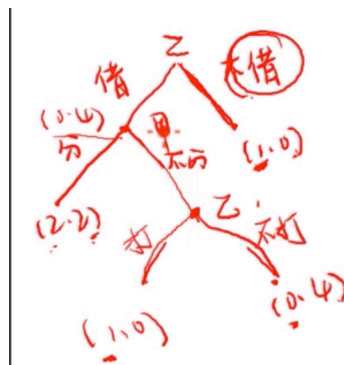
- To get an SPNE for finite horizon games, we could work back from terminal nodes.
- go to final “decision node” and assign action to the player that maximizes his payoff
- reduce game by trimming tree at this node and making terminal payoffs at this node, the payoffs when the player whose turn it was takes best action.
- keep working backwards
- This procedure is called backward induction solution (倒推归纳法). (E, A) is the only backward induction solution to the entrant & incumbent game.
- e.g. The kidnapper has two choices after receiving the ransom: release (R) or kill the victim (K).
- After release, the victim has two choices: identify the kidnapper (I) or refrain (NI)



- If the victim had been released, he would choose to identify the kidnapper. Thus, the kidnapper would get -5 if he releases the victim. It is then optimal for the kidnapper to kill the victim in the first place.
- This is the only backward induction solution of this game. The outcome from (R, NI) is socially optimal, but it is not consistent with backward induction. the victim can not commit to playing NI.
- 所以要让绑匪认为，告发之后我的收益没那么高。譬如，交换人质，或者提供一些自己的不雅照片，让他手里有些筹码，从而选择释放人质，让承诺变的可置信。
- 借钱的例子：甲和乙说要借他的钱，赚了分你一些。乙可以借或者不借，借了之后甲可以选择分或者不分。但是甲的承诺是不可置信的承诺。



- 解决方法：乙可以打他。



- 这种情况，甲会选择分，在乙会报复的情况下，子博弈精炼纳什均衡会是借且分。把承诺变的可置信了

2 Oligopoly

2.1 Basic points

- A number of firms compete in a certain market. Each of them has non-negligible effects on the market.
- Decisions of one firm influence and are influenced by other firms. In other words, firms strategically interact in the market.
- Examples: automobiles, airlines, aircraft, phone... Game theory provides a tool to study firms' behavior in these markets. Unless otherwise indicated, we shall consider only pure strategies in our analysis.
- 差异没有足够大到让消费者觉得是两件完全不一样的事情。厂商和厂商之间有一些同质化的东西。譬如钢铁行业、石油行业，都是寡头市场，卖的差不多。

2.2 Traditional Models

- 寡头市场：价格搜寻者
- Quantity competition: Simultaneous quantity setting: the Cournot model (古诺模型). Quantity leadership: the Stackelberg model. (有先后顺序之分)
- Price competition: Simultaneous price setting: the Bertrand model
- Product Differentiation: Hotelling model (霍特模型)

2.3 Games with Continuum of Actions

- The games we will deal within this chapter are different from what we have learned in that players choose actions from infinitely many possible actions
- Although we can no longer express a normal form game by a matrix, there is nothing conceptually new.
- Consider a two-player game.
- Each player i can choose an action from A_i . The set A_i might contain infinitely many actions, e.g. quantity, price.

- Player i has a payoff function $u_i(a_i, a_{-i})$ (if $i = 1$, then $-i$ means 2 and if $i = 2$ then $-i$ means 1)
- Given a_{-i} , let $BR(a_{-i})$ be i 's optimal choice(s).
- The correspondence (function when it is single-valued) BR_i is i 's best response correspondence (function)
- A strategy profile (a_1^*, a_2^*) is a Nash equilibrium of this game if it is a pair of mutual best responses: $a_1^* \in BR_1(a_2^*)$ $a_2^* \in BR_2(a_1^*)$
- The same idea of backward induction also applies when we talk about sequential move games.

2.4 Cournot Model

- Two firms $i = 1, 2$ compete in quantities in a market with (inverse) demand curve: $p(y) = a - by$
- Two firms simultaneously choose their own outputs. If firm i chooses output y_i , then the total output of this market is $y_1 + y_2$ and the market price is $a - b(y_1 + y_2)$ where $a > 0$ and $b > 0$
- For simplicity, assume there is no production cost for both firms.
- If firm 2 produces y_2 , then firm 1 faces the following profit-maximizing problem:

$$\max_{y_1 \geq 0} [a - b(y_1 + y_2)]y_1$$

- Firm 1's optimal output choice is (assume a is large enough so that we do not need to worry about the constraint $y \geq 0$)

$$y_1 = \frac{a - by_2}{2b}$$

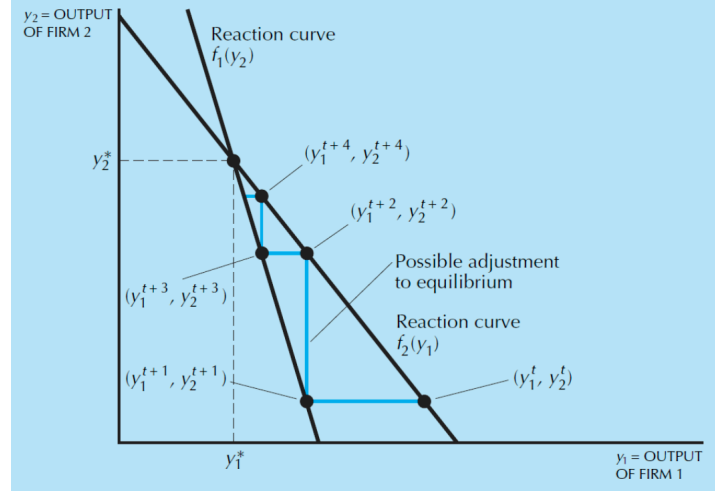
- Similarly, if firm 1 produces y_1 , we know the optimal output choice for firm 2 is

$$y_2 = \frac{a - by_1}{2b}$$

- These two functions are firm 1 and 2's best response functions (最优反应函数) respectively.
- Therefore, a pair of outputs (y_1^*, y_2^*) is a Nash equilibrium of this game if and only if it solves

$$\begin{aligned} y_2^* &= \frac{a - by_1^*}{2b} \\ y_1^* &= \frac{a - by_2^*}{2b} \end{aligned}$$

- $y_1^* = y_2^* = \frac{a}{3b}$
- This is the unique Nash equilibrium, also known as Cournot equilibrium. The profits for these two firms are $\pi_1^* = \pi_2^* = \frac{a^2}{9b}$



- The Cournot model can be naturally extended to multiple firms.
- Assume there are $n > 1$ firms. The demand function is as above. So if firm i produces y_i for $i = 1, \dots, n$, then the price is

$$a - b \sum y_i$$

- Assume again for simplicity, production cost is 0 for all firms.
- Adopting previous analysis, we can show firm i 's best response function is

$$y_i = \frac{a - b \sum_{j \neq i} y_j}{2b}$$

- Rearranging yields: $y_i + \sum_{j=1}^n y_j = \frac{a}{b}$
- A strategy profile $(y_1^*, \dots, y_n^*)$ is a Nash equilibrium if and only if it solves $y_i^* + \sum_{j=1}^n y_j^* = \frac{a}{b}$
- We get: $y_i^* = \frac{a}{(n+1)b}$
- 如果假设有成本 c , 那么就是 $y_i^* = \frac{a-c}{(n+1)b}$
- As the number of firms n becomes large, $n \rightarrow \infty$, the total output $\frac{na}{(n+1)b} \rightarrow \frac{a}{b}$, price $\rightarrow 0$
- That is the market becomes pure competition
- More generally, suppose that there are again two firms, 1 and 2, compete in quantities in a market with the inverse demand function being $p(\cdot)$

- The constant marginal cost of each firm is $c \geq 0$
- The profits of firms i are: $\pi_i = y_i p(y_i + y_j) - c y_i$
- Each firm chooses its own output to maximize its own profit, taking the other firms output as given
- $p(y_i + y_{-i}) + y_i p'(y_i + y_{-i}) c \leq 0$, with equality if $y_i > 0$

$$p(y_1^* + y_2^*) + y_1^* p'(y_1^* + y_2^*) - c = 0$$

$$p(y_1^* + y_2^*) + y_2^* p'(y_1^* + y_2^*) - c = 0$$

- Therefore,

$$p(y_1^* + y_2^*) + \frac{y_1^* + y_2^*}{2} p'(y_1^* + y_2^*) = c$$

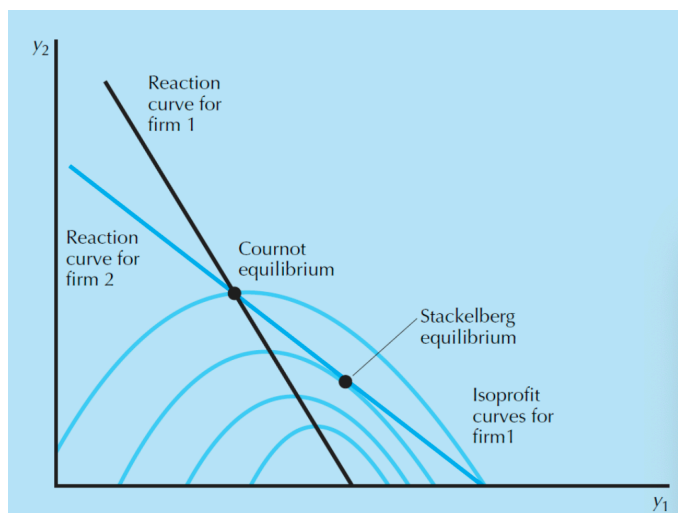
- 通常来说 p' 为负数, 从而 p 大于 c , 产量会更少
- $p(Y)(1 - \frac{1}{2|\epsilon|}) = MC$
- n 越大意味着每一个厂商面对的弹性 $n\epsilon$ 也会越来越大, 市场的均衡价格会趋向于 MC 。
- We can show that the market price is smaller than the monopoly price
- To show $p(y_1^* + y_2^*) < p^m$; we need to show that $y_1^* + y_2^* > y^m$
- if $y_1^* + y_2^* < y^m$, by increasing y_1 to $y^m - y_2^*$ the total profit of the two firms will be higher(此时两者若视为一个整体, 则 y^m 才是最大化一个垄断厂商利润的产量), while the market price will be lower
- 此时总产量变高了, 总价格会下跌, 而给定厂商 2 的产量不变, 价格下跌, 那么利润一定会减少, 从而推出厂商 1 的利润上升。this means that firm 2's profit will be lower, while firm 1's profit must therefore be higher
- contradicts the assumption that y_1^* is an equilibrium price
- if $y_1^* + y_2^* = y^m$, we would have $p(y^m) + \frac{y^m}{2} p'(y^m) = c$ contradicts with $p(y^m) + y^m p'(y^m) = c$ (这是 y^m 的定义)
- Hence $y_1^* + y_2^* > y^m$
- 拓展情况: 厂商非完全同质的情况, 若 C 不同, 若 C 更低, 则最优反应线应当向外移动, 这种情况下只能用两个最优反应线联立求解。
- 产品不同的情况, 需求函数不同了, 目标函数也要发生变化。
- With linear demand $p = a - by$, $y^m = \frac{a-c}{2b}$, $y^{Cournot} = \frac{2(a-c)}{3b}$, $y^C = \frac{a-c}{b}$

- In addition, the joint profit in the Cournot duopoly is lower than the monopoly profit.
- This is because when a firm increases its output, it reduces the other firms profit, which is not taken into account when a firm decides its own optimal output.
- 古诺利润最低，垄断的利润最高

2.5 Stackelberg Model

- We modify the Cournot game so that firm 1 chooses first and firm 2 chooses after observing firm 1's output.
- Firm 1 is the leader firm (领导者厂商) and firm 2 is the follower (跟随者厂商)，领导者厂商一般都是规模比较大的，跟随者是比较小的
- Assume the market demand is the same as the Cournot game: $p(y) = a - by$
- We again assume that the production cost is 0 for both firms.
- This is a sequential move game. We use backward induction.
- Assume firm 1 chooses y_1
- After observing y_1 , firm 2 faces its profit-maximization problem: $\max_{y_2 \geq 0} [a - b(y_1 + y_2)]y_2$, solution is $y_2 = \frac{a - by_1}{2b}$
- Firm 1 knows that firm 2 will choose quantity after observing its quantity choice.
- Backward induction also states that 2 will choose $y_2 = (a - by_1)/2b$ if firm 1 chooses y_1 .
- Therefore, firm 1 faces the profit-maximization problem $\max_{y_1 \geq 0} [a - b(y_1 + y_2)]y_1$
- Substituting y_2 by the constraint, firm 1's problem can be rewritten as $\max_{y_1 \geq 0} [a - b(y_1 + \frac{a - by_1}{2b})]y_1$
- This implies $y_1^* = \frac{a}{2b}$ and $\pi_1^* = \frac{a^2}{8b}$
- $y_2^* = \frac{a}{4b}$ and $\pi_2^* = \frac{a^2}{16b}$
- 厂商 1 的产量高，拿到的利润更高，存在先供优势
- $y_1^* + y_2^* = \frac{3a}{4b}$
- Firm 1, the leader, chooses the point on firm 2's reaction curve that touches firm 1's lowest possible isoprofit line

- 等利润线越往下走，利润是越高的，可以看极端情况：全部由 y_1 生产。等利润线的形状：向上凸，画一条水平线： y_2 的产量相同的情况，和黑色的线（1 的最优反应函数线）交点应当是最高利润点，从而向上凸。



- $\pi_{1S}^* > \pi_{1C}^*$: this is obvious because in the Stackelberg model firm 1 always has the opportunity to choose the quantity that it would choose in the Cournot model. this is usually called the first mover advantage (先动优势)
- $y_{1S}^* > y_{1C}^*$: Firm 1 always would like firm 2 to choose a lower quantity. Intuitively, by committing to a quantity higher than it would choose in the Cournot model, firm 1 forces firm 2 to choose a lower quantity.
- $y_{2S}^* < y_{2C}^*$: 厂商 2 始终在同一条线上行动，this is because of $y_{1S}^* > y_{1C}^*$ and the fact that firm 2's best response decreases in y_1
- $y_{1S}^* + y_{2S}^* > y_{1C}^* + y_{2C}^*$: because of $y_{1S}^* > y_{1C}^*$ and the fact that the slope of firm 2's best response function is larger than -1
- $\pi_{2S}^* < \pi_{2C}^*$: 产量更少了，但是总产量变多让市场价格变低了，从而利润一定会更低了
- 厂商 1 在古诺的情况下，可不可以通过威胁达到斯塔伯格的情况呢？这是不可置信的威胁，厂商 2 的角度来说知道，如果厂商 2 真的选了 y_{2S}^* ，厂商 1 最优选择不是 y_{1S}^* ，而是最优反应线上的选择。还是会回到古诺均衡。什么情况下会让厂商 2 相信这个威胁呢？政府扶植策略
- 比如说厂商 1 是波音，厂商 2 是空客，美国说要给波音提供补贴，政府让空客相信了波音的反应线要向右移动，从而交点可能在斯塔伯格的情况，从而在同时行动的时候也可以产生斯塔伯格的均衡。

2.6 Bertrand Model

- Price competition as opposed to quantity competition. There are two firms $i = 1, 2$ selling homogeneous products. There is one consumer with a unit demand.
- The two firms announce their prices simultaneously. Seeing the prices, the consumer simply buys from the firm which charges a lower price. When there is a tie in prices, the consumer randomizes between the two firms with equal probability. Constant marginal cost is $c > 0$ for both firms.
- The only consumer in this setting can also be (is usually) interpreted as a continuum of identical consumers with total unit mass.
- The randomization when there is a tie in prices is then interpreted as that half of the consumers buy from one firm and the other half buy from the other firm.
- Assume firm 2 announces a price p_2 , then firm 1's profit function can be written as

$$\pi_1(p_1, p_2) = \begin{cases} p_1 - c & \text{if } p_1 < p_2 \\ 0.5(p_1 - c) & \text{if } p_1 = p_2 \\ 0 & \text{if } p_1 > p_2 \end{cases}$$

- We can similarly write out firm 2's profit function
- Suppose (p_1^*, p_2^*) is a Nash equilibrium
- Is it possible that $p_1^* > p_2^*$, i.e., firm 2 gets all the demand and firm 1 gets nothing?
- If $p_2^* > c$, firm 1 can lower its price to p_1 such that $p_2^* > p_1 > c$. In this way, firm 1 can get positive profit
- If $p_2^* = c$, then firm 2 can raise its price to p_2 such that $p_1^* > p_2 > p_2^*$. In this way, firm 2 can get positive profit.
- If $c > p_2^*$, then firm 2 gets negative profits. It can get zero by announcing any price $p_2 > p_1^*$.
负利润变成零利润也是提高!
- 由于对称性, 小于也不可能
- What about $p_1^* = p_2^* = p_*$
- If $p_* < c$, then both firm get negative profits. Again one firm can announce a higher price to get zero profit.
- If $p_* > c$, then each of them gets $(p_* - c)/2 > 0$. But one firm can lower its price by an infinitely small amount $\epsilon > 0$ and obtain the whole demand with profits $(p_* - \epsilon - c)$. as long as $0 < \epsilon < \frac{p_* - c}{2}$ this firm can get higher profits.

- Now we are left with $p_1^* = p_2^* = c$.
- This is indeed a Nash equilibrium: given $p_2^* = c$, firm 1 can not gain more by raising its price. Firm 1 will get negative profits if it lowers its price. Hence $p_1^* = c$ is a best response to p_2^*
- similarly, given $p_1^* = c$, $p_2^* = c$ is a best response for firm 2.
- 十分反直觉：只有两个厂商，但是变成了完全竞争的同样情况，这就是过度竞争了，两个厂商没必要这样子竞争！在讨论寡头市场的时候，更多的合理的是古诺模型的刻画，有些改进 Bertrand 的情况。
- The Bertrand model serves as a benchmark to think of sharp small-number price competition. The striking result that with only two firms the competitive price is obtained is troubling to observers of most markets
- There are several ways to resolve this paradox: firms compete in quantities (the Cournot model), capacity constraints, differentiated products, dynamic models with repeated consumer purchases, repeated interactions between firms.

General Equilibrium

姜峻赫

2022.12.10

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1 Co-operative Behavior: Collusion

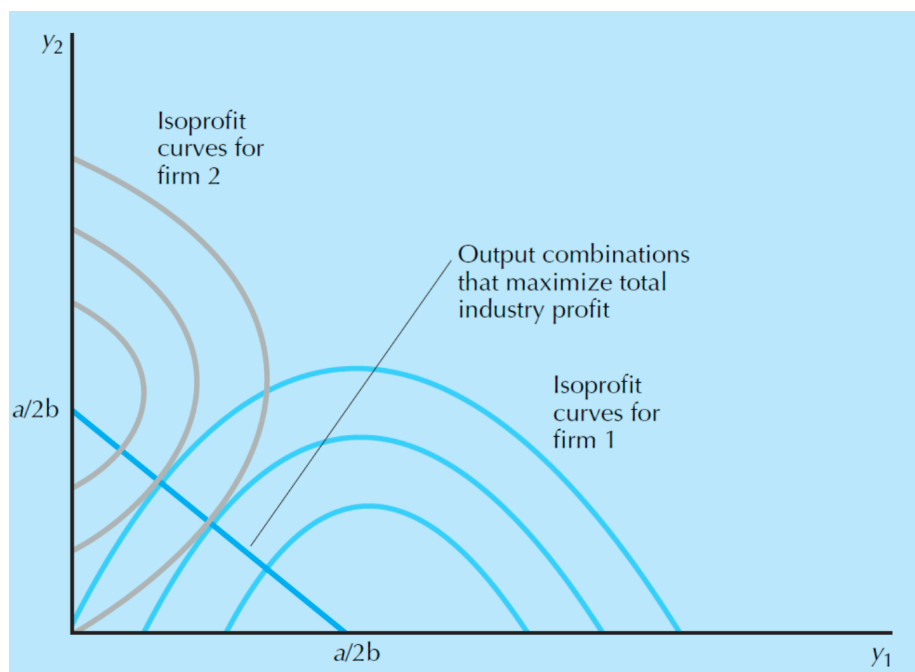
- Collusion is illegal in US, but not for international cartels.(OPEC)
- Goal of cartel: Joint profit maximization, can achieve (joint) monopoly profits ,must divide output, profits among cartel members
- Instability: successful cartel has $p > MC$, a firm gets huge profits if lowers own price while others hold price constant (cheat on agreement)
- 每个人都有偏离的动力:

$$p(y_1 + y_2) + p'(y_1 + y_2)(y_1 + y_2) = MC_1(y_1)$$

$$\frac{\partial \Pi_1}{\partial y_1} = p(y_1 + y_2) + p'(y_1 + y_2) - MC_1(y_1) = -p'(y_1 + y_2)y_2 > 0$$

- 此时厂商 1 单方面提高自己的产量，可以提高利润，存在背离合约的动力。

- 想达成稳定的 cartel，要用其他的措施来消除背离合约的动力，譬如制裁措施，让长远来看单期偏离无好处。



- 在合谋利润最大化的情况下：给定一方的选择，另一个厂商的选择的点连成线就如上。这里是直线是特例，用了直线需求曲线和边际成本为 0 的条件。应该是等利润线的切点的连线。

2 General Equilibrium

- Up until now we have been ignoring the effect of these other prices on the market equilibrium.
- When we discussed the equilibrium conditions in a particular market, we only looked at how demand and supply were affected by the price of the particular good we were examining
- This is called partial equilibrium analysis (局部均衡分析)
- In this chapter we will begin our study of general equilibrium analysis (一般均衡分析): how demand and supply conditions interact in several markets to determine the prices of many goods

2.1 Pure Exchange Economies

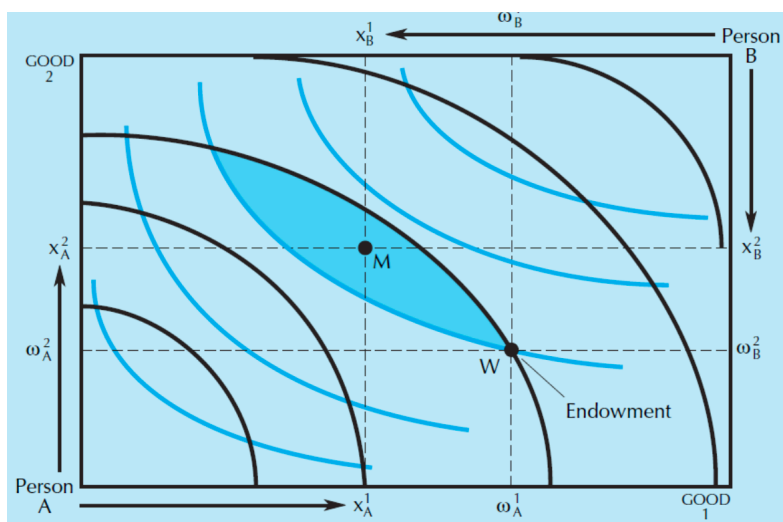
- To keep things simple, we start with pure exchange economy (纯交换经济)

- A pure exchange economy is an economy without production: people have fixed endowments of goods, they can trade these goods among themselves
- To further simplify, we will focus on an economy consisting of: 2 goods, denoted by 1 and 2; 2 agents, denoted by A and B
- Agent $i = A, B$ has $w_i = (w_i^1, w_i^2)$ initial endowments (初始禀赋) and utility function $u_i(x_i^1, x_i^2)$
- The total endowments in this economy are thus $w_A + w_B$ (this is a vector!)
- The agent will exchange some of these goods with each other to end up at a final allocation (最终配置)
- an allocation is a pair of vectors $x_A = (x_A^1, x_A^2)$ and $x_B = (x_B^1, x_B^2)$
- a feasible allocation (可行的配置) is an allocation such that

$$x_A + x_B = w_A + w_B$$

Edgeworth Box

- Edgeworth box (埃奇沃斯盒) can be used to analyze the exchange of two goods between two people



- 蓝色阴影部分叫做交换核 (exchange core)

- 在 w 点, 有不等式成立:

$$\frac{MU_A^1}{MU_A^2} < \frac{MU_B^1}{MU_B^2}$$

- 对于 A, 更喜欢 2, 对于 B, 更喜欢 1. 也就是存在交换的可能性, 两者都愿意交换! 在蓝色区域内是可以两个人交换的, 因为效用都变高了!

Utility Maximization

- Agents are assumed to take the market price as given and then to choose a consumption bundle out of their budget set to maximize their utilities.
- in this economy, i's income comes from "selling" his endowments at the market price p:

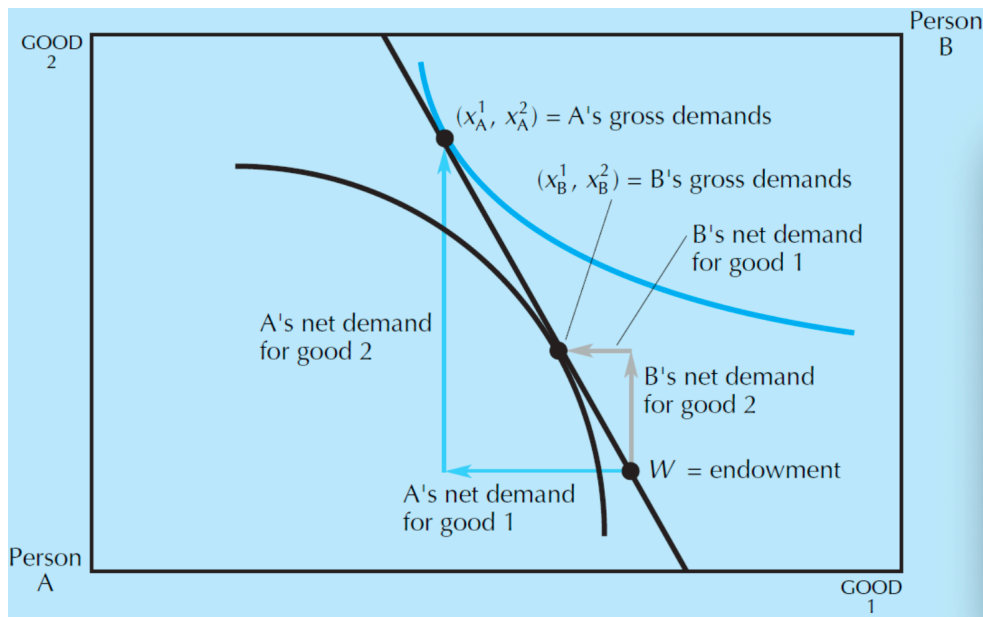
$$\max_{(x_i^1, x_i^2)} u_i(x_i^1, x_i^2) \quad s.t. \quad p^1 x_i^1 + p^2 x_i^2 \leq p^1 w_i^1 + p^2 w_i^2$$

- Let $d_i(p) = (d_i^1(p), d_i^2(p))$ denote a solution to the above utility maximization problem. $d_i(p)$ is agent i's (gross) demand given the market price p.
- The excess demand (超额需求)(or net demand 净需求) is then

$$e_i(p) = d_i(p) - w_i$$

- The sum of excess demand across agents is called the aggregate excess demand (总超额需求)

$$z(p) = e_A(p) + e_B(p)$$



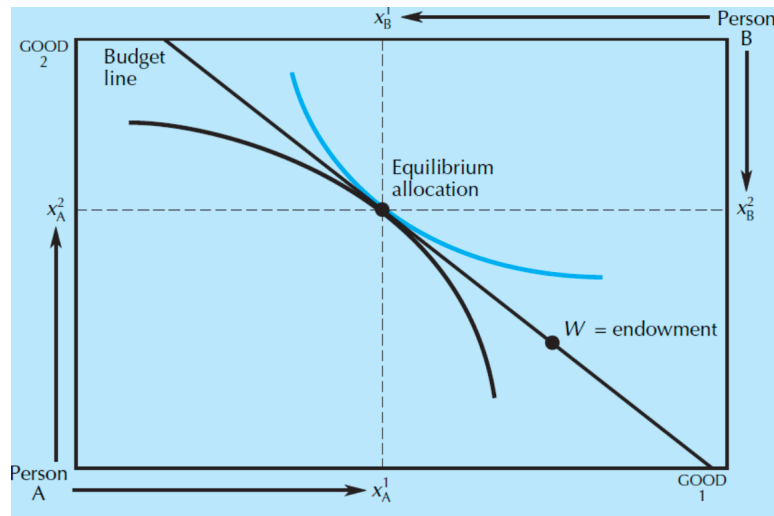
- As we can see from previous figure, for arbitrary market price p, it might be the case that

$$d_A(p) + d_B(p) \neq w_A + w_B$$

- Or equivalently, $z(p) \neq 0$, We are interested in those p's under which the above inequalities becomes equalities.
- 这张图里面, 对于商品 1 来说, 整个市场的供给来自于 A, 而 B 的需求没有那么多, 出现了供过于求的数量。而商品 2 来说, B 的供给不能满足 A 的需求
- 所以价格不会停留在这种情况下, 会改变价格比例。p1 会下降, p2 会上升

2.2 Competitive Equilibrium

- A competitive (Walrasian) equilibrium of this economy consists of a market price p^* and an allocation (x_A^*, x_B^*) such that: given price p^* , x_i^* maximizes i 's utility, i.e. $d_i(p^*) = x_i^*$; market clears: $x_A^* + x_B^* = w_A + w_B$
- Condition 1 states that agents behave optimally given the price; Condition 2 states that the price equates demand and supply.
- Condition 1 and 2 together is equivalent to $z(p^*) = 0$



- 此时达成了供需平衡，从而价格不会发生变化，这是市场出清条件带来的。
- The model is silent about how equilibrium price is formed. We only know that this equilibrium price mysteriously clear the markets.
- One way to think about the price formation is to consider a hypothetical auctioneer who adjusts the price so that markets clear.
- Price-taking is hardly the case for trade between two agents. One way to reconcile is to think about the two agents as two types of agents and assume each type has many many identical agents

2.3 A Numerical Example

- Assume the two consumers have utilities:

$$u_A(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$$

$$u_B(x_1, x_2) = x_1^\beta x_2^{1-\beta}$$

- Endowments: w_A and w_B , We try to find its competitive equilibrium.(用到结论，花在某物品上的金钱数量占比刚好是总量的指数倍数)

$$d_A^1(p) = \alpha \frac{p_1 w_A^1 + p_2 w_A^2}{p_1}$$

$$d_A^2(p) = (1 - \alpha) \frac{p_1 w_A^1 + p_2 w_A^2}{p_1}$$

$$d_B^1(p) = \beta \frac{p_1 w_B^1 + p_2 w_B^2}{p_1}$$

$$d_B^2(p) = (1 - \beta) \frac{p_1 w_B^1 + p_2 w_B^2}{p_1}$$

- Market clearing conditions states:

$$d_A^1(p) + d_B^1(p) = w_A^1 + w_B^1$$

$$d_A^2(p) + d_B^2(p) = w_A^2 + w_B^2$$

$$\Rightarrow (1 - \alpha)p_1 w_A^1 + (1 - \beta)p_1 w_B^1 = \alpha p_2 w_A^2 + \beta p_2 w_B^2$$

$$\frac{p_1^*}{p_2^*} = \frac{\alpha w_A^2 + \beta w_B^2}{(1 - \alpha)w_A^1 + (1 - \beta)w_B^1}$$

- Plugging this relative price back into the demand functions will pin down the equilibrium allocations.
- Two observations from previous example: If (p^*, x_A^*, x_B^*) is a competitive equilibrium, so is $(\lambda p^*, x_A^*, x_B^*)$ where λ is a positive scalar. If $p > 0$, i.e. prices are positive, then the fact that one market clears implies that the other clears as well.
- 此处 p 为向量，且只与 p 的比例有关（从而表现为直线斜率的变化），与具体大小无关。且一个市场出清意味着另一个市场出清。
- Therefore, when solving for equilibrium prices, we usually normalize the price of one good to be 1(计价物). This good is called a numeraire, the choice of the numeraire can be arbitrary
- The second observation is a simple implication of Walras' Law (瓦尔拉斯法则):

$$p_1 z_1(p) + p_2 z_2(p) = 0$$

$$p_1(e_A^1 + e_B^1) + p_2(e_A^2 + e_B^2) = 0$$

- where $z_1(p)$ and $z_2(p)$ are the aggregate excess demand for good 1 and 2 respectively.
- Walras' Law is nothing but an accounting identity, the sum of all agents' budget constraints,

$$p_1 x_i^1(p) + p_2 x_i^2(p) = p_1 w_i^1 + p_2 w_i^2$$

$$p_1(x_i^1(p) - w_i^1) + p_2(x_i^2(p) - w_i^2) = 0$$

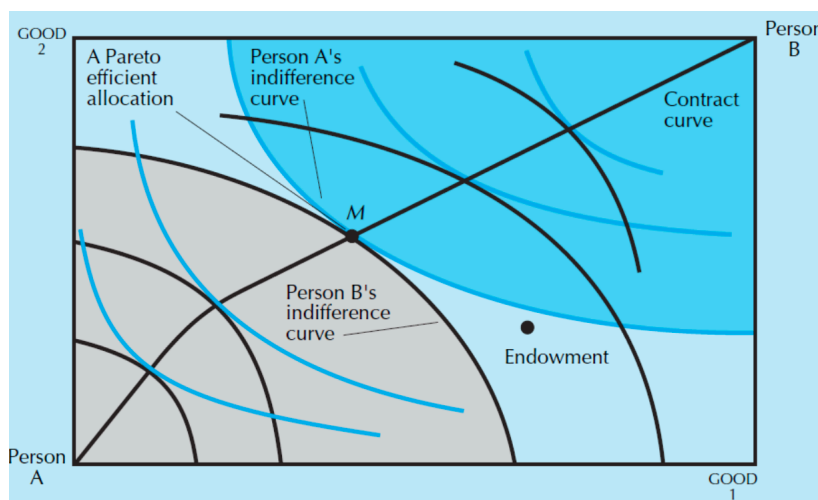
- 只是一个简单的变形！就有了瓦尔拉斯法则！从而有一个市场出清，另外一个一定也会出清。
- If there are n markets, $n-1$ market clear at a positive price, then the n th market must clear as well.
- One of the central questions in general equilibrium analysis is the existence of competitive equilibrium. In 1954, Kenneth Arrow (1972 Nobel Prize) and G'érard Debreu (1983 Nobel Prize) proved that under fairly general conditions on the preferences, a competitive equilibrium always exists.
- Roughly speaking, the conditions are continuity, convexity and monotonicity.

3 Pareto Efficiency

- Are competitive equilibria outcomes “desirable”? Pareto efficiency is a universally accepted concept in economics to describe social desirability
- Formally, in our 2 agents economy, we say a feasible allocation (x_A, x_B) Pareto dominates another feasible allocation $(\hat{x}_A, \hat{x}_B)$ if

$$u_i(x_i) > u_i(\hat{x}_i) \quad u_j(x_j) \geq u_j(\hat{x}_j)$$

- A feasible allocation (x_A, x_B) is Pareto efficient if it is not Pareto dominated by any other feasible allocation. In words, a feasible allocation is Pareto efficient if no one can be made strictly better off without hurting others.
- Illustrations of Pareto (in)efficient allocations and contract curve:



- At a Pareto efficient allocation such as M, each person is on his highest possible indifference curve, given the indifference curve of the other person.
- The indifference curves of the two agents must be tangent at any Pareto efficient allocation in the interior of the box.
- The set of all Pareto efficient points in the Edgeworth box is known as the Pareto set (帕累托集), or the contract curve (契约曲线). All “final contracts” for trade must lie on the Pareto set
- 契约曲线就是上图里面的黑线。有点像竞争性均衡的切点? 可以得出结论: 竞争性均衡就是帕累托有效率的点! (福利经济学第一定律)

4 Welfare Theorem

- First welfare theorem (福利经济学第一定理)
- if agents' preferences are monotonic (utility functions are increasing), then any competitive equilibrium allocation is Pareto efficient.
- As a result, this theorem is widely considered as a formal statement of “invisible hand”.
- Assume (p^*, x_A^*, x_B^*) is a competitive equilibrium. A feasible allocation (x_A, x_B) Pareto dominates (x_A^*, x_B^*)
- Assume w.o.l.g $u_A(x_A) > u_A(x_A^*)$ and $u_B(x_B) \geq u_B(x_B^*)$
- Then we know

$$p_1^* x_A^1 + p_2^* x_A^2 > p_1^* x_A^{1*} + p_2^* x_A^{2*} \quad \text{显示性偏好, 只能是买不起}$$

$$p_1^* x_B^1 + p_2^* x_B^2 \geq p_1^* x_B^{1*} + p_2^* x_B^{2*}$$

- 但是根据可行配置的定义 $(x_A^1 + x_B^1 = w_A^1 + w_B^1)$

$$p_1^*(w_A^1 + w_B^1) + p_2^*(w_A^2 + w_B^2) > p_1^*(w_A^1 + w_B^1) + p_2^*(w_A^2 + w_B^2)$$

- 矛盾了! 两边明明相等!
- 两个人自由交换的选择, 却是对社会最有效的一个结果。
- The second welfare theorem (福利经济学第二定理) asks the converse: can every Pareto efficient allocation be a competitive equilibrium if we adjust the initial endowment (e.g. by redistributing wealth)?
- Second welfare theorem: if agents' preferences are continuous, monotonic and convex, then for any Pareto efficient allocation there exists some initial endowment redistribution such that the efficient allocation is a competitive equilibrium allocation after redistribution.

4.1 One-consumer, One-producer Economy

- We extend our pure exchange economy to incorporating production. To start with, we first consider the simplest possible setting. There is one price-taking consumer and one price-taking firm.
- There are two goods: labor (or leisure) of the consumer; a consumption good produced by the firm
- The consumer has a utility function $u(c, l)$ where c stands for consumption good and l for leisure. He is endowed with L units of leisure initially
- The firm can hire the consumer and produce the consumption good. the production function is $f(L)$
- Firms in any economy must be owned by someone. Thus in our economy, the single consumer is the sole owner of the firm and receives all the profits earned by the firm. This economy is sometimes referred to as Robinson Crusoe economy (鲁滨逊·克鲁索经济).
- Given price p for the consumption good and wage w for labor, the firm seeks to maximize its profits by choosing how much labor to hire:

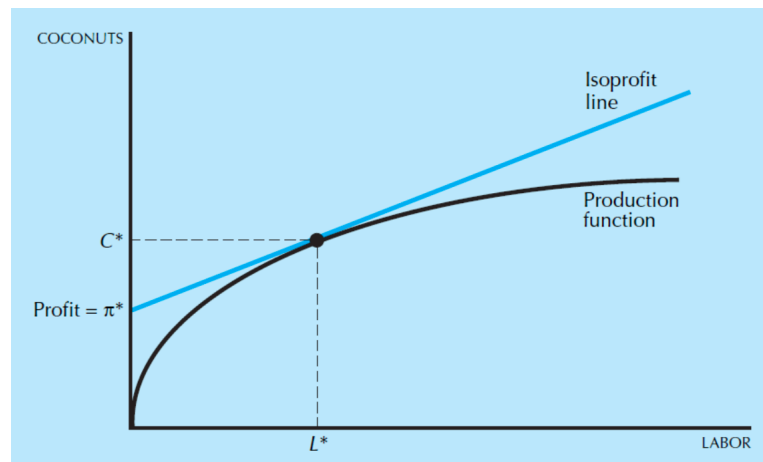
$$\max_{L \geq 0} pf(L) - wL$$

- Let $\pi(p, w)$ be the corresponding profits. We have

$$\pi(p, w) = pf(L) - wL$$

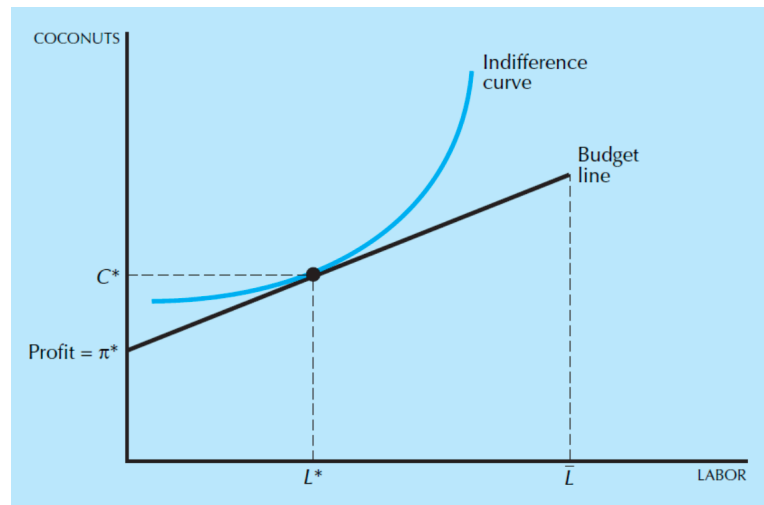
$$f(L) = \frac{\pi}{p} + \frac{wL}{p}$$

- The firm's problem can be illustrated as:



- The consumer decides how much leisure to enjoy and how much consumption good to buy to maximize his utility. If the consumer enjoys l units of leisure, then he supplies $L-l$ units of labor and earns a total wage $w(L-l)$
- Because the consumer also owns the profits made by the firm, his total income is $w(L-l) + \pi(p, w)$
- Thus the consumer's problem is:

$$\max_{(c,l)} u(c, l) \quad s.t. \quad pc \leq w(L-l) + \pi(p, w)$$



- 喜欢闲暇，不喜欢劳动，劳动需要消费品来保证！
- A competitive equilibrium of this economy consists of prices (p^*, w^*) and allocations (c^*, l^*, L^*) such that
- Given (p^*, w^*) , L^* solves firm's profit maximization:

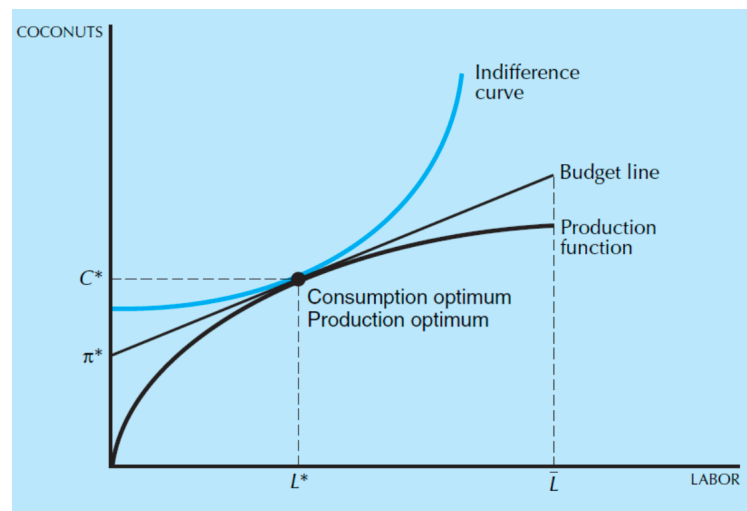
$$\max_{L \geq 0} p^* f(L) - w^* L$$

- Given (p^*, w^*) , (c^*, l^*) solves consumer's utility maximization:

$$\max_{(c,l)} u(c, l) \quad s.t. \quad pc \leq w(L-l) + \pi(p, w)$$

- Market clear:

$$c^* = f(L^*) \quad l^* = L - L^*$$



- 说白了就是无差异曲线和生产函数的切点。
- Moreover, a particular consumption-leisure bundle can be a competitive equilibrium if and only if it maximizes the consumer's utility subject to the economy's technological and endowment constraints, the first welfare theorems, the second welfare theorems.